

The Interweaving City:

Mixed-Income Housing and Multi-Layered Interaction for Social Cohesion

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Approved _____
Adviser

Acknowledgment

This thesis marks the completion of my studies in the Master of Science in Architecture program at the College of Architecture, Illinois Institute of Technology. Throughout this process, I have been fortunate to receive the support and guidance of many people, to whom I am deeply grateful.

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I am also grateful to my parents and family, who have supported me in every way possible throughout this journey. Their encouragement and belief in me have been a source of strength. A special thanks to my wife, who has been by my side through every challenge. Her patience and support have made this process much more manageable.

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Lastly, I acknowledge the many influences—academic, professional, and personal—that have shaped this research. This thesis reflects an ongoing interest in the role of architecture in fostering more inclusive and connected communities.

To all who have supported and encouraged me—thank you.

Thesis Statement

"The Interweaving City" explores strategies to foster social integration through multi-layered spatial design. It addresses the challenges faced by marginalized communities and critiques the limitations of existing urban policies, emphasizing the need for integrated approaches to bridge social, spatial, and economic divides. Through three projects at the urban, neighborhood, and block scales, this thesis proposes multi-layered design strategies to revitalize neglected areas, promote social integration across income levels, and encourage healthy, meaningful interactions among neighbors.

The projects include Exodus City, a floating urban design on Lake Michigan for displaced populations, which examines how layered spatial configurations can foster belonging and community; New Bronzeville, a functional-mixed development centered on the renovation of a stadium and the creation of a transportation hub, aiming to catalyze urban revitalization and reconnect fragmented neighborhoods; and Permeable Block, a study of clustered public and private spaces within a mixed-income residential block, designed to foster interaction and community integration.

This thesis ultimately positions architecture and urban design as powerful tools for creating inclusive environments that strengthen community ties and provide a shared sense of belonging.

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01 Global Issues

URBANIZATION, DISPLACEMENT, AND HOUSING INEQUALITY

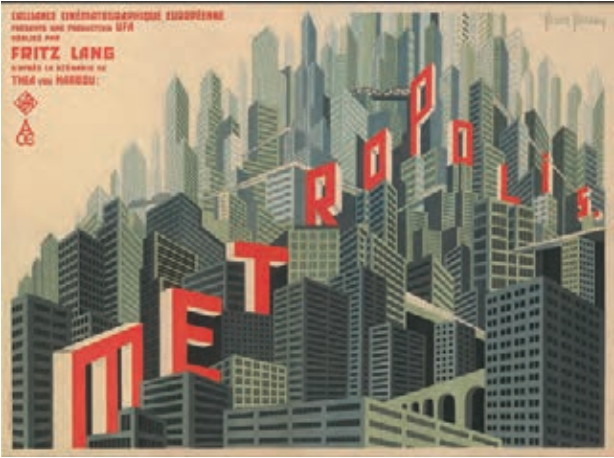
Rapid urbanization has led to overcrowding, rising housing costs, and uneven development between cities, exacerbating economic and social disparities. Meanwhile, conflict and political instability continue to displace millions, creating a growing refugee crisis with urgent resettlement needs. Simultaneously, the housing crisis worsens as home prices soar, supply fails to meet demand, and residential inequalities deepen, widening gaps in living conditions between communities.

An aerial photograph of New York City, showing a dense urban landscape with numerous skyscrapers and buildings. Central Park is visible in the lower-left quadrant, surrounded by greenery. The Hudson River is visible on the left side, and the city extends towards the horizon under a hazy sky. The word "URBANIZATION" is overlaid in large, semi-transparent white capital letters across the center of the image.

URBANIZATION

"By 2050, nearly 70% of the world's population is projected to live in urban areas, straining infrastructure, housing, and resources."

United Nations, World Urbanization Prospects (2018)



Metropolis Poster (1927 film)_©Boris Bilinski



The Tower of Babal_©Friedrich-Wilhelm-Murnau-Foundation, Kunsthistorisches Museum



The Underground City of workers_©UFA GmbH



The Upper City_©UFA GmbH

Metropolis

Definition and Characteristics of a Metropolis

A metropolis is a large city or conurbation that serves as a significant economic, political, and cultural hub for a country or region. It also acts as a critical node for regional and international connections, facilitating commerce and communication. Unlike smaller urban centers within a larger agglomeration, a metropolis is defined by its central role in governance and integration within the urban network.

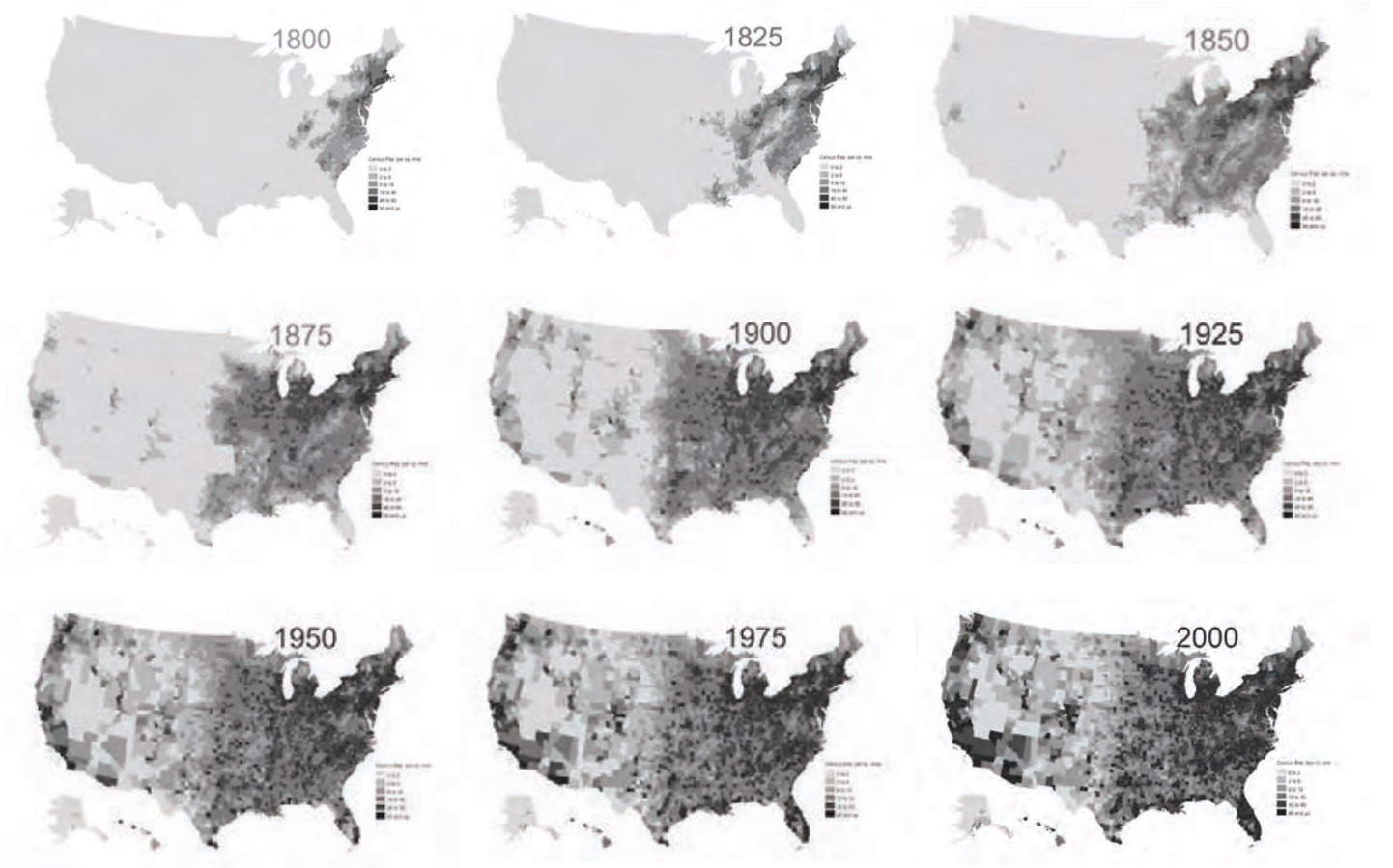
Historical Context and Etymology

The term "metropolis" derives from the Greek word "μητρόπολις"(metropolis), combining "μήτηρ"(mētēr), meaning "mother," and "πόλις"(polis), meaning "city" or "town." In ancient Greece, it referred to the mother city of a colony, maintaining cultural and political connections with its settlements. Overtime, the term evolved in post-classical Latin to denote the principal city of a province, particularly the seat of governance or the episcopal see of a metropolitan bishop.

In colonial contexts, the term indicated the "mother city" of a colony, highlighting its administrative and cultural primacy. Its usage in Middle English around the 14th century generalized the concept to encompass any large and influential city within a nation.

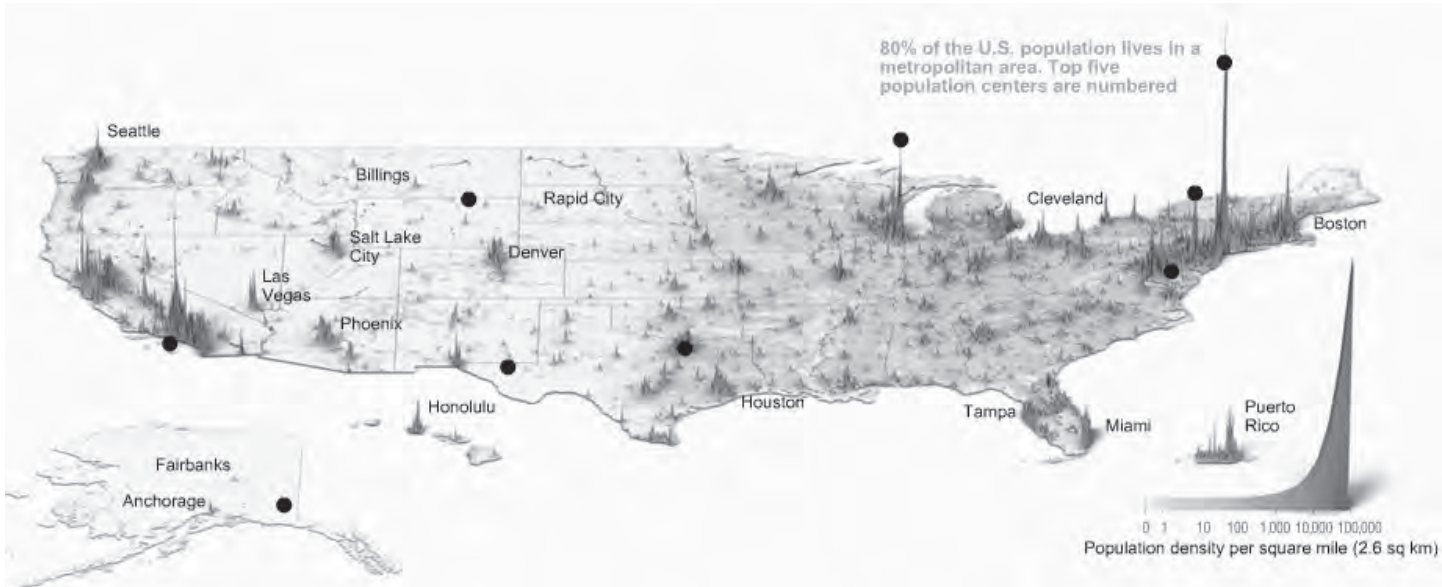
Urbanization: Growth and Challenges

The dynamics of urbanization have fundamentally reshaped the global spatial landscape. According to the UN-Habitat's Global State of Metropolis 2020, 1,934 metropolitan areas with populations exceeding 300,000 accommodate approximately 60% of the world's urban populace, representing nearly one-third of humanity. Urban centers, particularly those exceeding 5 million inhabitants, have experienced unprecedented growth, driven by the convergence of economic opportunities, higher income prospects, and the synergistic potential of technological advancements. However, this rapid expansion has engendered significant challenges, including overcrowding, infrastructure overextension, and the exacerbation of socio-economic inequalities. Addressing these complexities necessitates the adoption of equitable and inclusive urban planning paradigms that are both adaptive and resilient.

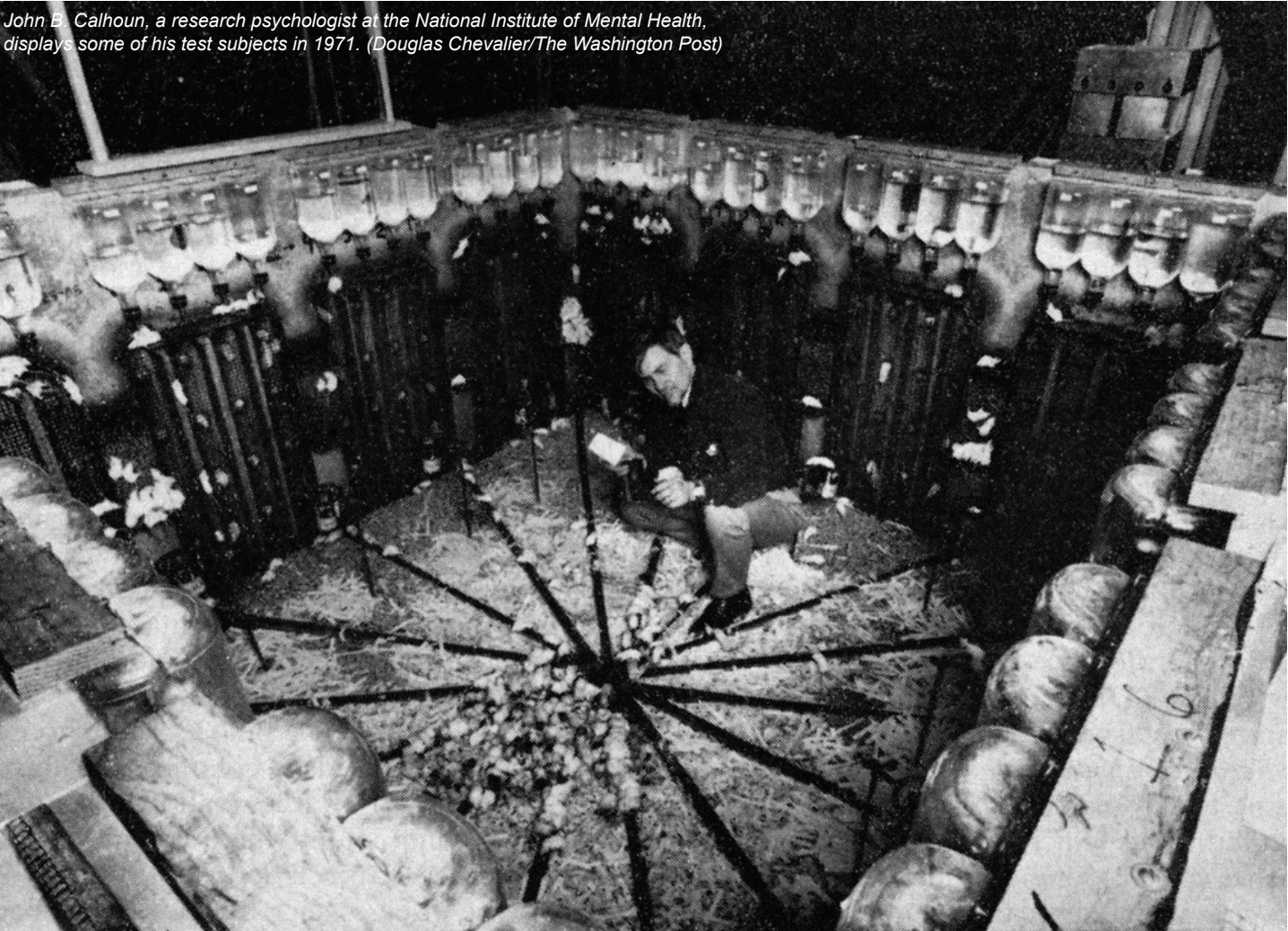


Population Density Over Time: Shifts in U.S. Urban Landscapes

Examining the evolution of U.S. population density provides insight into broader patterns of urban development. From 1790 to 2010, U.S. Census data and Jonathan Schroeder's county-level estimates reveal the migration and settlement trends shaping the nation's urban history. By the mid-20th century, the distribution of the U.S. population began to resemble modern patterns, with urban centers consolidating as economic and social hubs. However, population density shifts also exposed systemic issues, including racial segregation, poverty concentration, and inadequate housing. These challenges continue to shape the discourse on equitable urban development in America.



John B. Calhoun, a research psychologist at the National Institute of Mental Health, displays some of his test subjects in 1971. (Douglas Chevalier/The Washington Post)



Behavioural Sink; Mouse Utopia Experiment

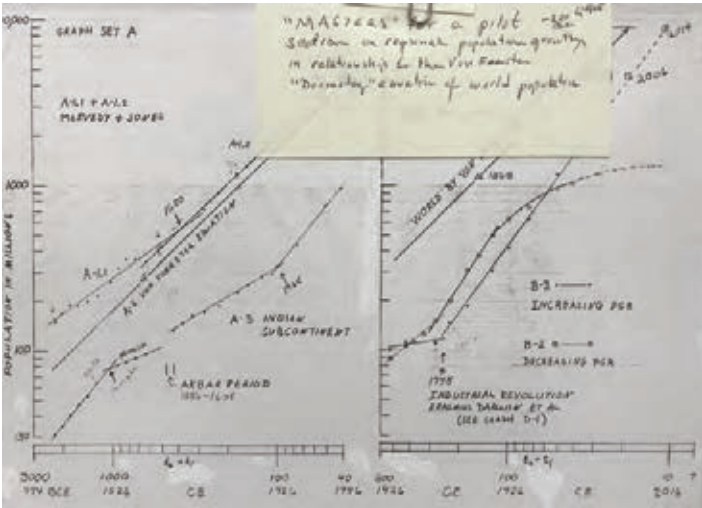
Behavior and Population Density: Lessons from the Animal World

The behavioral effects of population density have long intrigued researchers, as illustrated by John Calhoun's experiments from the 1950s to the 1970s. By constructing artificial rodent habitats with ample food and space, Calhoun sought to explore the social and behavioral consequences of overpopulation. Initially envisioned as utopias, these environments rapidly devolved into overcrowded, chaotic systems marked by aggression, social withdrawal, and population collapse. Calhoun termed these phenomena "behavioral sinks," signaling the dire consequences of density-induced social breakdown.

Despite initial interpretations tying these findings to human urbanization and overpopulation fears, later analyses revealed that the challenges in Calhoun's "Universe 25" were less about sheer density and more about inequitable resource distribution and isolation. The lessons of these experiments emphasize the importance of equitable design in fostering harmonious social environments, both for humans and other species.

At the peak population, most mice spent every living second in the company of hundreds of other mice. They gathered in the main squares, waiting to be fed and occasionally attacking each other. Few females carried pregnancies to term, and the ones that did seemed to simply forget about their babies. They'd move half their litter away from danger and forget the rest. Sometimes they'd drop and abandon a baby while they were carrying it. The few secluded spaces housed a population Calhoun called, "the beautiful ones." Generally guarded by one male, the females - and few males - inside the space didn't breed or fight or do anything but eat and groom and sleep. When the population started declining the beautiful ones were spared from violence and death, but had completely lost touch with social behaviors, including having sex or caring for their young.

Ultimately, "[r]ats may suffer from crowding; human beings can cope," Ramsden says. "Calhoun's research was seen not only as questionable, but also as dangerous." Another researcher, Jonathan Freedman, turned to studying actual people - they were just high school and universtiy students, but definitely human. His work suggested a different interpretation. Moral decay could arise "not from density, but from excessive social interaction," Ramsden says. "Not all of Calhoun's rats had gone berserk. Those who managed to control space led relatively normal lives."





Delirious New York Rem Koolhaas, 1978

Delirious New York: A Retroactive Manifesto for Manhattan is a 1978 book, written by Dutch architect Rem Koolhaas. The book serves as a retroactive manifesto for Manhattan between 1850 and 1960, analyzing the development of architecture and urban design throughout New York's history from the founding of New Amsterdam by the Dutch, to the design of the Headquarters of the United Nations by Le Corbusier. Rem Koolhaas describes the concept of 'Manhattanism', the theory of the creation and functioning of the city of New York, at length in the book.

The cover image of the first edition of the book was designed by Madelon Vriesendorp. The painting 'Flagrant Délit' depicts the Chrysler Building and the Empire State Building laying in bed, with 30 Rockefeller Plaza intruding on them. The gridiron street pattern of Manhattan is shown through the window, with the rooftops of skyscrapers being faces looking at the ordeal. Furthermore, the nightlight near the Empire State Building is the torch of the Statue of Liberty and a used condom in the shape of a Goodyear Blimp can be seen lying on the bed, referencing the zeppelin docking station built on top of the tower.

Termed as a Retroactive Manifesto by Rem Koolhaas, the book is a database; mapping the history of the most famous city in the world, and how its architecture has grown and influenced its world image. Koolhaas in the beginning points out the inherent flaws in manifestos: a lack of evidence; and states that New York is filled with a mountain of evidence, but has no manifesto recording its progress and history, thereby his motivation to pursue the creation of this very text.

Broken down into various phases of growth into the city we now know as New York; the entire book is categorized into chapters that cover different segments or blocks, covering facts about the city of New York: this narrative method emulates the planning strategy of New York; like the way the city is laid out in blocks, so are the chapters in the book.

From Manhattan to Coney Island, as well as the Empire State Building and Rockefeller Centre; Rem Koolhaas covers an urbanistic perspective and the life-cycle trajectory regarding each major area in the city of New York. From conceptualization to planning, along with the emergence of certain structures, which are now synonymous with the cultural identity of New York. The manifesto can proficiently provide a story-like narrative to a non-fictional, historic excerpt about the architecture and design of the most active city in the world. Despite its inception dating back in 1978, the content matter of the book stays relevant, 43 years later; and gives a comprehensive account of the making of the city we now know as New York.

Starting from the prehistory chapter; Koolhaas develops a historical account of how Manhattan came to be. From its 24-dollar purchase to Peter Minuit in 1626, to the conceptualization and implementation of New Amsterdam; a blueprint for a framework of a utopian Europe. Koolhaas lays out the facts chronologically while giving his perspectives either admonishing it or supporting it.

The Appendix provides a non-canonical conclusion to what New York would have been, by giving a plethora of evidence on unbuilt projects. The second coming of Manhattanism, this time as an explicit doctrine that can claim its place among contemporary urbanisms. Impressive documentation of original materials and unpublished projects provides evidence for this architectural manifesto.

What can we learn from Retroactive Manifesto?

CARPET

Central Park is not only the major recreational facility of Manhattan but also the record of its progress: a taxidermic preservation of nature that exhibits forever the drama of culture outdistancing nature. Like the Grid, it is a colossal leap of faith; the contrast it describes - between the built and the unbuilt - hardly exists at the time of its creation. (p.21)

CONTRAST

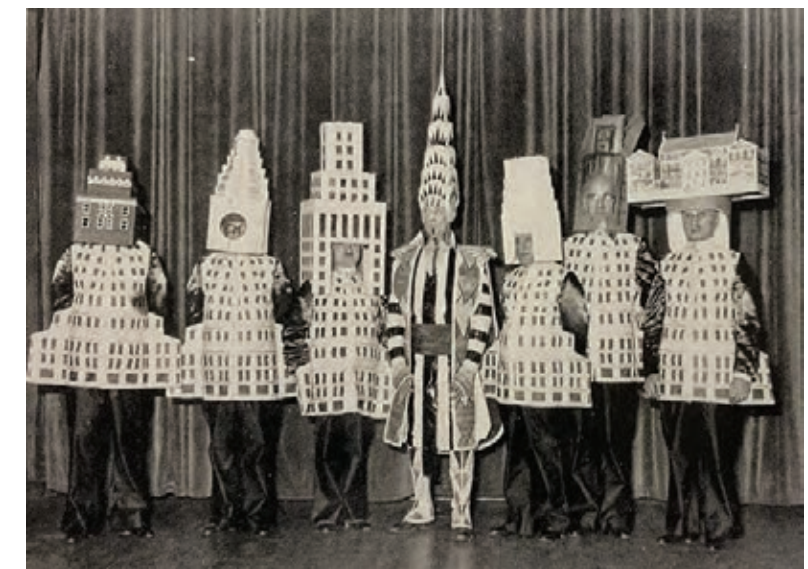
In many ways, the history of Manhattanism as a separate, identifiable architecture is a dialectic between these two forms, with the needle wanting to become a globe and the globe trying, from time to time, to turn into a needle - a cross-fertilization that results in a series of successful hybrids in which the needle's capacity for attracting attention and its territorial modesty are matched with the consummate receptivity of the sphere. (p.27)

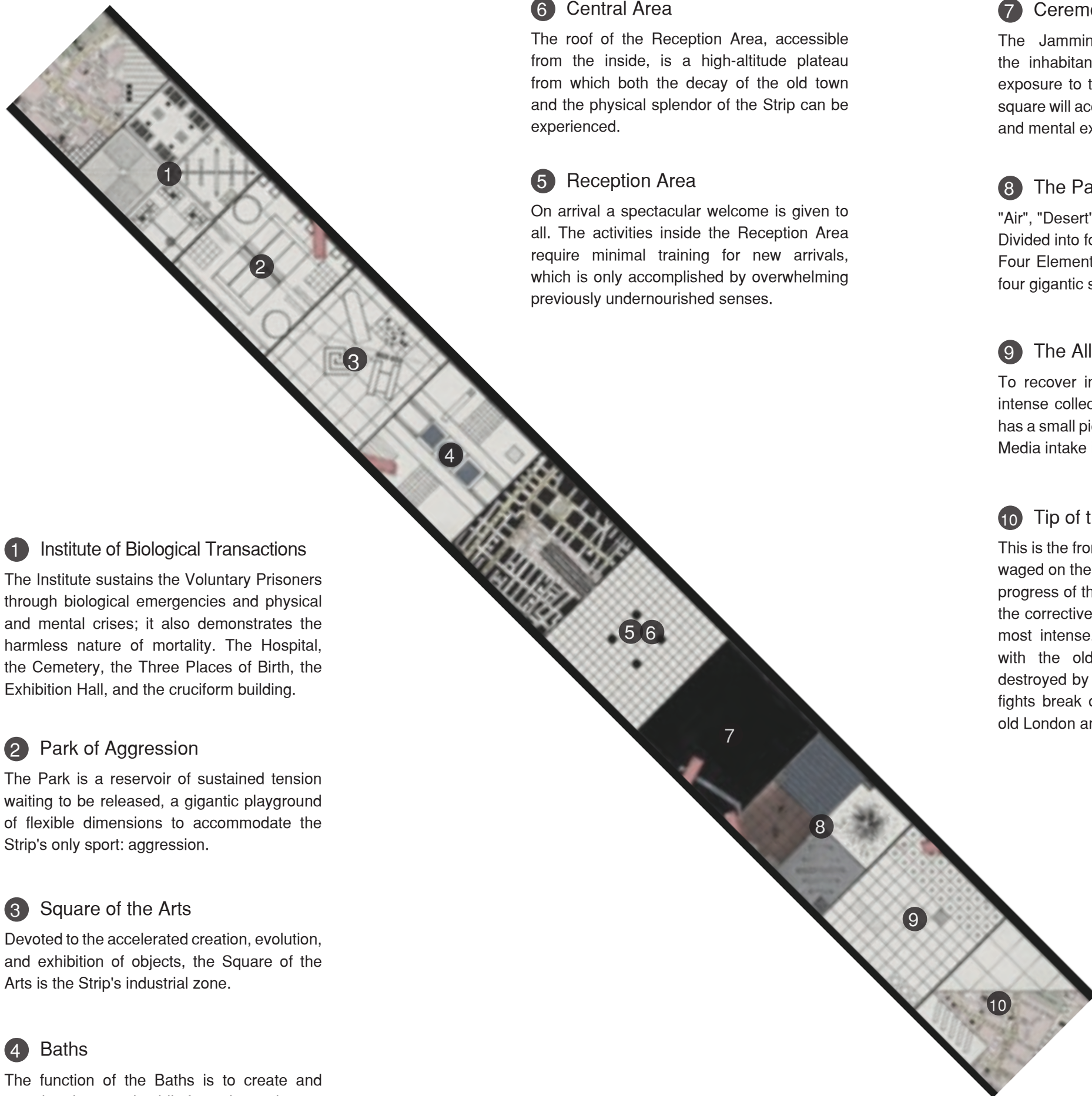
SHORTAGE

In less than a decade they have invented and established an urbanism based on the new Technology of the Fantastic: a permanent conspiracy against the realities of the external world. It defines completely new relationships between site, program, form and technology. The site has now become a miniature state; the program its ideology; and architecture the arrangement of the technological apparatus that compensates for the loss of real physicality. (p. 61)

SOLITUDES

The three lowest floors of the Waldorf form one of the most elaborate manifestations yet of Manhattan's concept of a modern Venice. Even though all their spaces are easily accessible, they are not exactly public; they form a sequence of theatrical "living rooms" - an interior for the Waldorf's houseguests that admits visitors but excludes the general population. These living rooms constitute one of the bloated private realms that together form Manhattan's Venetian system of solitudes. (p.145)





1 Institute of Biological Transactions

The Institute sustains the Voluntary Prisoners through biological emergencies and physical and mental crises; it also demonstrates the harmless nature of mortality. The Hospital, the Cemetery, the Three Places of Birth, the Exhibition Hall, and the cruciform building.

2 Park of Aggression

The Park is a reservoir of sustained tension waiting to be released, a gigantic playground of flexible dimensions to accommodate the Strip's only sport: aggression.

3 Square of the Arts

Devoted to the accelerated creation, evolution, and exhibition of objects, the Square of the Arts is the Strip's industrial zone.

4 Baths

The function of the Baths is to create and recycle private and public fantasies, to invent, test, and possibly introduce new forms of behavior. The building is a social condenser. It brings hidden motivations, desires, and impulses to the surface to be refined for recognition, provocation, and development.

6 Central Area

The roof of the Reception Area, accessible from the inside, is a high-altitude plateau from which both the decay of the old town and the physical splendor of the Strip can be experienced.

5 Reception Area

On arrival a spectacular welcome is given to all. The activities inside the Reception Area require minimal training for new arrivals, which is only accomplished by overwhelming previously undernourished senses.

7 Ceremonial Square

The Jamming Station tower will protect the inhabitants of the Strip from electronic exposure to the rest of the world. This black square will accommodate a mixture of physical and mental exercises, a conceptual Olympics.

8 The Park of the Four Elements

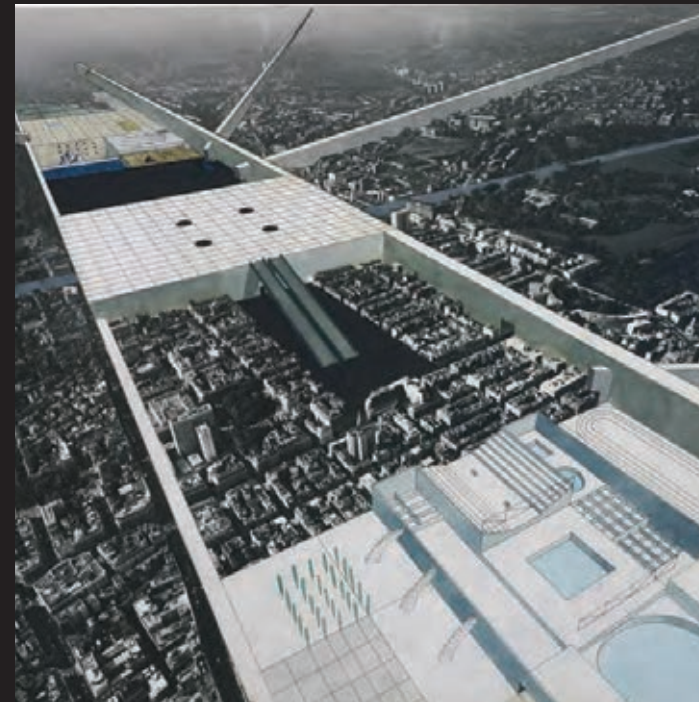
"Air", "Desert", "Water", and "Earth"
Divided into four square areas, the Park of the Four Elements disappears into the ground in four gigantic steps.

9 The Allotments

To recover in privacy from the demands of intense collectivism, each Voluntary Prisoner has a small piece of land for private cultivation. Media intake in this area is nil.

10 Tip of the Strip

This is the frontline of the architectural warfare waged on the old London. Here, the merciless progress of the Strip performs a daily miracle; the corrective rage of the architecture is at its most intense. In a continuous confrontation with the old city, existing structures are destroyed by the new architecture, and trivial fights break out between the inmates of the old London and the Voluntary Prisoners.



Exodus Rem Koolhaas, Elia Zenghelis, Madelon Vriesendorp, and Zoe Zenghelis, 1972

This drawing, part of a series of eighteen drawings, watercolors, and collages titled Exodus, or the Voluntary Prisoners of Architecture, pictures a walled city within the city of London. Tall barriers cut through the urban fabric, an intervention designed to create a new urban culture invigorated by architectural innovation and political subversion. Echoing the work of Italian radical architecture groups of the 1960s and '70s, such as Superstudio and Archizoom, the series is a combination of vivid architectural imagery and "text/script," in Koolhaas's words, that incorporates narration and poetry. The dense pictographic storyboard, reflecting Koolhaas's earlier stints as a journalist and a screenwriter, is intended to be read as a factual and fictional scenario for the contemporary metropolis. The title of the project alludes to West Berlin's situation during the Cold War as a restricted enclave within East Germany, encircled by a forbidding wall—in effect, a prison on the scale of a metropolis, in which people sought refuge voluntarily.



DISPLACEMENT

"110,000,000 people were forcibly displaced around the world due to persecution, war, conflict, generalized violence, or human rights violations."

2023 UNHCR Report



A refugee

A refugee is a person who has been forced to flee their home country due to persecution because of their race, religion, nationality, political opinion or membership in a particular social group. A refugee is often afforded some sort of legal protection, either by their host country's government, the United Nations High Commissioner for Refugees (UNHCR) or both. In the United States, refugees are hand-selected by the U.S. government and are screened in advance. They are subject to background checks and security screenings by multiple U.S. agencies. Only after everything is approved are they brought to the U.S. to reside permanently.

An asylum seeker

An asylum seeker is a person who has fled persecution in their home country and is seeking safe haven in a different country, but has not yet received any legal recognition or status. In several countries, including the U.S., asylum seekers are sometimes detained while waiting for their case to be heard.

Internally displaced person

An internally displaced person, or IDP, is a person who fled their home but has not crossed an international border to find sanctuary. Even if they fled for reasons similar to those driving refugees (armed conflict, generalized violence, human rights violations), IDPs legally remain under the protection of their own government – even though that government might be the cause of their flight.

Migrant

A migrant is a person who chooses to move from their home for any variety of reasons, but not necessarily because of a direct threat of persecution or death. Migrant is an umbrella category that can include refugees but can also include people moving to improve their lives by finding work or education, those seeking family reunion and others.

Refugees: The Crisis of Displacement

The global refugee crisis has reached an unprecedented scale, with the United Nations High Commissioner for Refugees (UNHCR) reporting approximately 122.6 million forcibly displaced individuals worldwide as of mid-2024. This represents a dramatic rise from the 117.3 million recorded at the end of 2023, showcasing the growing severity of forced displacement across the globe.

The primary causes driving this crisis include conflicts, climate-related disasters, and economic instability. In Sudan, ongoing violence displaced more than 5.8 million people in 2023 alone. Similarly, the war in Ukraine has created one of the largest contemporary displacement crises, with millions forced to flee both within the country and abroad.

Prolonged conflicts in regions such as Myanmar further exacerbate the displacement of vulnerable populations.

Climate shocks and economic turmoil also play a significant role, particularly in regions like East Africa and Latin America. Venezuela exemplifies this trend, where over 7.2 million people have fled the country since 2015 due to economic hardship and failing public infrastructure. Many of these individuals have sought refuge in neighboring countries, such as Colombia, Ecuador, and Peru, placing substantial pressure on regional resources.

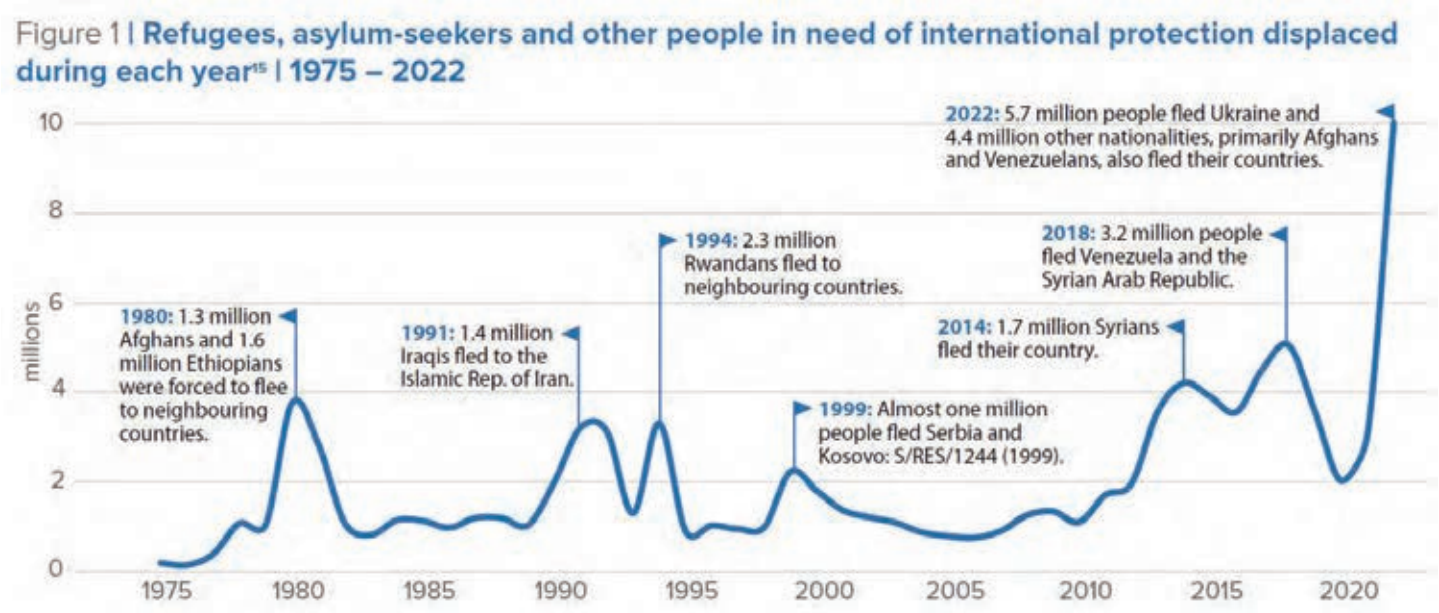
The displaced population comprises several groups, including approximately 37.8 million refugees who have

crossed international borders, asylum seekers awaiting legal determination of their status, and over 62.5 million internally displaced persons (IDPs) who remain within their own countries. Notably, Syria and Ukraine are among the countries with the highest numbers of IDPs, with 6.8 million and 5.4 million people displaced internally, respectively.

Low-and-middle-income countries continue to host the majority of displaced individuals, accommodating 76% of the world's refugees. Countries such as Türkiye, Iran, and Colombia rank among the top hosts, often bearing disproportionate burdens in terms of resources and infrastructure. In contrast, displaced individuals arriving at the southern border of the United States come from a wide range of countries,

including Cuba, Haiti, and Nicaragua, as well as more distant nations such as Cameroon, Ethiopia, and Ukraine.

These statistics underscore the immense scale and complexity of forced displacement, driven by intersecting global challenges of conflict, climate change, and socio-economic instability. The data highlights the regional variations in displacement patterns and the critical need for continued international collaboration to address the crisis effectively.



52%
originate from just three countries

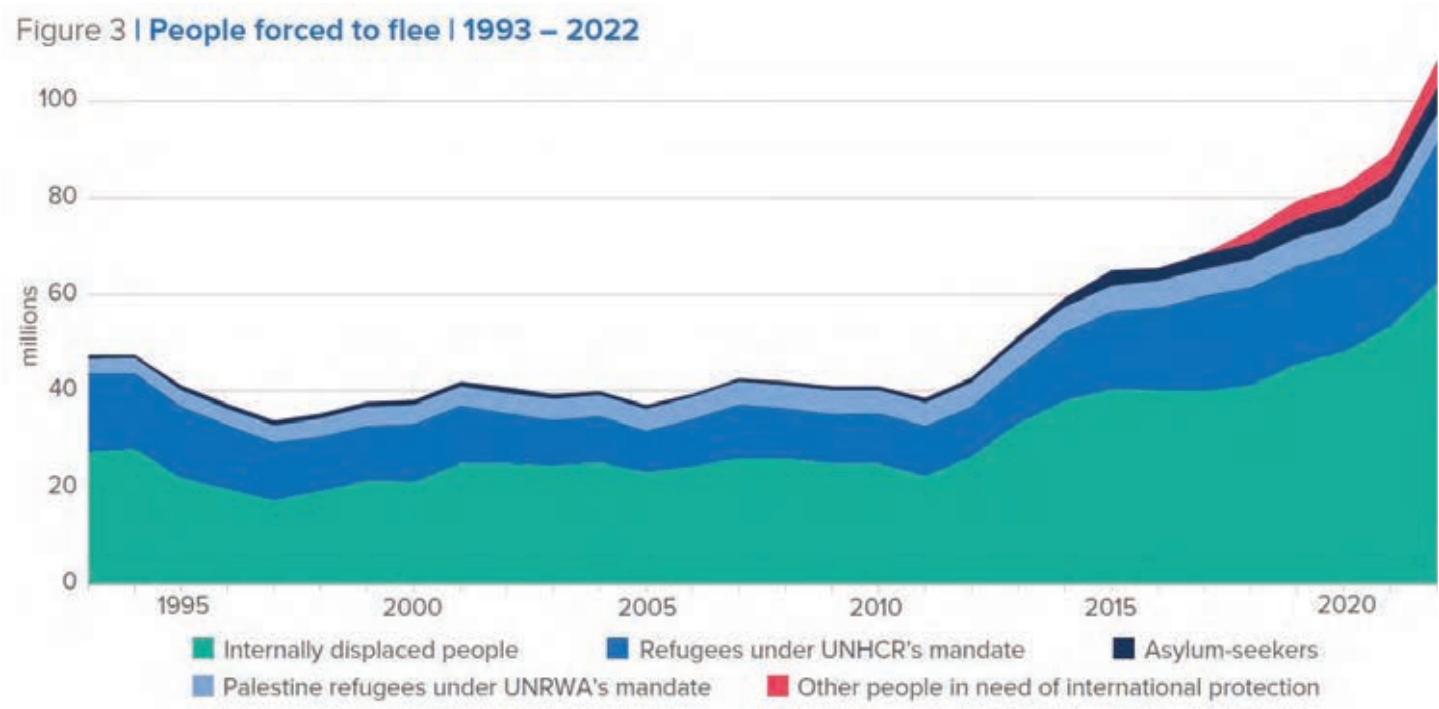
Over half of all refugees under UNHCR's mandate and other people in need of international protection come from just three countries.

Syrian Arab Republic	6.8 million
Ukraine	5.7 million
Afghanistan	5.7 million

38%
hosted in five countries

Türkiye hosts the largest number of refugees, with 3.6 million people, followed by the Islamic Republic of Iran with 3.4 million people. Colombia is third with 2.5 million, including other people in need of international protection.

Türkiye	3.6 million
Islamic Republic of Iran	3.4 million
Colombia	2.5 million
Germany	2.1 million



43.3 million
are children

At the end of 2022, of the 108.4 million forcibly displaced people, an estimated 43.3 million (40 percent) are children below 18 years of age.

1.9 million
children were born as refugees

Between 2018 and 2022, an average of 385,000 children were born as refugees per year.

453,600
refugees returned or were resettled

Some 339,300 refugees returned to their countries of origin during 2022 while 114,300 were resettled (with or without UNHCR's assistance).

4.4
million stateless people

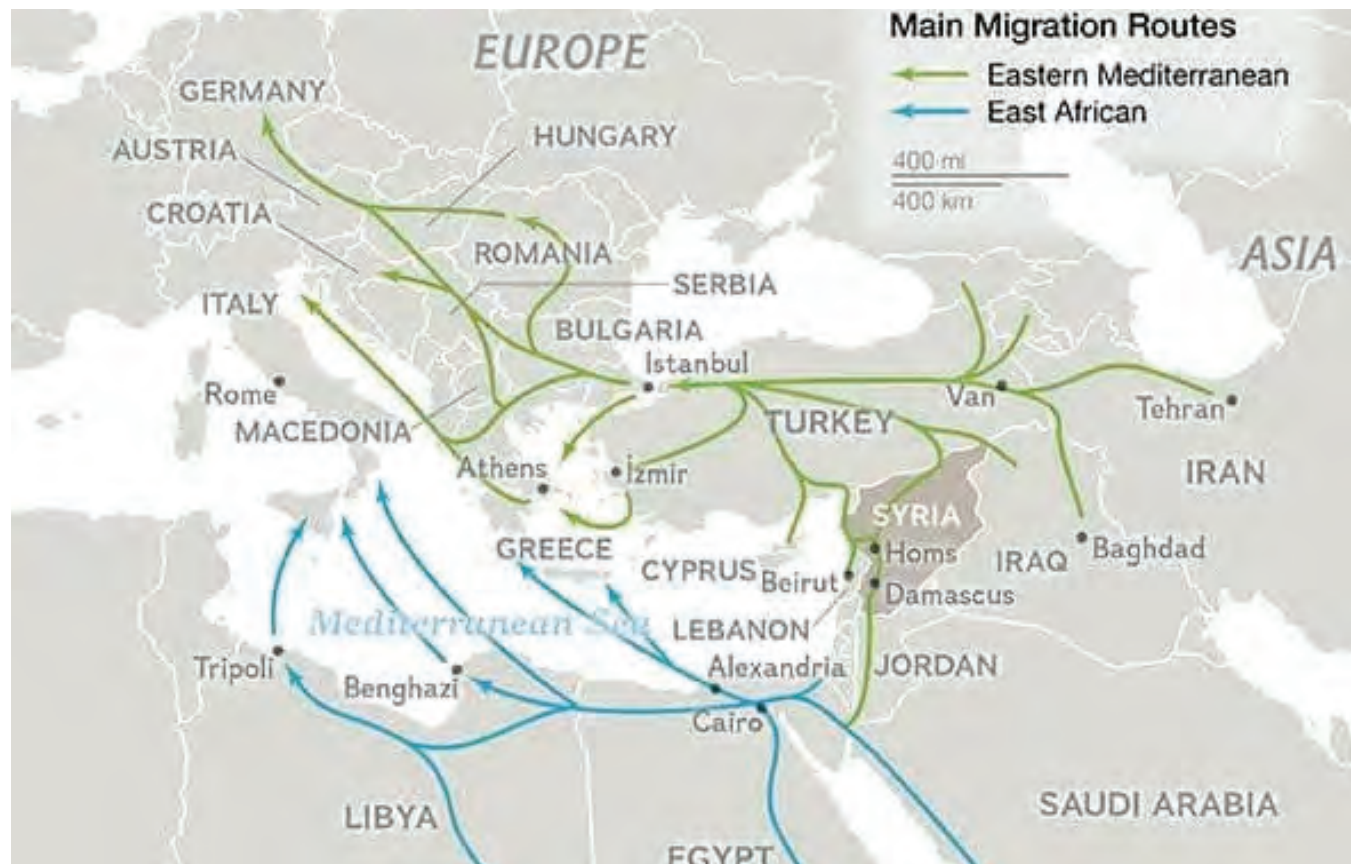
Data on some 4.4 million stateless people residing in 95 countries was reported at end-2022. The true global figure is estimated to be significantly higher.

76%
hosted in low- and middle-income countries

Low- and middle-income countries host 76 per cent of the world's refugees and other people in need of international protection. The Least Developed Countries provide asylum to 20 per cent of the total.

70%
hosted in neighbouring countries

70 per cent of refugees and other people in need of international protection lived in countries neighbouring their countries of origin.



Eastern Mediterranean Route

- Nearly 90% of migrants reaching Europe via this route originate from countries like Syria, Afghanistan, and Eritrea.
- In 2023, over 3,000 deaths or disappearances were reported, making it one of the deadliest migration paths.



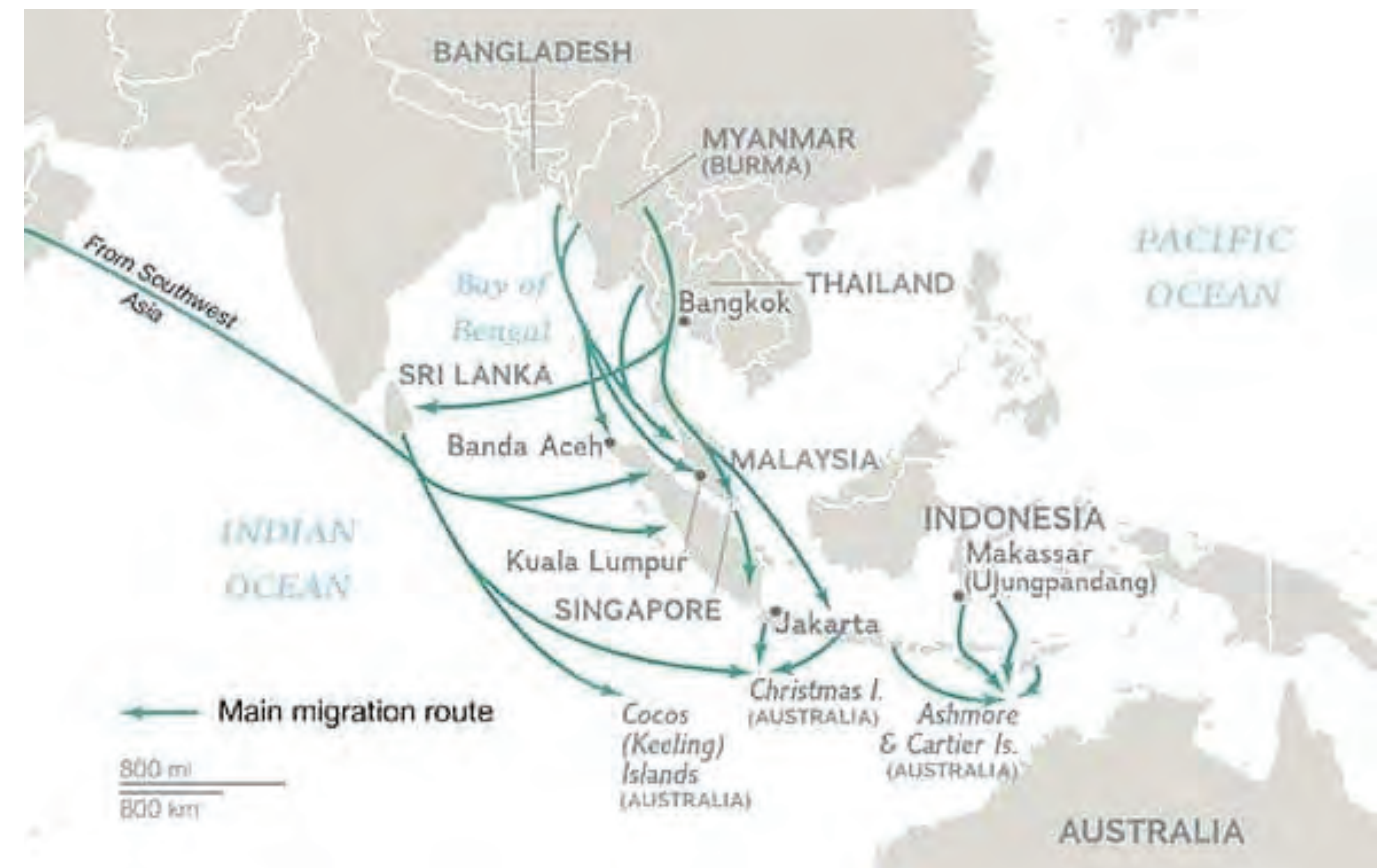
Mediterranean Sea Route

- This route connects migrants from Ethiopia, Somalia, and Eritrea to the Gulf via Yemen, despite ongoing conflict in the region.
- Over 700 deaths have been recorded since 2014 in the Gulf of Aden, highlighting the dangers of the migration path.



Central American Route

- Migrants from Mexico and Central America travel northward toward the U.S. border to escape poverty and violence.
- Increasing control of smuggling routes by drug cartels exposes migrants to extreme risks, including exploitation and violence.



Southeast Asian Route

- Migrants, including Rohingya Muslims fleeing Myanmar, often traverse overcrowded boats through the Andaman Sea and the Strait of Malacca.
- Restrictive migration policies and the absence of legal frameworks compound the hardships faced along this route.



HOUSING CRISIS

"More than 1.6 billion people — over 20% of the global population — lack adequate housing, with 150 million classified as homeless."

United Nations Human Settlements Programme (UN-Habitat, 2022)

02

Housing Policies and Strategies

Housing policies and strategies are essential for shaping affordable, inclusive, and resilient urban environments. As cities face rising housing costs, socioeconomic disparities, and spatial segregation, effective policies must balance funding mechanisms, mixed-tenure development, and community-driven approaches. This chapter examines key strategies, including affordable housing models, mixed-use integration, and interaction-focused design, to explore how urban planning can create equitable and sustainable housing solutions for diverse populations.

Affordable Housing Funding in the U.S.

The provision of affordable housing in the United States is supported through a complex framework of federal, state, and local programs aimed at addressing housing insecurity for low-income households. These programs combine policy initiatives and economic mechanisms to increase the availability of affordable housing.

1. Low-Income Housing Tax Credit (LIHTC):

The LIHTC program, established in 1986, is a pivotal federal initiative to incentivize private investment in affordable housing. Administered by state and local housing authorities, the program allocates tax credits to developers who build or rehabilitate affordable housing units. Investors purchase these credits, providing upfront capital for development, and in return, receive tax benefits over a ten-year period. Properties financed through LIHTC are required to meet affordability standards for a minimum of 15 to 30 years, ensuring long-term benefits for low-income tenants.



2. Federal Housing Assistance Programs:

Programs such as Section 8 vouchers and public housing initiatives provide direct support to low-income families. Section 8 allows eligible households to rent housing in the private market, with the government subsidizing a portion of their rent. Public housing, managed by local housing authorities, offers government-owned affordable units to qualifying residents. Both programs are designed to alleviate the financial burden of housing costs while maintaining decent living conditions.

3. State and Local Government Initiatives:

State and local governments play a critical role in expanding affordable housing through bond issuance, grants, and tax incentives. For example, New York State has recently

allocated over \$236 million to finance more than 700 affordable housing units, supporting the construction of energy-efficient developments and the rehabilitation of existing properties. These initiatives often target vulnerable populations, including seniors, veterans, and individuals experiencing homelessness.

These funding mechanisms reflect a multifaceted approach to addressing housing affordability, combining public and private efforts to create sustainable solutions for low-income communities. By understanding the economic and policy frameworks behind these programs, stakeholders can better address the challenges of affordable housing in the United States.

Current Trends in Affordable Housing

The affordable housing market in the United States is experiencing several key trends driven by socio-economic and market forces. These trends provide insights into the evolving challenges and opportunities in housing policy and development.

Shift Towards Smaller Housing Units:

Rising housing costs and economic pressures have led to a growing preference for smaller housing units. For instance, new housing developments in cities like Seattle have seen average home sizes shrink by 18% over the past five years, reflecting a broader shift in consumer demand for more affordable and compact living spaces.

Increasing Housing Inventory:

Real estate analysts predict that housing inventories will rise in the coming years, particularly by 2025. This trend is expected to provide potential homeowners and renters with a wider array of options, contributing to a more balanced housing market.

Mortgage Rate Volatility:

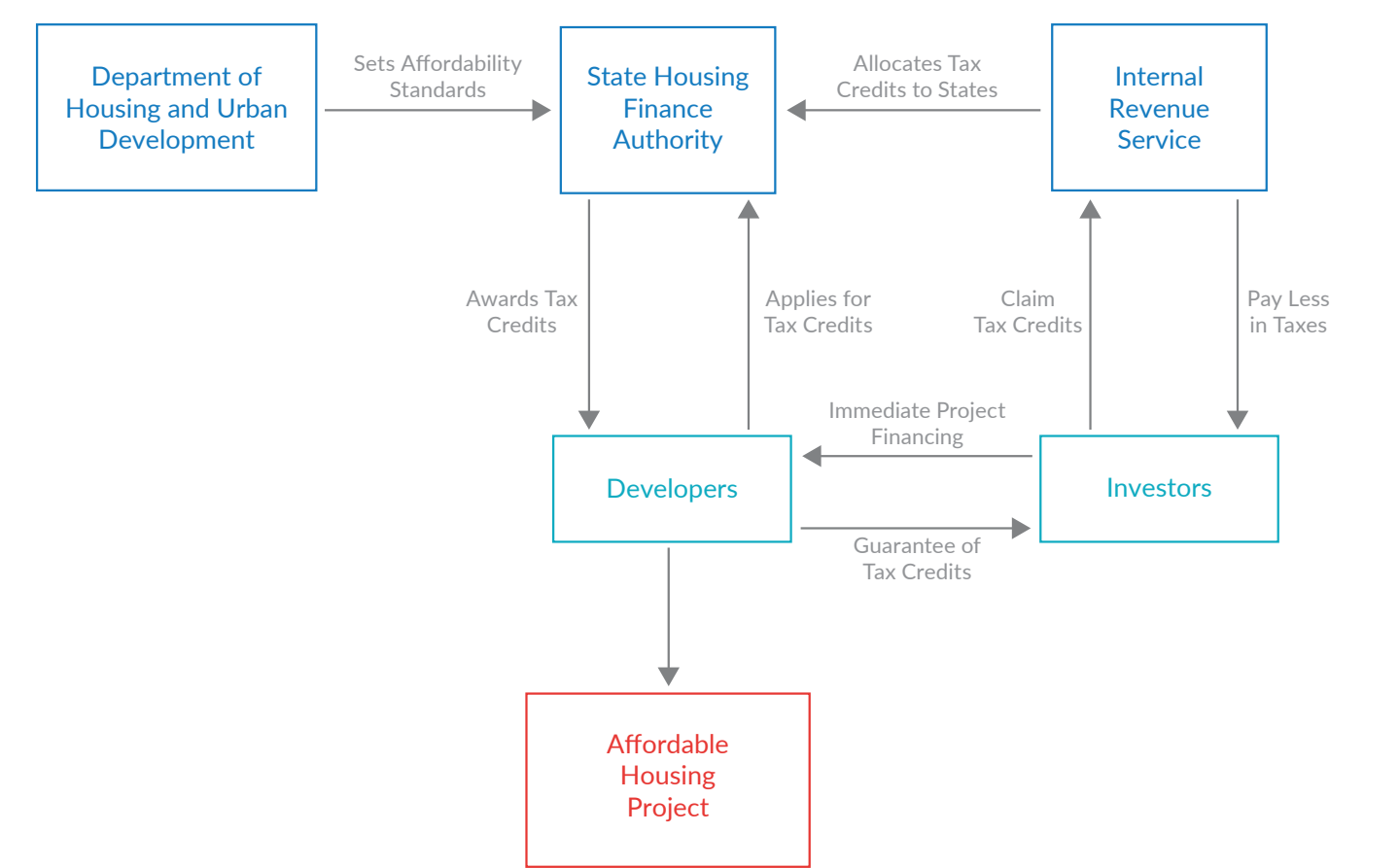
Fluctuations in mortgage rates continue to create uncertainty for prospective homebuyers, influencing the affordability and accessibility of housing. These changes are a critical factor in shaping housing demand and market stability.

Evolving Rental Market Dynamics:

High living costs have pushed many Americans to opt for long-term rentals over homeownership. This trend is coupled with a rise in rental prices and the increasing availability of rental properties offering flexible lease terms and additional amenities.

These trends highlight the dynamic nature of the affordable housing market and underscore the need for innovative policy responses and development strategies to address ongoing challenges.

Structure of Low-Income Housing Tax Credit



Everett Stamm, Tax Foundation, Aug 2020

The Low-Income Housing Tax Credit (LIHTC) program represents a crucial mechanism for incentivizing private-sector development of affordable housing in the United States. Below is a detailed breakdown of how the program operates:

1. Federal Oversight:

The U.S. Department of Housing and Urban Development (HUD) sets nationwide affordability guidelines, while the Internal Revenue Service (IRS) determines each state's annual tax credit allocation. These allocations are based on population and other factors.

2. State-Level Customization:

State Housing Finance Agencies (HFAs) adapt federal guidelines to meet local housing needs. They design competitive processes and criteria for allocating the credits to developers.

3. Developer Applications:

Private developers propose affordable housing projects and submit detailed applications to their state's HFA, outlining design plans, financial models, and tenant affordability commitments.

4. Project Approval:

HFAs evaluate applications and select projects for tax credit

allocation. Selection criteria often prioritize developments that address critical housing shortages or target specific vulnerable populations.

5. Investor Partnerships:

Approved developers partner with institutional investors, such as banks or syndicates, to exchange future tax credit benefits for immediate capital (equity). This financial arrangement ensures developers can initiate construction without significant delays.

6. Construction and Service Initiation:

Upon completion and activation of the housing project, the IRS officially allocates the tax credits to developers. These projects must adhere to affordability and maintenance standards for a specified duration (typically 15 to 30 years).

7. Credit Transfer:

Developers transfer the allocated tax credits to their investors as agreed, providing the investors with long-term tax relief.

8. Tenant Housing:

Subsidized housing units are offered at reduced rental rates to qualified low-income tenants, meeting the program's affordability goals while maintaining high standards for living conditions.

Mixed-tenure development

Diverse Housing Options: These developments offer a range of housing types and tenures, including privately owned homes, market-rate rentals, and subsidized housing, catering to a broad spectrum of income levels.

Integrated Design: A key principle is "tenure blindness," where housing units are designed to be indistinguishable in quality and appearance, regardless of tenure, to foster inclusivity and reduce stigma.

Community Amenities: Mixed-tenure projects often include shared amenities such as parks, community centers, and retail spaces, encouraging interaction among residents.

Advantages of Mixed-Tenure Developments:

- 1. **Social Cohesion:** By housing diverse income groups together, these developments aim to reduce socio-economic segregation and promote community interaction.
- 2. **Economic Stability:** The presence of higher-income residents can attract investment and enhance local services, benefiting the entire community.
- 3. **Improved Housing Quality:** Integrating different tenures can lead to better maintenance and management standards, as there is a collective interest in the upkeep of the area.
- 4. **Access to Opportunities:** Residents of affordable housing gain proximity to employment opportunities, quality education, and healthcare services available in more affluent neighborhoods.

Limitations of Mixed-Tenure Developments:

- 1. **Management Challenges:** Coordinating the needs and expectations of diverse resident groups requires sophisticated management strategies to address potential conflicts and ensure equitable access to shared resources.
- 2. **Market Dependencies:** The success of these developments can be influenced by local housing market conditions, which may affect demand and the financial viability of the project.
- 3. **Social Integration Assumptions:** Evidence suggests that merely placing diverse income groups together does not automatically lead to increased social interaction or improved outcomes for lower-income residents.
- 4. **Design and Implementation Complexities:** Achieving true tenure blindness and ensuring equal quality across housing types can be challenging, potentially leading to perceptions of inequality if not properly managed.

In the context of U.S. urban planning, mixed-tenure developments are often implemented through policies like inclusionary zoning, which mandates or incentivizes the inclusion of affordable housing units within new developments. This approach seeks to create economically diverse communities and mitigate the negative effects of concentrated poverty.



Unit by Unit Integration

The Nicholson, Melbourne

29% affordable housing

199-apartment complex with commercial ground floor is located in the inner-Melbourne. Homeground, a homeless support service and registered community housing provider, own 58 of the apartments: 40 for community housing and 18 social housing to support tenants transitioning out of homelessness.



Floor by Floor Integration

One Riverside Park, NYC

25% affordable housing

The Upper West Side development includes 219 apartments and an additional 55 apartments rented through the NYC Housing Partnership. The inclusion of affordable housing gave the developer a number of tax and planning concessions. However, the two tenure types occupy separate sections of the building, and are serviced by separate facilities and separate entrances.



Building by Building Integration

Kensington Housing Redevelopment

47% affordable housing

429 public housing dwellings, 15 community dwellings and 497 private dwellings; 47:53 split of subsidized and market housing. The objective of integrating the tenure types has been successful, with the construction quality and building-by-building distribution making the tenures indistinguishable.

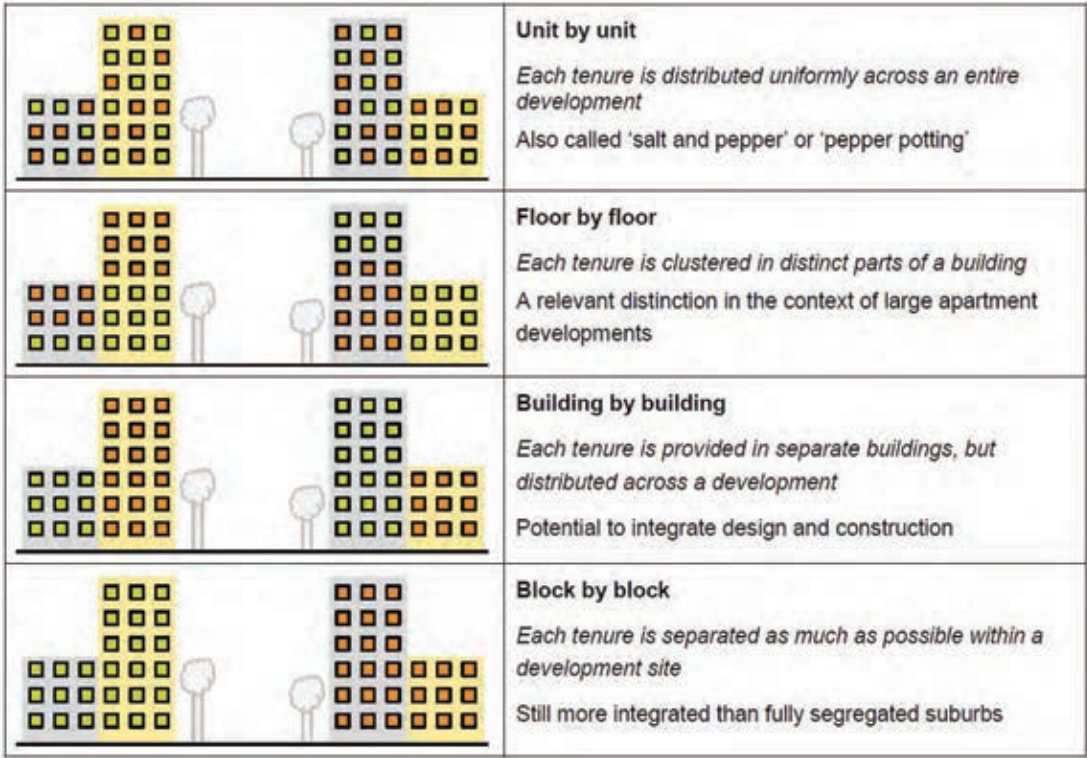


Block by Block Integration

Washington Park, Sydney

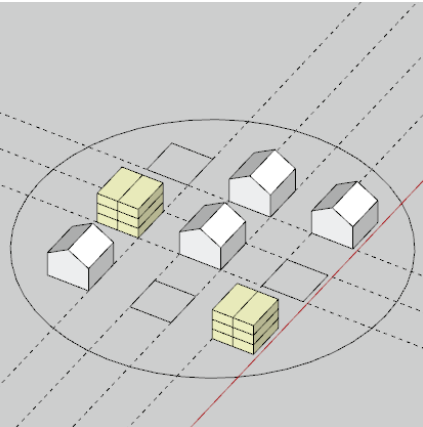
85% affordable housing

A seniors housing complex comprises the vast majority of the social housing onsite (127 of the 150 social housing apartments). The market housing, expected to deliver around 450 to 500 apartments, is on separate blocks in the other quarters of the site.



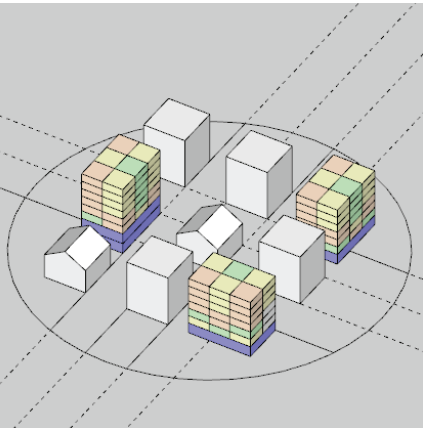
Degree of Integration

Affordable and Inclusive Urban Housing Strategies



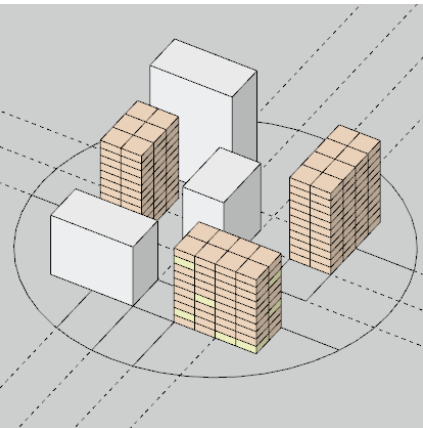
Subsidized Affordable Housing

Development Entity	Public Housing Authorities (PHA)
Target Population	Low-income households
Advantages	Strong affordability for vulnerable groups
Disadvantages	Often segregated, reliant on government subsidies



Mixed-Income Housing

Development Entity	Public-Private Partnerships (PPP)
Target Population	Low-, middle-, and high-income households
Advantages	Social integration, long-term economic sustainability
Disadvantages	Managing social dynamics between income groups



Inclusionary Zoning Policies

Development Entity	Private Developers (incentivized by policy)
Target Population	Low- and moderate-income households
Advantages	Increases affordable housing supply
Disadvantages	Often a small percentage of affordable units

This section explores three critical housing strategies aimed at addressing affordability, equity, and integration within urban areas. Subsidized affordable housing provides essential support for low-income residents through government aid, ensuring access to stable and affordable living environments. Mixed-income housing blends various income levels within a single development, fostering economic diversity, social cohesion, and reducing income segregation. Inclusionary zoning policies require or incentivize developers to incorporate affordable housing units within market-rate projects, promoting equitable access to urban amenities and countering the effects of gentrification. Together, these strategies contribute to creating sustainable, inclusive, and socially integrated communities in rapidly urbanizing cities.

Subsidized affordable housing refers to housing units that are made affordable to low- and moderate-income individuals and families through direct government support. This support typically comes in the form of subsidies, tax credits, or vouchers provided by federal, state, or local governments. Programs like the Low-Income Housing Tax Credit (LIHTC) and Section 8 in the United States are key mechanisms for creating and maintaining such housing.

A defining characteristic of subsidized housing is its affordability criteria, often set at a percentage of the Area Median Income (AMI), such as 30%, 50%, or 80%. Tenants typically pay a portion of their income (e.g., 30%) toward rent, with the subsidy covering the remaining cost. These units can be found in standalone developments or integrated within mixed-use projects.

Advantages of subsidized housing include its ability to provide immediate relief for vulnerable populations, such as seniors, disabled individuals, and low-income families. It helps combat homelessness, stabilize communities, and support economic mobility by ensuring access to safe and affordable living conditions. Moreover, such housing can spur economic activity by creating construction and maintenance jobs.

However, subsidized housing has limitations. Critics argue that it can perpetuate income segregation if units are concentrated in certain areas. Additionally, reliance on public funding can make projects vulnerable to political and budgetary constraints. The stigma surrounding subsidized housing can also pose challenges for integration into broader communities.

Subsidized affordable housing is typically initiated by government agencies, non-profits, or private developers who receive incentives to build and manage these units. The beneficiaries are individuals and families who qualify based on income thresholds. Despite its challenges, subsidized housing remains a vital tool in addressing housing insecurity, promoting social equity, and ensuring that the most vulnerable populations have access to stable living environments.

Mixed-income housing refers to developments that include units for households with a variety of income levels, from low- to moderate-income families to market-rate tenants. The goal is to foster economic diversity, reduce income segregation, and encourage community integration. Mixed-income housing can be found in standalone buildings, residential complexes, or mixed-use developments.

The key feature of mixed-income housing is its intentional blending of affordability levels. For example, a development might allocate 20% of its units to low-income households, 30% to moderate-income households, and the remainder to market-rate tenants. Financing for these projects often involves a combination of public subsidies, private investment, and tax incentives like the Low-Income Housing Tax Credit (LIHTC).

Advantages of mixed-income housing include promoting social cohesion and reducing the stigma often associated with low-income housing. By integrating diverse residents within a single development, these projects help combat economic segregation and provide opportunities for upward mobility. They can also strengthen local economies by attracting a mix of tenants who contribute to commercial activity.

However, mixed-income housing also faces challenges. High-income tenants may resist living alongside low-income residents, and there is often a delicate balance between

ensuring affordability and maintaining financial viability for developers. Critics also argue that mixed-income housing might not adequately address the needs of the lowest-income households due to limited deeply affordable units.

Mixed-income housing is typically driven by private developers in partnership with public agencies, often as part of inclusionary zoning policies. The target demographic includes a range of income groups, making it an effective tool for creating economically diverse communities. While not a panacea for housing inequality, mixed-income housing offers a promising model for fostering inclusivity and economic integration in urban areas.

Inclusionary zoning (IZ) policies require or incentivize developers to include a certain percentage of affordable housing units in new residential developments. These policies aim to increase the availability of affordable housing in high-demand areas while fostering economically diverse communities.

A common characteristic of IZ policies is their flexibility. Developers may be mandated to build affordable units on-site, offer them off-site, or contribute to a housing trust fund. In return, municipalities often provide incentives such as density bonuses, expedited permitting, or tax abatements to offset the costs of providing affordable units.

The advantages of inclusionary zoning include its ability to generate affordable housing in desirable areas where low-income households might otherwise be excluded. It helps counteract the effects of gentrification and promotes economic integration. Additionally, IZ policies often operate without requiring direct public subsidies, making them a cost-effective strategy for municipalities.

However, IZ policies have limitations. Critics argue that the added costs to developers can discourage construction or lead to higher market-rate rents, reducing the overall housing supply. Enforcement and compliance can also vary, limiting the effectiveness of these policies. Furthermore, the affordable units created through IZ often target moderate-income households, potentially leaving the lowest-income residents underserved.

Inclusionary zoning is typically implemented at the municipal level and targets developers of market-rate housing. The beneficiaries are often households earning between 50% and 80% of the Area Median Income (AMI). While IZ policies alone cannot solve housing affordability issues, they are an important tool for promoting mixed-income communities and ensuring equitable access to urban amenities.

Residents and Units in Mixed-Use Housing

Resident Types

High-Income: Above 120% of AMI*
Middle-Income: 80% to 120%
Low-Income: Below 80%
Very Low-Income: Less than 50%
Extremely Low-Income: Less than 30%

**Income levels in mixed-income housing are typically defined based on percentages of the Area Median Income (AMI). The AMI is calculated by the U.S. Department of Housing and Urban Development (HUD).*

In mixed-use housing, the relationship between resident types and unit types is carefully planned to address diverse income levels and household needs, fostering inclusivity and economic diversity. By accommodating a range of income brackets within the same development, mixed-use projects ensure equitable access to urban amenities while promoting social integration.

Units in mixed-use housing are often categorized by income level, such as low-income, moderate-income, and market-rate units. These categories correspond to percentages of the Area Median Income (AMI). For example:

1. Low-income units typically serve households earning 30-50% of the AMI. These units often receive government subsidies, such as tax credits or vouchers, to maintain affordability. They may be smaller in size, such as studios or one-bedroom apartments, to meet cost and space efficiency goals.

2. Moderate-income units cater to households earning 50-80% of the AMI. These units are often designed for working professionals or families who need affordability without qualifying as low-income. Unit types can range from one- to three-bedroom layouts, providing flexibility for small to mid-sized households.

3. Market-rate units are priced at full market value and target higher-income residents. These units are usually larger and feature premium amenities, offering options like two- to four-bedroom apartments or penthouses.

The combination of these unit types within a single development encourages interaction among diverse resident types, creating opportunities for economic and social exchange. Amenities like shared lounges, courtyards, and co-working spaces further enhance community engagement.

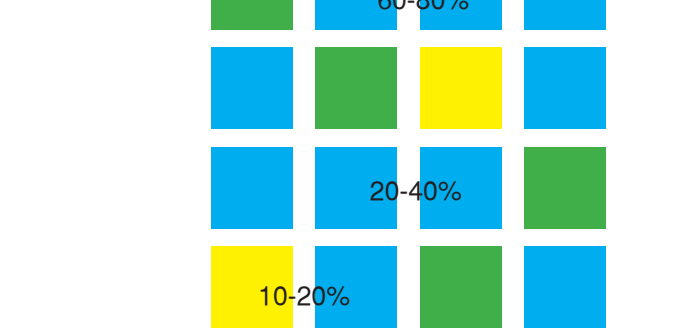
By aligning unit types with the needs of varied income groups, mixed-use housing projects create equitable and dynamic urban environments. This thoughtful approach

Unit Types

Market-rate units
Affordable units
Public units

not only addresses housing affordability but also promotes social cohesion, making mixed-use housing a sustainable and inclusive model for urban living.

Allocation

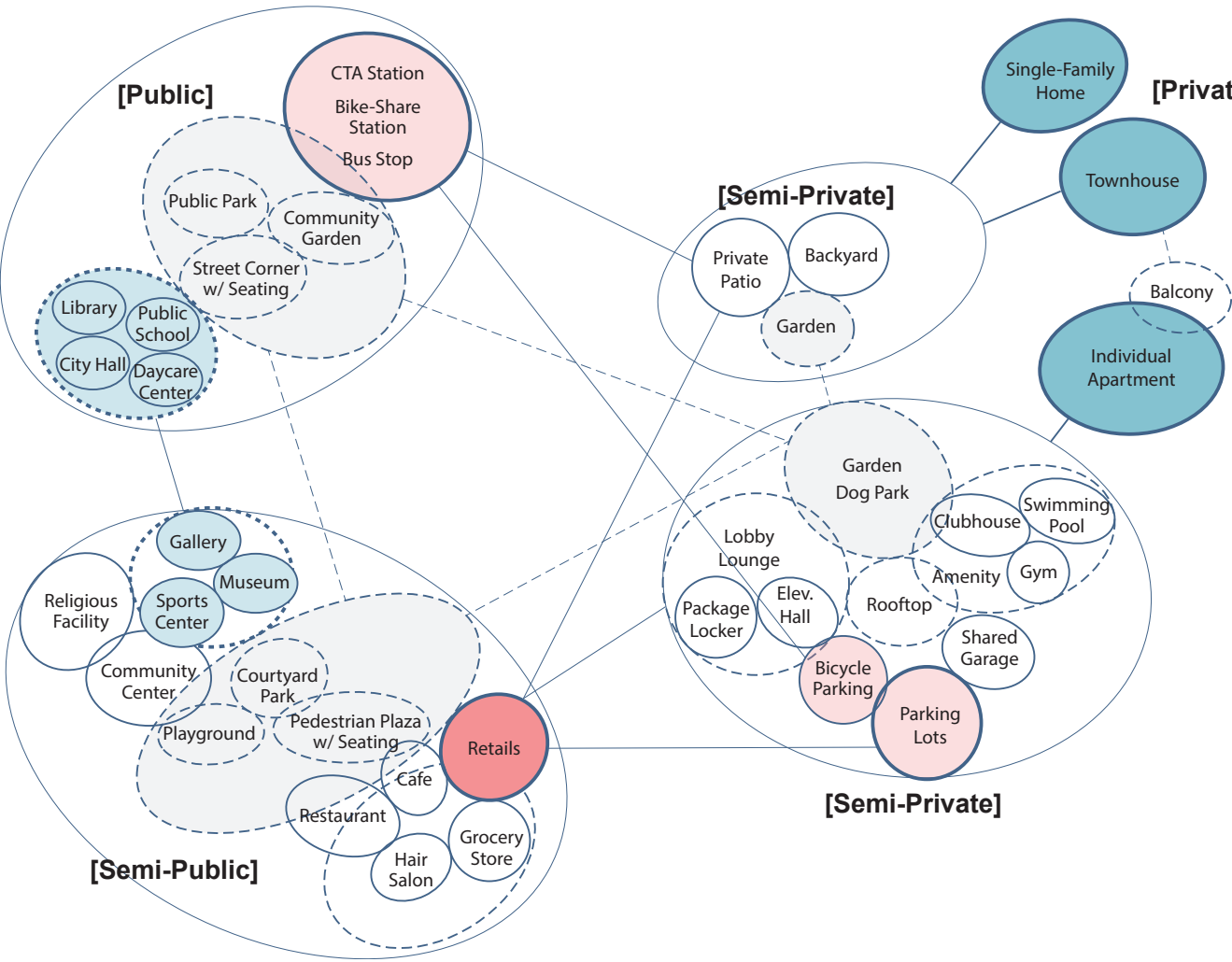


In mixed-use housing projects, the allocation of market-rate units, affordable units, and public units balances financial feasibility and social equity. Typically, 60-80% market-rate units are included to generate revenue, offsetting the costs of affordable and public units. These units cater to higher-income residents and often include premium amenities, ensuring the financial stability of the development.

Affordable units, which make up 20-40% of a project, serve households earning 30-80% of the Area Median Income (AMI). They are funded through government programs like tax credits or subsidies and address the housing needs of low- and moderate-income residents, fostering inclusivity.

Public units, though less common, may be allocated for vulnerable populations, such as seniors or individuals experiencing homelessness. These units represent a commitment to social equity.

This balanced approach promotes economic diversity, supports inclusivity, and ensures the long-term financial sustainability of mixed-use developments.



Spaces for Neighborly Interaction

To enhance social stability and strengthen interclass connections, the spatial hierarchy of public, semi-public, semi-private, and private spaces must be carefully analyzed, restructured, and clustered. This approach is rooted in Herman Hertzberger's theory, which emphasizes the fluid transition between shared and individual spaces, fostering social interaction while maintaining personal boundaries.

By strategically organizing spaces such as public plazas, shared courtyards, communal lounges, and private residences urban environments can encourage gradual interaction and a sense of belonging. Public spaces promote inclusivity, semi-public areas facilitate casual encounters, and semi-private zones offer controlled interaction before transitioning into private dwellings.

This spatial reconfiguration reduces social fragmentation by enabling natural exchanges between different socioeconomic groups in daily life. By designing environments where collective and personal realms coexist harmoniously, communities can cultivate stronger relationships, enhance urban cohesion, and promote long-term social resilience in mixed-income and diverse neighborhoods.

Mapping Daily Crosspoints: Structuring Interaction Through Space

Understanding how people's behaviors and lifestyles intersect at different times of the day is essential for designing socially cohesive environments. By simulating the movement patterns of various age groups and professions, we can identify key crosspoints where interactions naturally occur. These moments of overlap can be reinforced through intentional spatial design, fostering multi-generational engagement and stronger community networks.

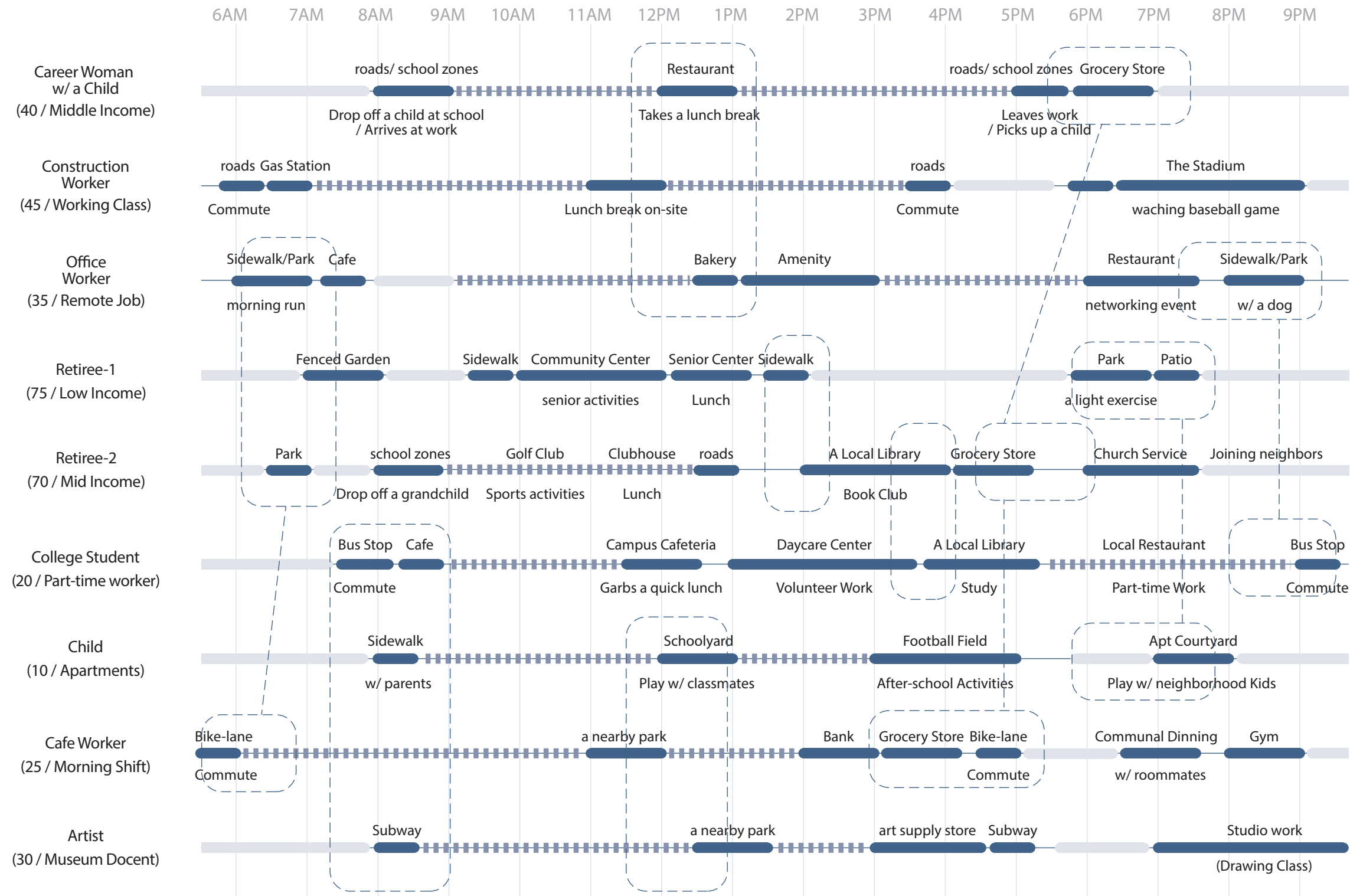
In the morning, an office worker walking to work may share a pedestrian path with a retired senior on a morning stroll, while a cafe worker cycling to an early shift moves through the same urban fabric. A parent walking their child to daycare may cross paths with a jogger heading to a park or a vendor setting up at the morning farmers' market.

By midday, seniors attending a book club at the library may interact with a college student studying before a part-time job, while freelancers and remote workers use shared workspaces alongside small business owners. In another setting, a community kitchen or food cooperative might bring together volunteers and local residents, reinforcing a shared sense of purpose.

In the afternoon, children engaging in outdoor activities in a shared garden could overlap with adults tending to their urban farms, creating an informal educational moment. High school students finishing classes may gather in public plazas, sharing space with local artisans selling handmade goods. Meanwhile, service workers and delivery cyclists activate the streets, contributing to the urban rhythm.

Evening hours introduce a different set of interactions commuters returning home may engage with local shop owners closing their businesses, while families gathering in parks, residents exercising in open-air fitness spaces, and musicians performing in community squares contribute to a socially rich nighttime environment.

By structuring spaces to facilitate these interactions, urban environments can encourage unintentional yet meaningful exchanges, breaking down generational and social divides. Flexible, multi-use spaces, strategically placed along daily movement routes, can promote continuous social engagement, transforming cities into dynamic, interactive environments where diverse individuals connect throughout their daily lives.



Mixed-Use Developments References



Taylor St. Apartments and Little Italy Branch Library

Chicago, IL / SOM

The Taylor Street Apartments and Little Italy Branch Library, designed by SOM, represents an exemplary mixed-use housing project that seamlessly integrates residential living with a public community resource. Located in Chicago's Near West Side, this development combines 73 affordable housing units with a 16,000-square-foot public library, addressing both housing and cultural needs in a dense urban environment.

The residential component features a mix of one- to three-bedroom units, accommodating families of various sizes. These apartments provide affordable housing options in a neighborhood experiencing increasing demand for equitable housing solutions. Positioned above the public library, the residential units benefit from easy access to educational and community programs, fostering a sense of connection between residents and the surrounding neighborhood.

The public library serves as a vibrant community hub, offering a range of services such as early childhood education, digital literacy workshops, and spaces for meetings and

gatherings. This multifunctional facility bridges the gap between residential and public spaces, creating an inclusive environment that supports both individual growth and community engagement.

Sustainability and thoughtful urban integration are key features of the project. The design incorporates energy-efficient systems, maximizes natural light, and includes outdoor spaces to encourage interaction among residents and visitors. By colocating housing and public services, the project optimizes land use and promotes a walkable, resource-rich urban setting.

The Taylor Street Apartments and Library has revitalized the surrounding neighborhood by enhancing access to affordable housing and essential services. This project demonstrates the potential of mixed-use developments to create vibrant, interconnected spaces that address social, economic, and cultural needs, setting a standard for equitable and community-focused urban design.



A Farm-Plus Housing

Westbrook, ME / Vertical Harvest

Vertical Harvest, located in Westbrook, Maine, is an innovative project that combines housing with urban farming, redefining the possibilities of mixed-use development. This farm-plus housing model integrates a hydroponic vertical farm with affordable housing, addressing both the need for sustainable food production and equitable urban living spaces.

The residential component of Vertical Harvest includes 50 affordable housing units, designed to accommodate individuals and families. These units cater to a diverse demographic, including residents with disabilities, ensuring inclusivity. The integration of a working vertical farm into the same development creates an environment where food production, housing, and employment coexist in a mutually beneficial relationship.

The hydroponic farm spans multiple levels, utilizing vertical growing techniques to maximize yield while minimizing land use. It produces fresh, locally grown produce year-round, supplying food to the local community and regional

markets. The farm also provides employment opportunities for residents, particularly those with disabilities, fostering economic empowerment and community engagement. This dual-purpose model not only addresses urban food insecurity but also strengthens the local economy by reducing reliance on long supply chains.

What sets Vertical Harvest apart is its innovative approach to sustainability and social equity. By combining housing and food production, the project reduces transportation-related emissions, enhances local food accessibility, and promotes self-sufficiency. Additionally, the design fosters a sense of community by encouraging interaction between residents, employees, and the broader Westbrook population.

Vertical Harvest exemplifies how mixed-use developments can address multiple urban challenges simultaneously, making it a pioneering example of sustainable and inclusive urban design.

Mixed-Use Developments References



West End Square 50

Washington, DC / TEN Arquitectos

West End Square 50, designed by TEN Arquitectos, is a prime example of mixed-use urban development that combines essential public services, recreational facilities, and affordable housing. Located in Washington, D.C., the project reflects innovative design solutions that optimize urban space while meeting diverse community needs.

The building's ground floor features a fire station, ensuring critical public safety services are seamlessly integrated into the urban environment. Above, the second and third floors house a squash club, which includes courts, a pub, and a pro shop, offering recreational amenities for residents and visitors. A public terrace overlooking M Street enhances the sense of openness and connection to the surrounding neighborhood.

The upper levels of the building are dedicated to 55 affordable housing units, focusing on providing homes for low-income seniors. These residential units are thoughtfully designed to promote independence and comfort while ensuring access to essential community services within the same building.

The mix of public, recreational, and residential spaces demonstrates a harmonious balance between functionality and community engagement.

One of the key design features of West End Square 50 is the two-story glass box that encloses the squash club, creating a visually striking architectural element that highlights the building's transparency and openness. This integration of public and private functions fosters a vibrant urban space that encourages interaction and accessibility.

West End Square 50 revitalizes its urban context by providing critical services, recreational opportunities, and affordable housing within a single development. It exemplifies how mixed-use projects can creatively address urban challenges, offering a model for sustainable, inclusive, and community-focused urban design.



Northtown Library and Apartments

Chicago, IL / Perkins&Will

The Northtown Branch Library and Apartments, designed by Perkins&Will, is an innovative mixed-use project in Chicago that integrates a public library with affordable senior housing. Located in the West Ridge neighborhood, this development reflects a community-centered approach, addressing the dual needs of education and equitable housing within a single structure.

The project features a 16,000-square-foot library on the ground floor, which serves as a hub for learning, social engagement, and community activities. Designed with flexibility and inclusivity in mind, the library offers a variety of programs, including early literacy workshops, digital education, and spaces for community events. Its open, light-filled design creates a welcoming environment for all visitors, while modern amenities cater to the evolving needs of library users.

Above the library, the project includes 44 affordable housing units for low-income seniors. These apartments are thoughtfully designed to provide comfort and independence,

with easy access to the library and other neighborhood amenities. Shared residential spaces, such as lounges and outdoor terraces, encourage social interaction among residents, fostering a sense of community.

A key feature of the Northtown Branch Library and Apartments is its emphasis on sustainability and urban integration. The building incorporates energy-efficient systems, green roofs, and environmentally conscious materials, aligning with Chicago's broader goals for sustainable development.

This mixed-use project has revitalized its surrounding neighborhood by creating a vibrant, multifunctional space that benefits both residents and the broader community. By combining public and residential functions, Perkins&Will has demonstrated how thoughtful design can enhance urban life, addressing both social and environmental challenges.

Mixed-Use Developments References



Independence Library and Apartments

Chicago, IL / John Ronan Architects

The Independence Branch Library and Apartments, designed by John Ronan Architects, is a mixed-use project in Chicago that seamlessly integrates a public library with affordable senior housing. Located in the Irving Park neighborhood, the development addresses both the educational needs of the community and the demand for equitable housing solutions.

The 16,000-square-foot library occupies the ground floor and serves as a vibrant community hub. With its light-filled interiors, flexible spaces, and modern amenities, the library supports a variety of uses, including reading areas, meeting rooms, and technology access points. It also offers diverse programs, such as literacy workshops, community events, and digital skills training, making it a key resource for residents of all ages.

Above the library, the project includes 44 affordable housing units designed for low-income seniors. These apartments are thoughtfully arranged to ensure privacy, comfort, and accessibility. Shared spaces, such as terraces and community rooms, encourage interaction among residents

and foster a supportive living environment. The building's design ensures a seamless connection between the library and the residential units, providing residents with easy access to educational and cultural resources.

The project emphasizes sustainability and urban integration, incorporating green roofs, energy-efficient systems, and sustainable materials. Its design reflects a commitment to creating a resource-rich, environmentally conscious space that benefits both residents and the broader community.

The Independence Branch Library and Apartments has revitalized its neighborhood, demonstrating how mixed-use developments can address critical urban challenges. By combining housing and public amenities in an innovative, community-centered design, John Ronan Architects has created a model for inclusive and sustainable urban living.



Five88

San Francisco, CA / David Baker Architects

Five88, designed by David Baker Architects, is an affordable housing development in San Francisco that exemplifies thoughtful urban design by integrating retail and community spaces into its framework. While it does not feature public programs like libraries or cultural centers, its integration of retail and communal areas highlights a meaningful approach to creating vibrant, resident-focused environments.

Located in the Mission Bay neighborhood, Five88 includes 200 affordable rental units, providing housing for individuals and families earning between 50% and 60% of the area median income. The building's ground floor features retail spaces, which activate the street frontage, fostering interaction between residents and the broader community. This commercial integration not only provides convenient access to essential goods and services but also supports local businesses, contributing to the economic vitality of the neighborhood.

A defining feature of Five88 is its central courtyard, which serves as the heart of the community. This open-air space

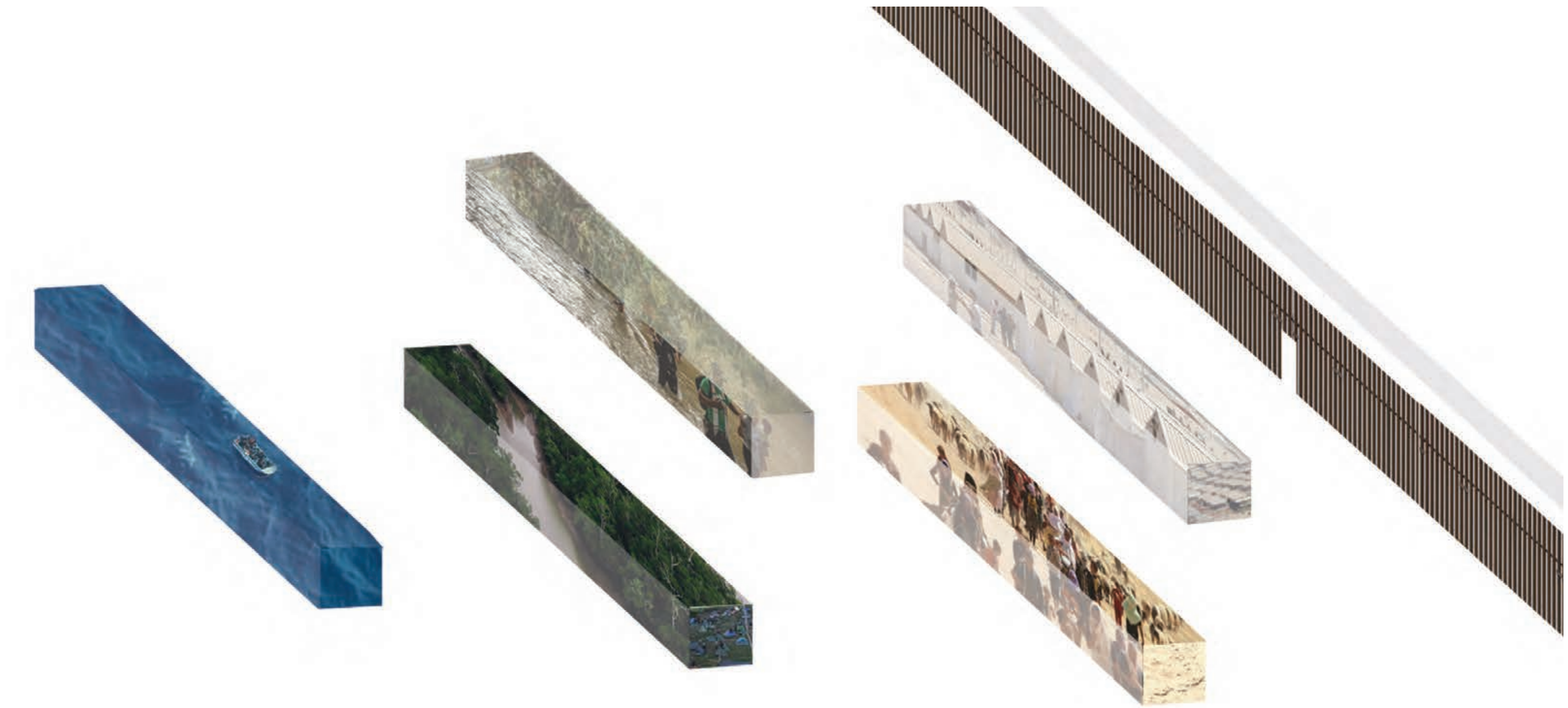
provides residents with a secure, tranquil environment for relaxation, social gatherings, and outdoor activities. Surrounding the courtyard, the building includes shared community spaces, such as lounges and multi-purpose rooms, designed to encourage interaction among neighbors and promote a sense of belonging.

The building's design emphasizes livability and sustainability, incorporating energy-efficient systems, ample natural light, and thoughtful layouts to enhance residents' quality of life. Five88 is an excellent example of how affordable housing can be both functional and enriching, blending retail, residential, and communal elements to create a holistic urban living experience. By prioritizing connection and community, the project demonstrates the value of integrating diverse uses in housing developments, even without traditional public programs.

03 Exodus City

A MULTI-LAYERED CITY

Over 100 million people worldwide are displaced due to wars and conflicts, including refugees, immigrants, and the homeless. This project focuses on designing a floating city on Lake Michigan as a new typology for urban design and temporary housing. By examining the physical dimensions of cities and the composition of communities, the research proposes a multi-layered city that fosters a sense of belonging and encourages community building for socially isolated individuals.



The Gulf of Urabá

Migrants traveling through Necoclí to cross the Gulf of Urabá face substantial delays due to limited transportation. Commercial boats operating in the region can only accommodate about 500 passengers per day, forcing many to wait days or even weeks for their opportunity to continue toward the Darién Gap.

The Mediterranean

Despite intensified efforts to curb smuggling activities in the southern Mediterranean, the migration flow between Libya and Italy remains steady. This ongoing movement highlights the persistent challenges of addressing migration in this region, even amidst increased crackdowns.

The Darién Gap; A Jungle Crossing

Spanning over sixty miles of dense rainforest, steep mountains, and expansive swamps, the Darién Gap is a treacherous, roadless expanse along the border of Colombia and Panama. It serves as the only overland route connecting Central and South America, posing significant dangers to those who attempt the journey.

Smugglers & Criminal Groups

Migrant families, particularly those with children, are vulnerable to exploitation by criminal gangs. These groups exploit their precarious circumstances through acts of violence, including sexual abuse, trafficking, and extortion, taking advantage of the migrants' most desperate moments.

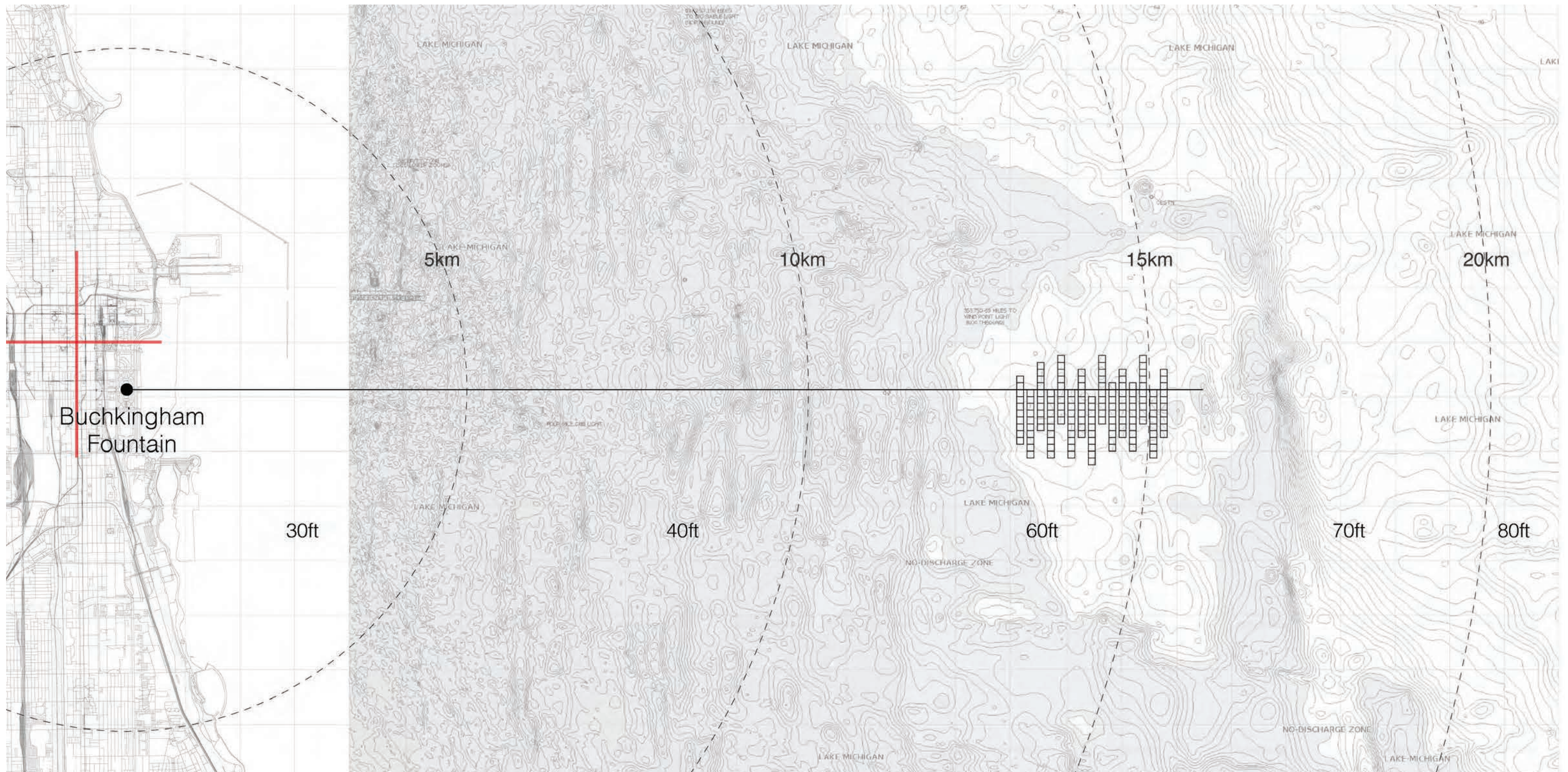
The Overlooked Struggles of Refugees

Refugees face not only physical dangers during their journeys but also a profound lack of public understanding and empathy. Many people fail to recognize that these migrations are driven by life-threatening circumstances such as war, persecution, and poverty, often misinterpreting them as voluntary or economically motivated. This lack of awareness dehumanizes refugees, reducing them to statistics rather than individuals seeking safety and dignity.

Media coverage often exacerbates misconceptions by focusing on crises and political debates, overshadowing the resilience and potential contributions of refugees. Such narratives weaken public support for meaningful policies

that could address both immediate needs and systemic issues like poverty and conflict. In host communities, limited education about refugees leads to fear and xenophobia, making integration more challenging. Bridging this gap requires promoting understanding and emphasizing the shared humanity of refugees and local populations.

Ultimately, addressing the struggles of refugees demands not just improved policies but also a transformation in how society perceives them. By shifting the narrative and fostering empathy, societies can create environments where refugees are valued and supported in rebuilding their lives.



A Vision for a Floating City on Lake Michigan

Lake Michigan, one of North America's five Great Lakes, significantly influences the climate of its surrounding regions. The lake's vast water volume moderates temperature fluctuations, creating a microclimate that is cooler in summer and milder in winter compared to inland areas. This climatic stability, along with consistent wind patterns and humidity levels, positions the lake as an ideal site for innovative urban experiments. Such natural advantages make the Lake Michigan basin a compelling choice for exploring a floating city concept, combining resilience with sustainability.

The proposed location for the floating city is around 9.3 miles (15 kilometers) from Chicago's Buckingham Fountain. Situated in a basin area with a depth of about 60 feet, this site offers multiple benefits. The stability of the lakebed reduces risks of subsidence, ensuring structural security, while the moderated microclimate minimizes extreme weather impacts. These characteristics create an optimal setting for testing modular floating structures, enabling innovative urban design and sustainable development.

A symbolic connection to Buckingham Fountain further enhances the cultural and historical relevance of the proposed city. Built in 1927, the fountain is one of Chicago's most iconic landmarks and a tribute to Lake Michigan itself. It symbolizes the vitality of the city's relationship with its

water resources while representing resilience, progress, and cultural pride. Aligning the floating city with Buckingham Fountain underscores the vision of continuity between Chicago's rich heritage and its aspirations for innovation. It situates the project as a reflection of how historical identity can inspire forward-thinking urban planning.

The concept of the floating city also draws parallels to Venice's Festa del Redentore, a centuries-old festival commemorating the end of the devastating plague of 1576–1577. A central feature of this festival is the construction of a temporary bridge, the Ponte Votivo, made by connecting boats to form a floating walkway across the Giudecca Canal. This bridge symbolizes unity and communal resilience while serving as a functional and cultural centerpiece

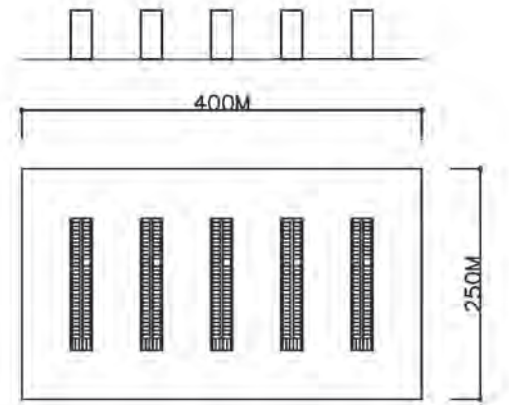
of the event. Similarly, the proposed floating city could adopt modular, interconnected floating platforms, echoing the festival's spirit of adaptability and collective effort.

By integrating the natural advantages of Lake Michigan with cultural landmarks like Buckingham Fountain, this floating city project offers a vision of sustainable urban innovation. It demonstrates how ecological and cultural contexts can inform future-oriented solutions to urban challenges. Like the traditions that inspire it, the project bridges past struggles and future ambitions, offering a model for how humanity can coexist with its environment while embracing creativity and community.



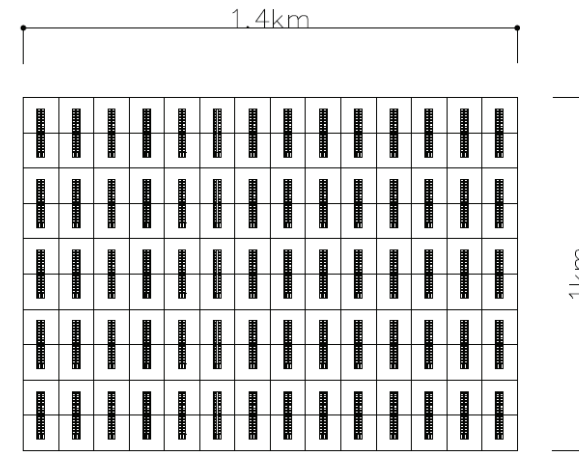
step 1)

1 Unite d'Habitation
= 1,500 inhabitants



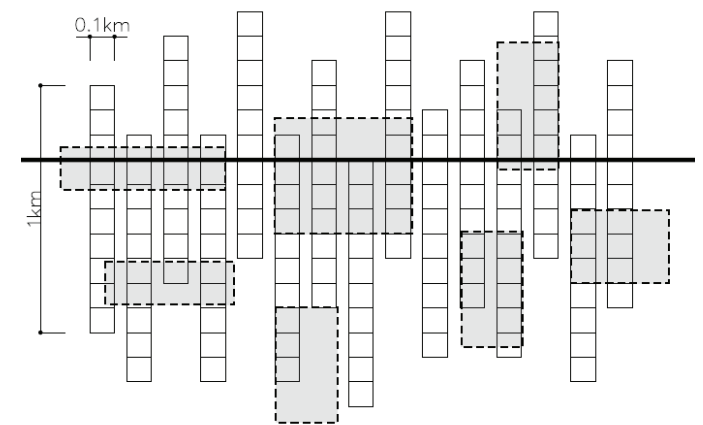
step 2)

5 Unite d'Habitation
= 7,500 inhabitants
= 1-Neighborhood



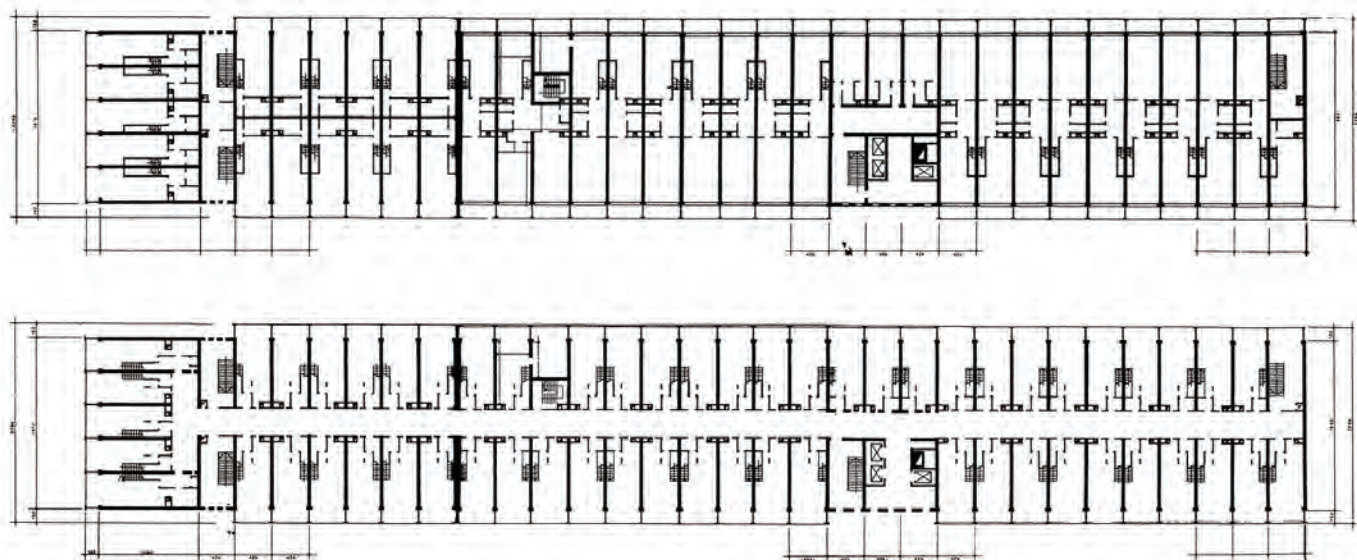
step 3)

Size: 1.4 x 1.0 km



step 4)

Divide and Link



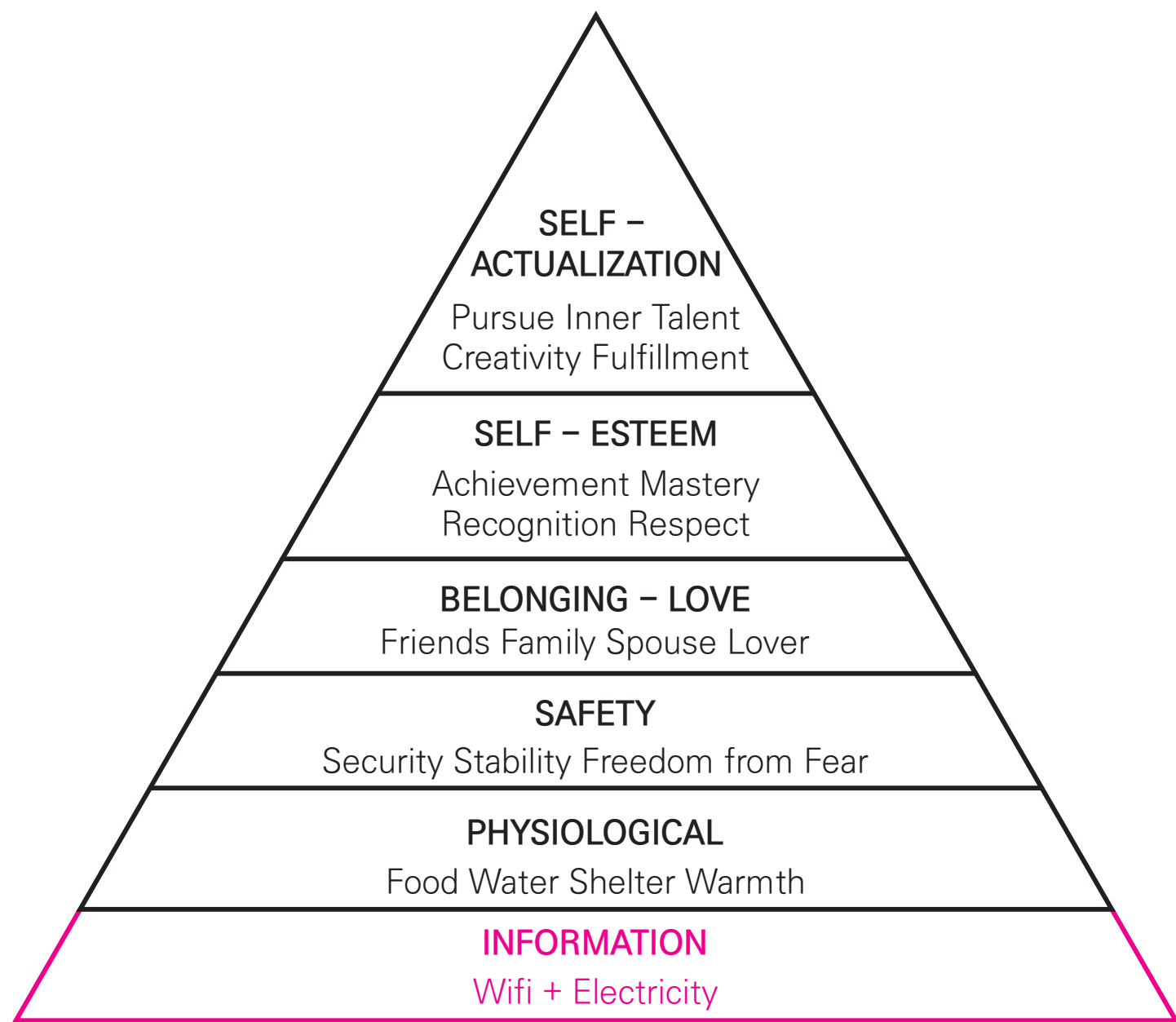
Optimizing Size and Capacity

Determining the optimal urban density for a floating city designed to accommodate 100,000 residents is essential for creating a sustainable and functional environment. Urban density influences walkability, resource efficiency, infrastructure design, and overall quality of life. Striking the right balance ensures efficient use of space while maintaining a livable and adaptable urban setting.

Using Le Corbusier's Unité d'Habitation as a reference provides valuable insights for this calculation. Completed in 1952 in Marseille, this iconic residential structure was designed as a "vertical city" that integrates housing, communal spaces, and essential amenities within a single

building. Accommodating approximately 1,500 residents, it emphasizes efficient spatial use while maintaining access to natural light and ventilation. The building's historical significance lies in its innovative approach to modernist architecture and urban living, providing a model for high-density housing without compromising quality.

Based on the Unité d'Habitation and ensuring adequate lighting, this city for 100,000 residents would require an estimated area of 1.4 km by 1.0 km. However, considering the unique constraints of a floating city—such as the need for modular separation, safety, and essential infrastructure—additional space for connectivity and functionality is crucial.



Maslow's Hierarchy of Needs + Refugee

Maslow's Hierarchy of Needs is a psychological framework that categorizes human needs into five levels, arranged in a pyramid. At the base are physiological needs like food, water, and shelter, essential for survival. The next level addresses safety needs, including physical security and stability. Above this are social needs, such as belonging and relationships. The fourth level is esteem needs, which involve respect, recognition, and self-worth. At the top is self-actualization, the pursuit of personal growth and fulfilling one's potential. Maslow's theory emphasizes that basic needs must be met before individuals can focus on higher-level goals.

However, in today's interconnected world, access to information has become a critical need, especially for

vulnerable groups like refugees. Reliable access to Wi-Fi and electricity ensures refugees can obtain real-time updates on safe routes, potential risks, humanitarian aid availability, and evolving international policies. This information can be a matter of life and death, directly impacting their survival and future.

For refugees, the ability to stay informed transcends convenience; it is integral to their safety and autonomy. By adding information access as a fundamental need, the framework acknowledges the modern realities where knowledge empowers individuals to navigate crises and improve their circumstances. Information, thus, is essential for survival in today's globalized society.

Public Safety	Perception of Safety	Community	Climate Change	Water Management	Renewable Energy	Immigration	Language & Cultural Adaptation	Emigration
Landuse	Safety & Security	Awareness	Environmental Protection	Sustainability	Resource Efficiency	Shelter & Housing	Refugee	Psychological Support
Police Station	Fire Station	Lively Street	Agriculture	Food Processing Plant	Fish Farming	Social Connection	Educational Opportunity	UNHCR
Shipping Industry	Commerce	Trade	Safety & Security	Sustainability	Refugee	Public Transit	Water Transport	Tram
R&D	Local Economy	Tourism	Local Economy	Exodus City	Transport & Mobility	Time & Distance	Transport & Mobility	Walkable City
Office	Agriculture	Construction	Culture & Entertainment	Housing	Publicness	Private Transit	Parking System	CO2 Reduction
Leisure	Public Art	Media	Temporary Housing	Density	Permanet Housing	Accessi-bility	Integration	Social Inclusion
Sports Center	Culture & Entertainment	Beach & Park	Afford ability	Housing	Community Engage-ment	Education	Publicness	Publicity
Festival	Museum & Gallery	Theater	Private & Public	Amenity	Quality	Library	Sqaure	Park

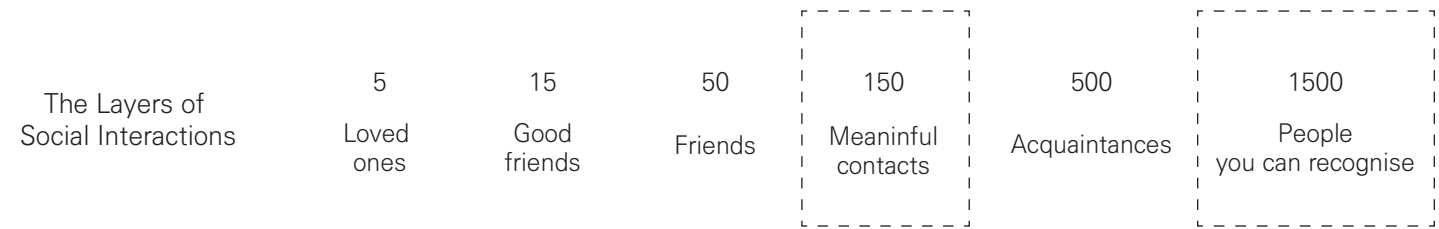
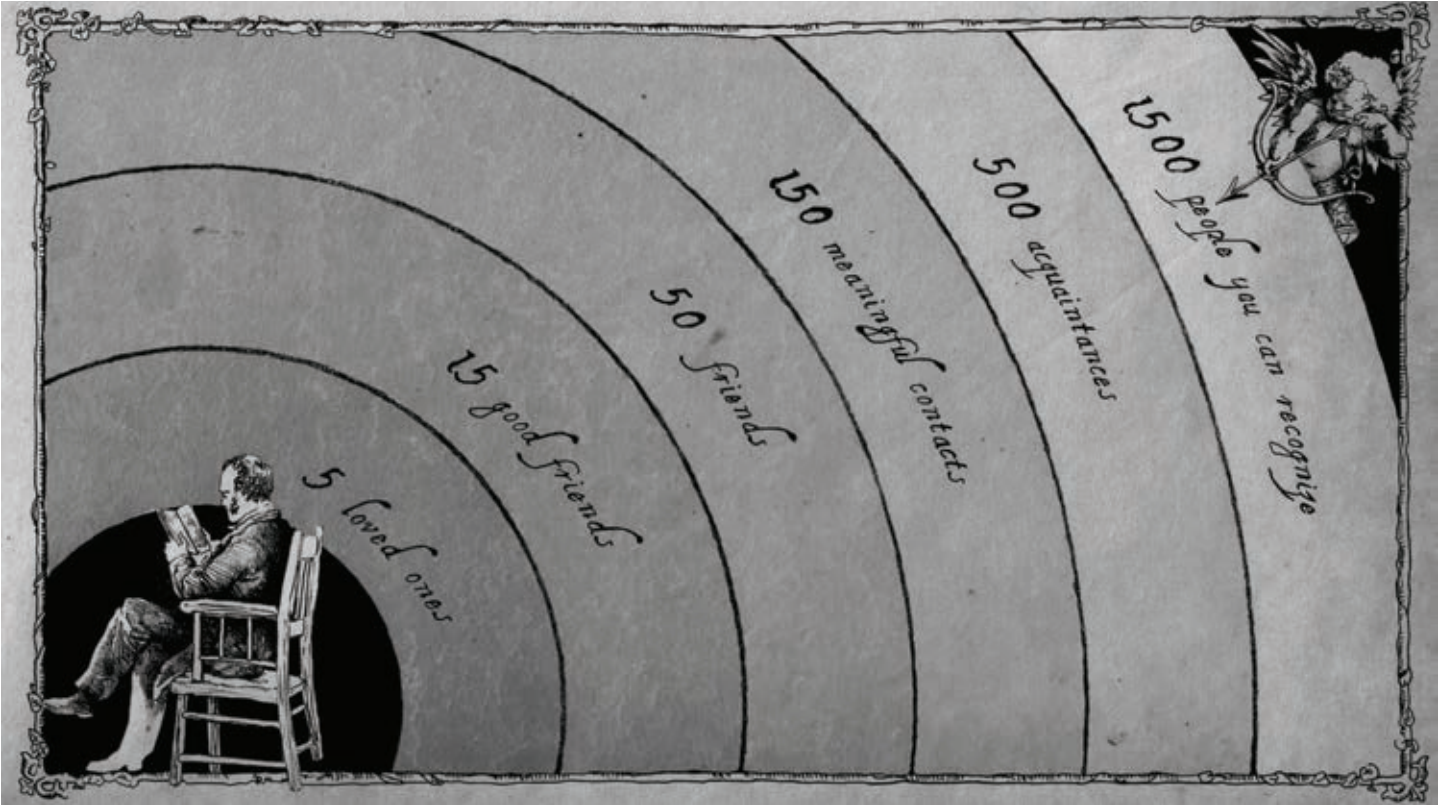
A Framework for Urban Development

To design Exodus City, eight core elements were identified: safety & security, local economy, culture & entertainment, housing, publicness, transport & mobility, refugee integration, and sustainability. Each of these elements was analyzed through eight subcategories, creating a comprehensive and balanced urban framework.

The study focused on ensuring that these elements work cohesively to address the complex needs of the city. For instance, safety and security are tied to transport systems, while sustainability influences housing design and public spaces. Refugee integration, a critical aspect of the city, connects to local economy initiatives and public facilities, promoting social cohesion and economic participation.

The subcategories serve as detailed parameters to guide the planning process, such as exploring renewable energy under sustainability or fostering inclusivity in publicness. By layering these elements and subcategories, the city's framework supports functionality, resilience, and adaptability.

This approach provides a holistic understanding of how Exodus City can integrate diverse urban components, ensuring a livable, equitable, and sustainable environment. The interplay of these eight elements forms the foundation for a dynamic urban ecosystem, capable of addressing current challenges while preparing for future needs.



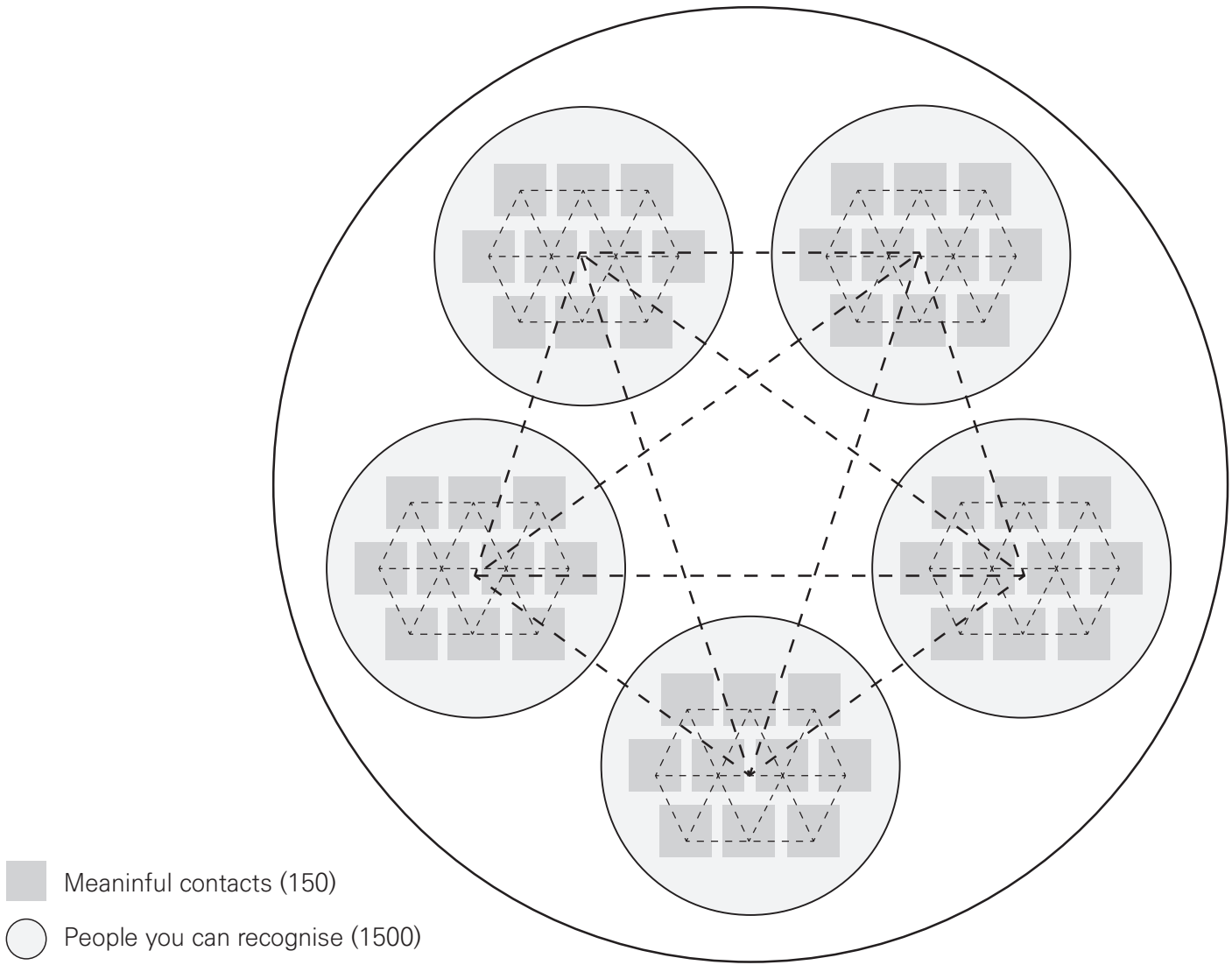
The Theory of Dunbar's Number

British anthropologist Robin Dunbar proposed that the "magic number" for cohesive social groups is 150. His conclusion stems from studies on non-human primates, where he observed a correlation between brain size and group size. Using neuroimaging and behavioral studies on grooming—a key social behavior among primates—Dunbar identified that the neocortex, responsible for cognition and language, plays a critical role in determining the complexity a social group can sustain. This neocortical ratio limits the number of meaningful relationships an individual can maintain, influencing group cohesion.

Dunbar's findings extended to humans, using historical, anthropological, and psychological data to explore how human groups form and function. From hunter-gatherer societies to modern examples like offices, military units, and even Christmas card lists, 150 consistently emerged as the maximum size for effective and cohesive social networks. Beyond this threshold, groups tend to fragment or lose cohesion.

In the context of urbanisation, Dunbar's theory highlights the importance of smaller, village-like networks within cities to mitigate alienation and social tensions. By fostering human-scale connections, urban residents can maintain meaningful relationships and community bonds, ensuring social networks remain functional even within sprawling metropolitan environments. This underscores the enduring relevance of Dunbar's "magic number" in modern society.

"According to Dunbar's theory, people can 'handle' up to about 150 relationships – whether in early hunter-gatherer societies or the modern workplace."

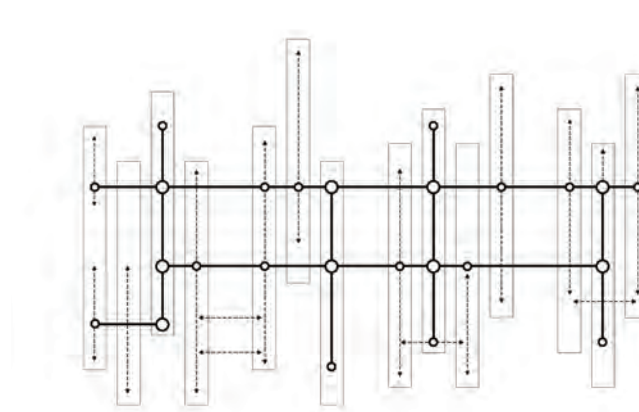
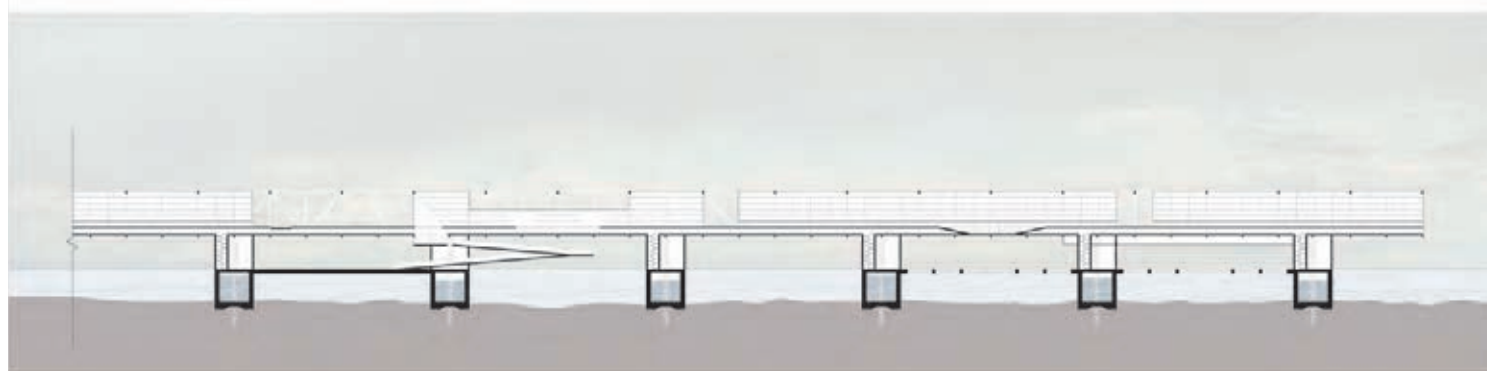
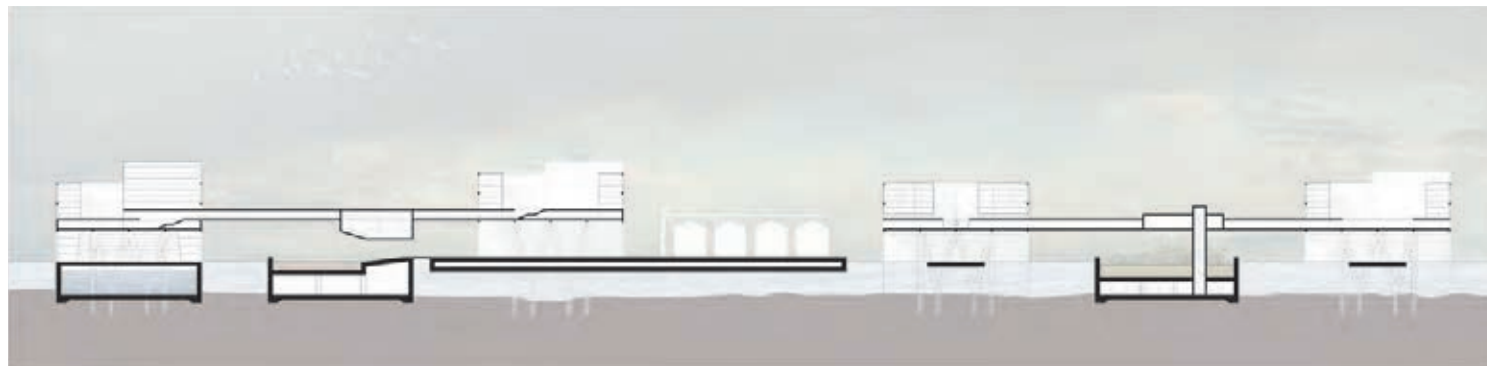
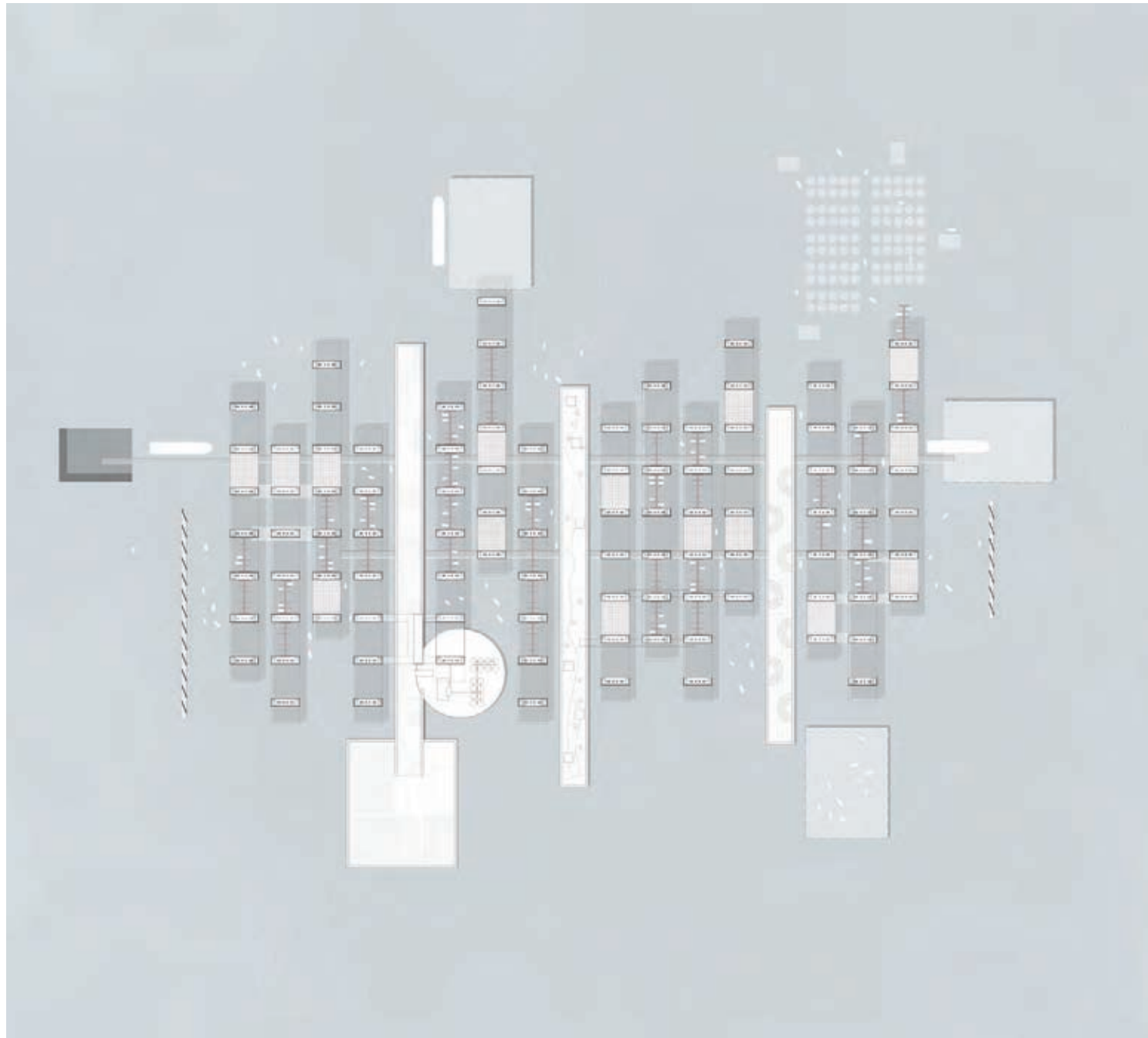


Community Scale and Structure

Based on Dunbar's theory, the range of neighbors in a community is set at 150 people, forming the foundation of a cohesive social structure. By gradually expanding this scale, multiple groups of 150 can be combined to form larger communities. For instance, 10 groups of 150 individuals result in a network of 1,500 identifiable neighbors. This layered community scale and structure provides a framework for reducing social discord and supporting a stable urban environment.

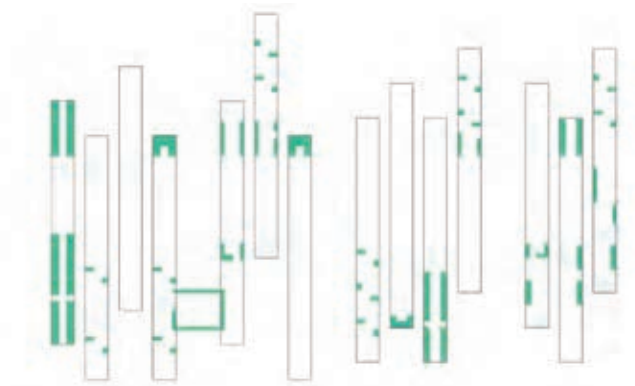
The benefits of this strategy extend beyond reducing confusion or fragmentation. It fosters a sense of belonging and strengthens relationships among neighbors. This is especially crucial for vulnerable populations, such as refugees, who often experience isolation, lack of belonging, and limited access to information. Being part of an organized, familiar community can provide a support system that addresses their social and informational needs, enhancing their ability to integrate and thrive.

This approach also creates opportunities for better resource sharing, trust-building, and efficient communication. By organizing urban structures into human-scaled communities, cities can promote social cohesion while ensuring inclusivity. Dunbar's insights into the limits of social networks highlight the importance of designing communities that are not just functional but also emotionally supportive, creating environments where all individuals feel valued and connected.



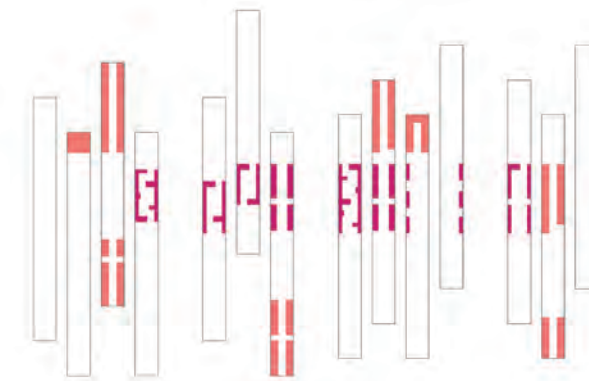
Transit system; tram line

A circular tram system is a vital component of the floating city's design, promoting sustainability and connectivity. Unlike car-dominated road networks, the tram system is eco-friendly, reducing carbon emissions and minimizing the need for expansive road infrastructure. This efficient transit network ensures seamless movement across the city, connecting residential, commercial, and recreational areas. By integrating the tram into the city's layout, the design encourages public transportation use, reduces traffic congestion, and fosters an organic, interconnected urban environment.



Public spaces and Squares

Public spaces and squares are distributed throughout residential neighborhoods to provide accessible areas for relaxation, interaction, and recreation. These spaces include parks, community farming areas, and shared facilities, designed to enhance residents' well-being and promote neighborly connections. By offering a variety of outdoor and communal areas, the city creates opportunities for informal gatherings and strengthens the sense of community. These public spaces also contribute to a visually and environmentally appealing urban landscape, improving the overall quality of life.

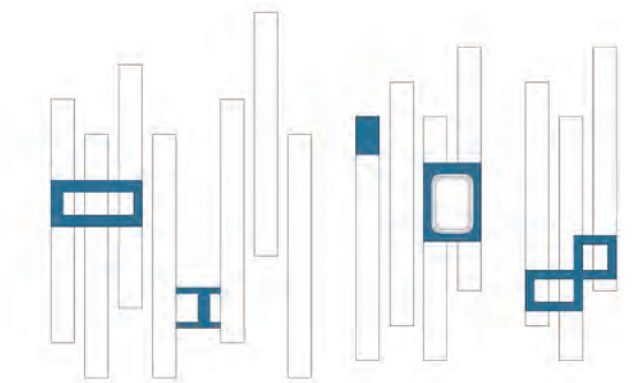


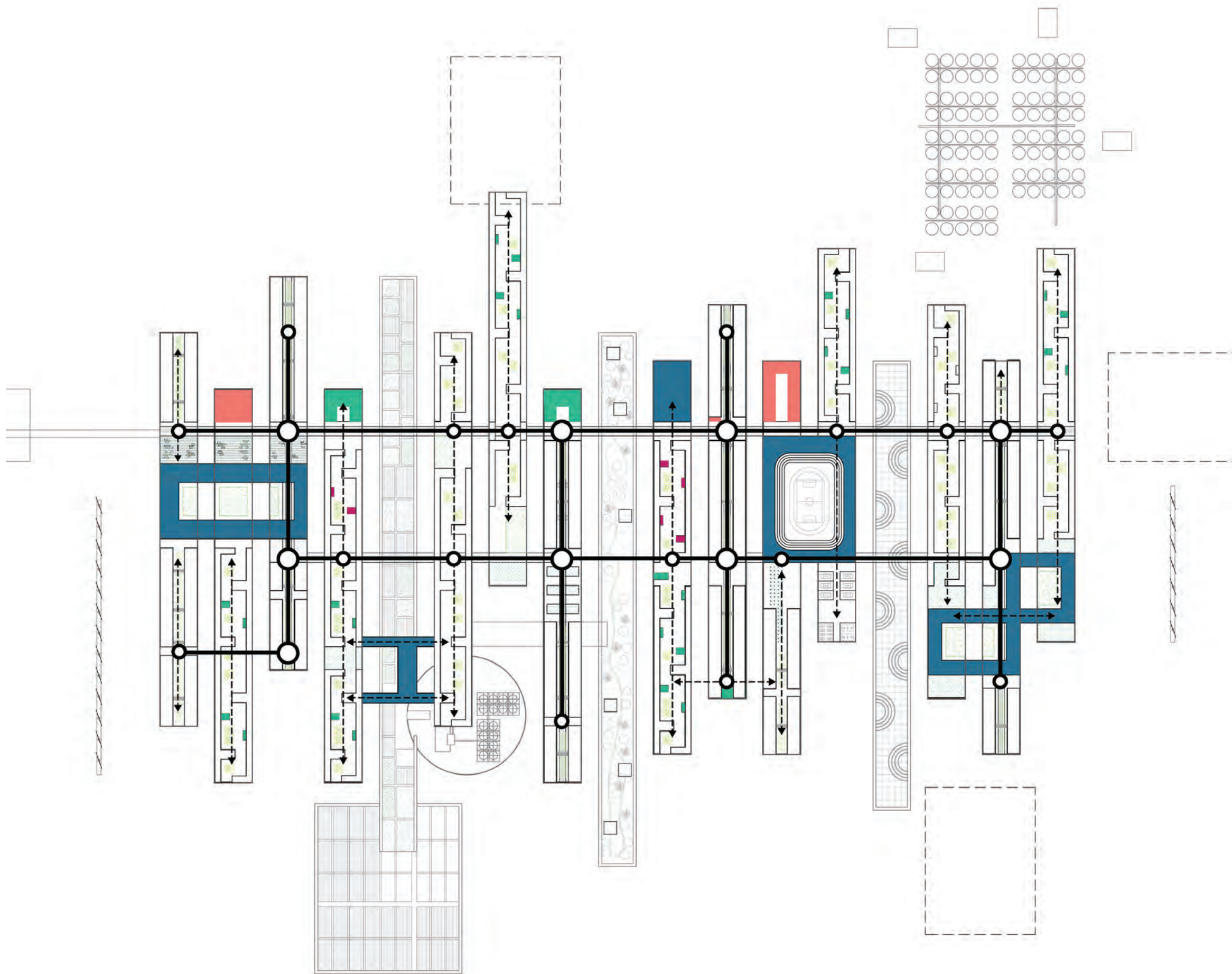
Commercial and Office Districts

Commercial and office facilities are strategically positioned along the tram line in the central areas of the city. These zones provide employment opportunities and essential services, enabling residents to access workplaces and retail spaces conveniently. The mixed-use housing strategy, where residential units are placed above commercial and office spaces, enhances urban density while ensuring safety through increased foot traffic and activity throughout the day. This integration of work and living spaces strengthens social interactions, supports local economies, and creates a vibrant urban core.

Major Elements

The strategic placement of major elements such as stadiums, theaters, and cultural facilities enriches the city's cultural and entertainment offerings. These landmarks serve as hubs for recreational, educational, and social activities, providing residents with access to diverse programs and events. Beyond their functional roles, these elements contribute to the city's identity and vibrancy, acting as focal points for community engagement. Their presence ensures that the city remains dynamic and inclusive, catering to a wide range of interests and fostering a well-rounded urban experience.

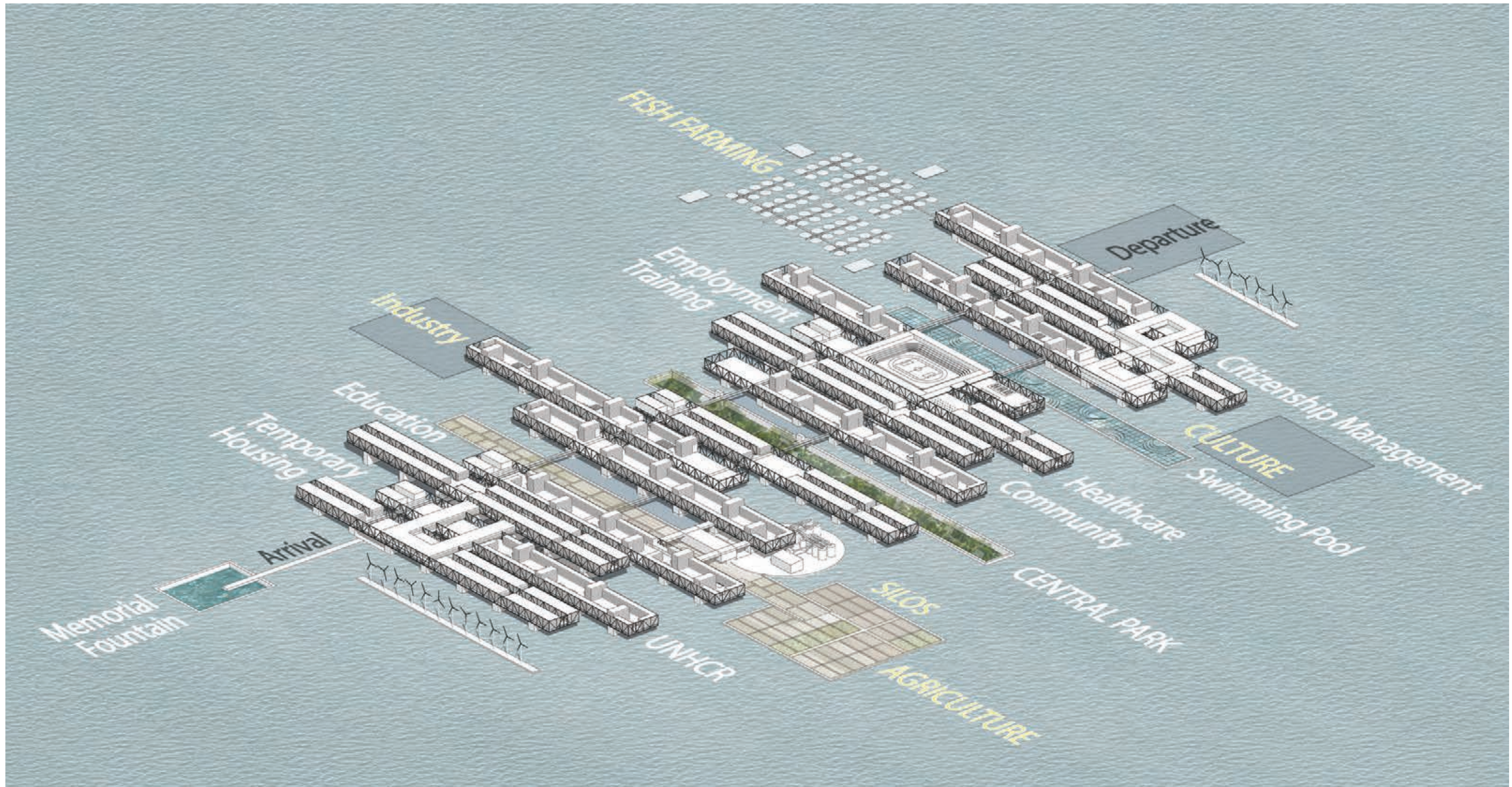




Rem Koolhaas's architectural theory, epitomized by the formula "a maximum of program and a minimum of architecture," redefines urban design as a platform for interaction and adaptability rather than a rigid structural framework. This approach envisions a city where architecture serves as an enabler of dynamic and diverse social functions, fostering environments that prioritize collective activity and evolving societal needs.

In this vein, Exodus City reflects a modern interpretation of the "social condenser," a concept introduced by Russian Constructivists in the 1920s. These pioneers believed architecture could actively shape society, promoting cooperation and a more collective way of life. Similarly, Exodus City seeks to integrate diverse programs and spaces that encourage interaction, innovation, and shared experiences, adapting to emerging social forms and challenges.

The city's character is defined by its collective, cultural, flexible, diverse, and adaptable nature. These principles ensure that the urban environment remains inclusive and resilient, accommodating the spontaneity of its residents while maintaining a structured framework. The concept of open exclusivity—inviting diversity while retaining cohesion—underscores its design. By enabling proactive spontaneity, where spaces evolve organically through use, Exodus City becomes a living, adaptable model for future urbanism. Its success hinges on creating an environment that not only accommodates but actively cultivates social interaction and collective growth.



Exodus City: A Platform for the Displaced

Arriving at Exodus City is an emotional moment for many displaced people. The city serves as a transitional platform, designed to provide not only shelter but also a structured environment for adaptation. Supported by organizations like the UNHCR, new arrivals are first assigned temporary accommodations and undergo medical and psychological assessments. These initial steps are critical in addressing their immediate physical and mental well-being.

The living spaces in Exodus City are intentionally designed to create comfort and encourage interaction. While initial interactions among residents may be marked by hesitation or distrust, the shared spaces dining halls, courtyards, and community rooms foster gradual engagement. Over time, these settings help establish a sense of belonging as neighbors begin exchanging information, sharing stories, and building relationships.

The city operates on a structured routine that balances education, work, and leisure. Language classes are held in classrooms designed for natural light and collaborative

learning. Workshops scattered across the city enable residents to develop practical skills aligned with their abilities and interests, such as tailoring, carpentry, or digital literacy. These programs are essential in equipping residents with the tools to rebuild their lives and regain a sense of purpose.

Recreational spaces play a key role in creating a holistic living experience. Parks, green spaces, and access to Lake Michigan offer opportunities for relaxation and reflection. Public amenities like theaters, sports stadiums, and cultural centers provide moments of joy and a chance to participate in shared activities. These elements not only improve quality

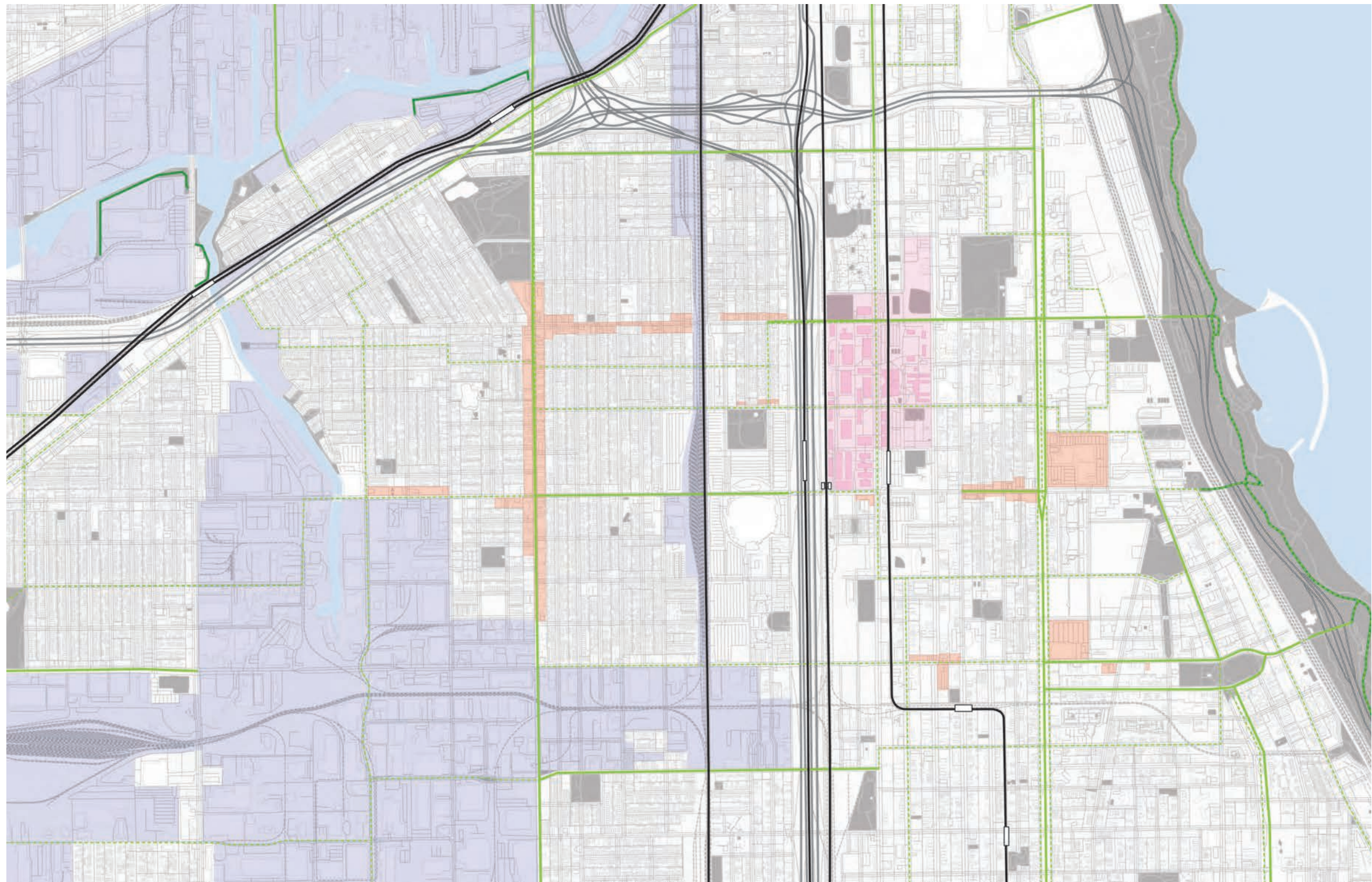
of life but also strengthen social bonds within the community.

The goal of Exodus City is to prepare its residents for the next phase of their lives, whether that involves resettlement or reintegration. By providing a supportive environment that addresses social, economic, and emotional needs, the city fosters resilience and confidence. As residents leave its protective boundaries, they carry with them not only practical skills but also a renewed sense of hope and belonging, ready to embrace the challenges and opportunities ahead.

04 New Bronzeville

A FUNCTIONAL-MIXED NEIGHBORHOOD

This project proposes a functional-mixed development that includes a renovated stadium and a new transportation hub, creating more appealing spaces for both residents and visitors. By repurposing nearby vacant lots and abandoned buildings, the project also promotes the provision of affordable housing and neighborhood revitalization.



- Industry
- Retail
- Education(IIT)
- Bike Lane
- Dan Ryan Expressway
- Railway
- Train Station

Site Analysis

Chicago's transportation network, including the Dan Ryan Expressway, Red Line, Green Line, and Amtrak, serves as the city's lifeblood, facilitating connections across its urban fabric. Yet, these same systems act as significant barriers, particularly between the historic neighborhoods of Bridgeport and Bronzeville. The Dan Ryan Expressway, with its immense width of approximately 500 feet, creates not only a physical but also a psychological divide, reinforcing a sense of disconnection between these communities.

Bridgeport, historically known as a working-class neighborhood, has deep roots in Chicago's industrial and political history. Generations of immigrants, particularly

Irish and Eastern Europeans, worked in its factories and stockyards, contributing significantly to Chicago's growth as an industrial hub. Today, Bridgeport has evolved into a diverse neighborhood celebrated for its cultural richness, vibrant arts scene, and commercial corridors like Halsted Street. However, the benefits of this vibrancy rarely extend into adjacent Bronzeville, leaving a critical gap in connectivity and collaboration.

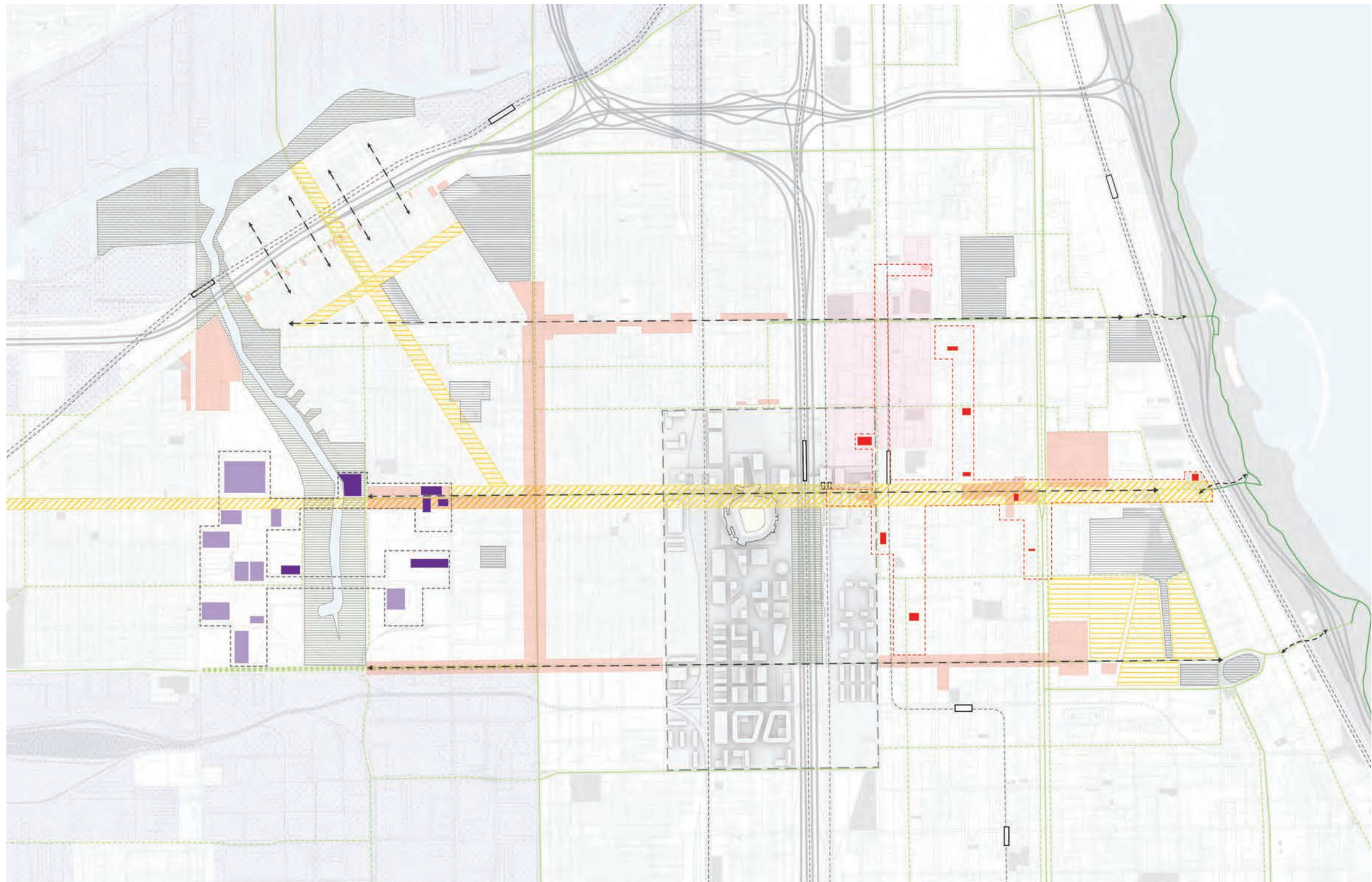
Bronzeville, on the other hand, is a landmark of African American history and culture, often referred to as the "Black Metropolis." During the Great Migration in the early 20th century, it became a thriving center of Black-

owned businesses, arts, and activism. Icons like the Regal Theater and the Chicago Defender newspaper helped shape its cultural identity. However, decades of systemic disinvestment and urban neglect have left Bronzeville physically and economically isolated. Despite ongoing revitalization efforts, it remains disconnected from nearby areas like Bridgeport.

Adding to this divide is the industrial corridor along the Chicago River. Once a hub of economic activity, the corridor has declined, leaving underutilized spaces that contribute to stagnation in the area. These deteriorating industrial zones amplify the sense of separation and create visual and

functional barriers.

Recognizing the rich histories and cultural contributions of Bridgeport and Bronzeville underscores the importance of bridging these divides. Thoughtful urban design and planning strategies that address physical barriers and prioritize inclusivity can create opportunities for collaboration and connectivity. By transforming underutilized infrastructure into cohesive urban spaces, these neighborhoods can honor their legacies while fostering a dynamic, unified future for the region.



- Art museum
- Historical Building
- Industry
- Retail
- Education(IIT)
- Bike Lane
- Dan Ryan Expressway
- Railway
- Train Station

Urban Framework

The redevelopment of the area surrounding White Sox Stadium presents a significant opportunity to revitalize Chicago's South Side by addressing existing barriers and fostering connectivity, inclusivity, and economic growth. This urban framework integrates innovative strategies to create a dynamic and sustainable environment.

A primary strategy involves constructing a deck park over the Dan Ryan Expressway, providing much-needed outdoor spaces while mitigating the psychological and physical divide caused by the highway. Multi-level platforms will further enhance the urban fabric by offering dynamic public spaces and improving accessibility. Enhancing the pedestrian

environment along 35th Street and Pershing Road with vibrant east-west commercial corridors will strengthen local economies and encourage community interaction.

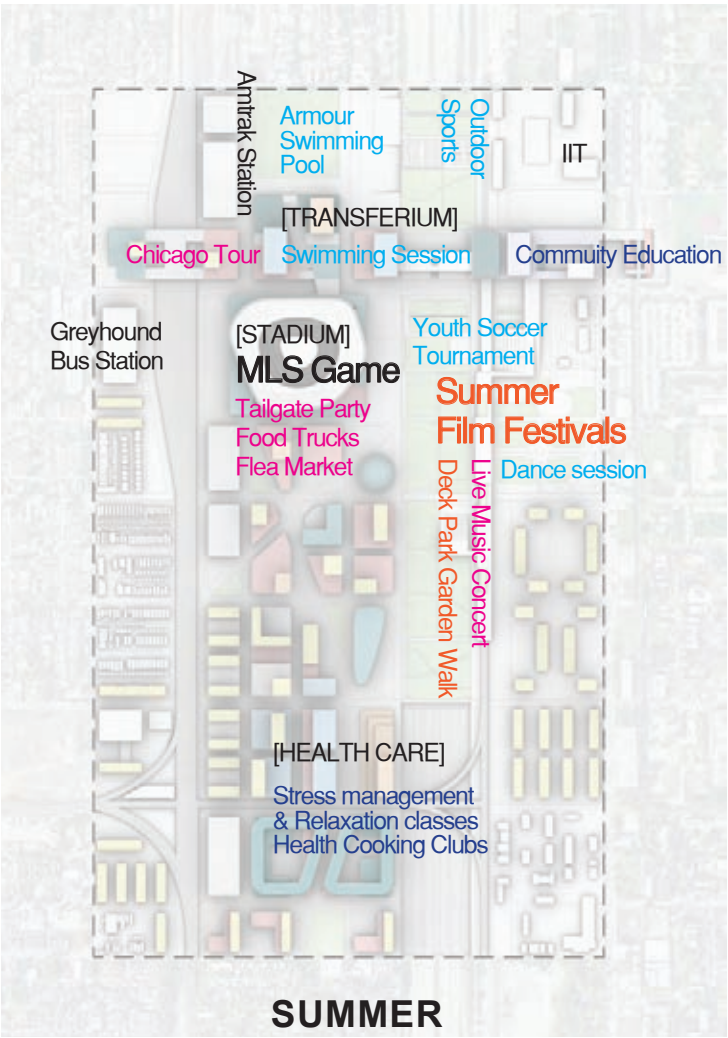
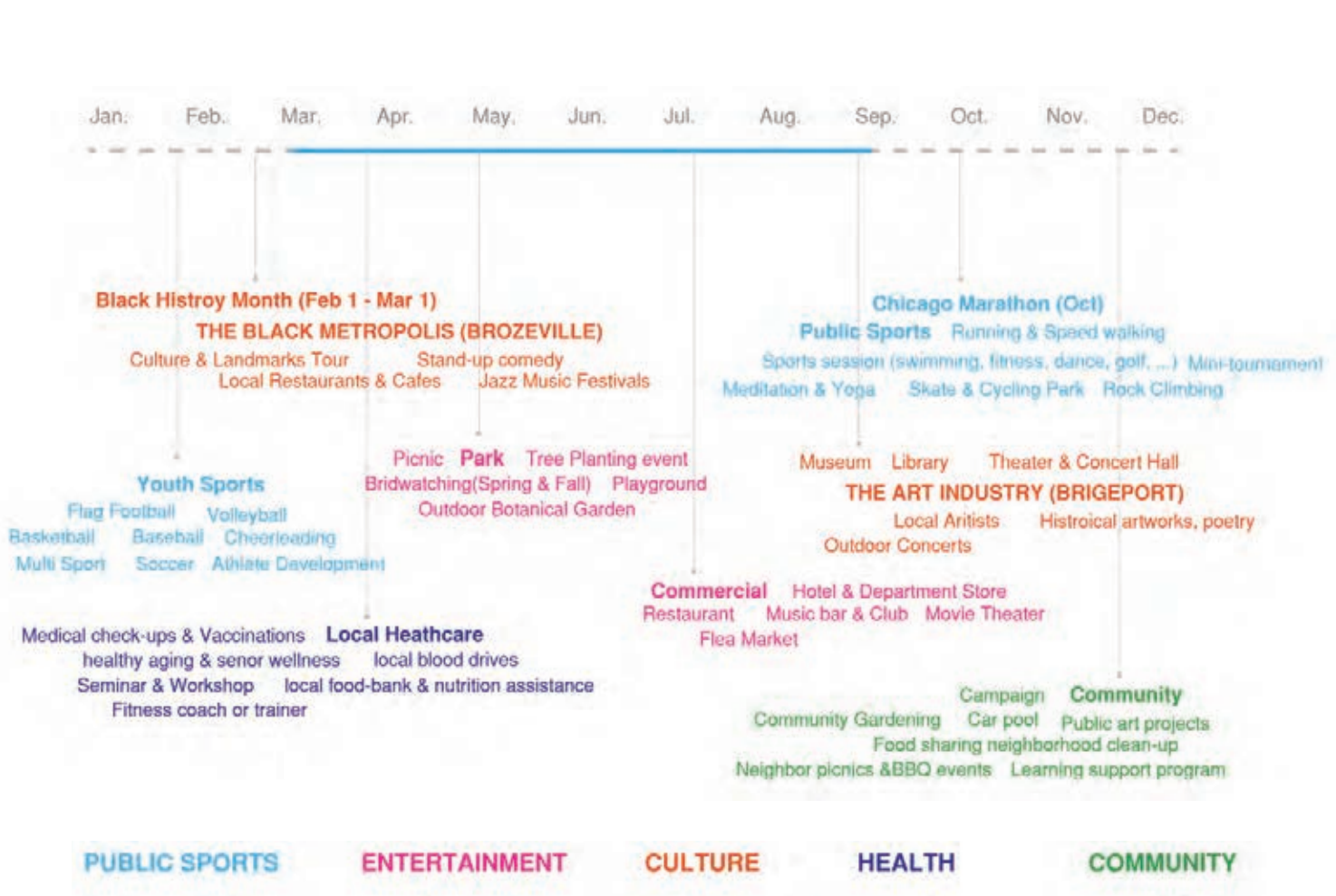
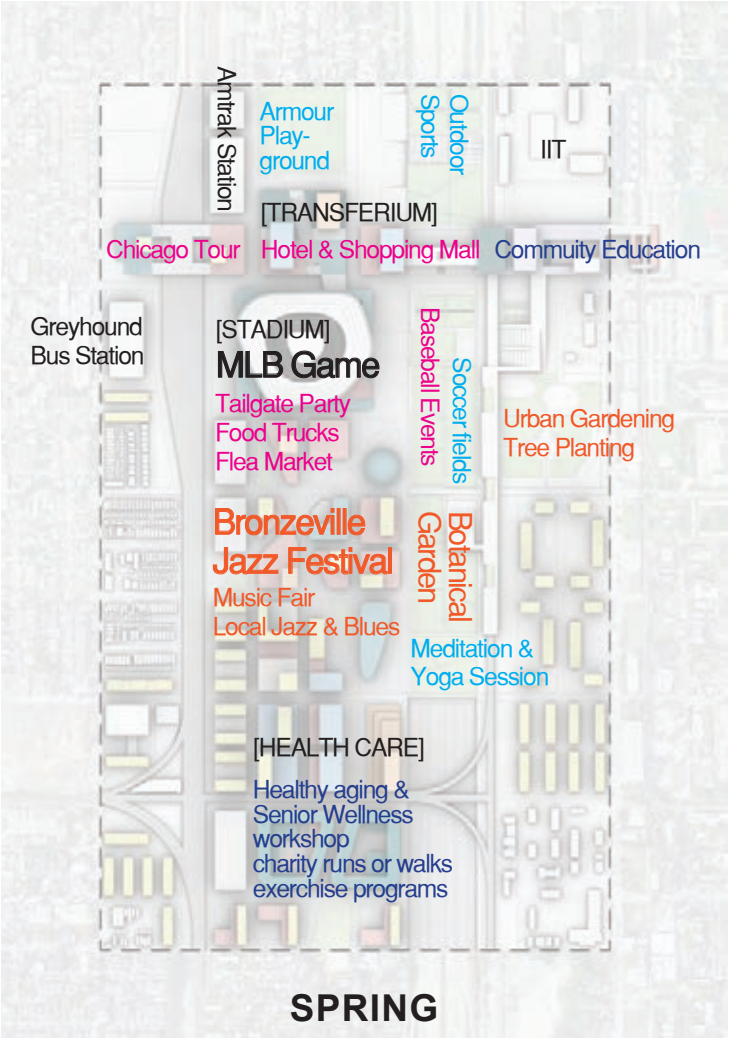
In Bridgeport, the adaptive reuse of former industrial zones into art museums has demonstrated the potential for transforming underutilized spaces. The framework proposes developing these areas into a cohesive cultural district. The Chicago River waterfront will also be revitalized, creating an accessible and vibrant area featuring cultural and recreational amenities. These enhancements will turn the riverfront into a key destination, promoting environmental sustainability and attracting visitors.

In Bronzeville, the preservation and promotion of historic buildings from the Black Metropolis era will celebrate the neighborhood's cultural legacy while fostering economic revitalization. By incorporating these historic sites into tourism-focused initiatives, the framework not only preserves Bronzeville's heritage but also attracts new investments and creates opportunities for local residents.

The extensive parking lots around the stadium will be reimagined as a new commercial district, transforming the area into a fresh central hub for Chicago. This mixed-use development will integrate transportation infrastructure with office spaces, retail centers, cultural venues, and healthcare

facilities. This reimagined district will drive economic growth, support local businesses, and promote walkability, creating a vibrant urban core.

By bridging divides, enhancing connectivity, and promoting sustainable development, this framework unlocks the potential for long-term revitalization. Through cultural preservation, adaptive reuse, and innovative urban planning, the proposal positions this area as a dynamic and integral part of Chicago's urban fabric, benefiting both local communities and the broader city.



Annual Programming: Activating Baseball Stadium and Surrounding Areas

The harsh winter weather and the off-season of Major League Baseball leave the White Sox Stadium parking lots underutilized, contributing little to the local community. To support community engagement and economic revitalization, sustained programming across all seasons is essential. This approach ensures consistent activity, attracting both residents and visitors while maximizing the site's potential.

During the winter season, programming could highlight the cultural richness of the area, particularly through events that celebrate Black History Month in connection with Bronzeville's Black Metropolis heritage. Cultural and landmark tours, as well as jazz and blues concerts, would draw attention to the neighborhood's history. Additionally, events such as New Year's concerts and Christmas markets held at the stadium could encourage outdoor activities and attract diverse audiences.

In the spring season, the start of the MLB season provides opportunities for tailgate parties, food trucks, and flea markets to enhance the game-day experience. Seasonal activities like the Bronzeville Jazz Festival and a botanical garden showcase could further enrich the community's offerings, fostering local pride and engagement.

The summer season would be perfect for summer film festivals hosted in the proposed Deck Park, offering outdoor entertainment for families. To promote public health, the city could organize sports sessions and health check-up campaigns that support resident well-being while activating public spaces.

For the fall season, aligning with the international appeal of the Chicago Marathon could be a transformative opportunity. By incorporating the stadium area into the marathon route and offering associated programs like vendor markets, live music, and sports exhibitions, the event's reach could extend beyond downtown. This would distribute tourism and economic activity to the South Side, introducing new vitality to the area.

This vision emphasizes that urban revitalization is not just about developing physical infrastructure but also fostering meaningful community participation through intangible events. By designing year-round programming that reflects the local identity and needs, the stadium and its surrounding areas can become a thriving hub for cultural, social, and economic activity. This holistic approach ensures that the site remains vibrant, inclusive, and central to the community throughout the year.



Mixed-Use Development & Transit-Oriented Design

Jane Jacobs (1961) argued that cities thrive when they integrate a mix of functions and encourage co-functioning between them. This principle remains particularly relevant in the context of transit-oriented development (TOD), where transportation infrastructure and mixed-use urban planning combine to create sustainable, vibrant communities. Along 35th Street in Chicago, the intersection of key transportation lines presents a unique opportunity to realize Jacobs' vision.

35th Street is uniquely positioned as a potential transit hub, integrating the Green Line, Red Line, Metra, and a proposed Amtrak station, all connected near the I-90 expressway. This nexus of transit options has the potential to transform the

area into a vibrant, accessible urban corridor. By prioritizing TOD principles, the redevelopment of 35th Street could significantly reduce carbon emissions by encouraging public transit use, discouraging car dependency, and fostering a walkable city. This approach directly supports sustainability goals while enhancing connectivity within the region.

With robust infrastructure in place, the surrounding area can accommodate affordable housing, commercial spaces, and office facilities, ensuring a balance between residential, economic, and recreational functions. This mixed-use strategy aligns with Jacobs' insights by reducing travel distances and creating a diverse urban environment where

daily needs are met within a short radius. The presence of affordable housing, in particular, ensures inclusivity and social equity, while commercial and office facilities foster economic vitality.

The development of 35th Street as a transit-oriented, mixed-use hub would also enhance the intensity of street life and safety. Jacobs' concept of "eyes on the street" highlights how a mix of functions generates continuous activity, fostering natural surveillance and encouraging community interactions. A pedestrian-friendly environment with consistent activity from transit users, shoppers, and residents would transform 35th Street into an urban center.

Additionally, integrating public spaces, green areas, and pedestrian-friendly design along 35th Street would further enhance its appeal, creating opportunities for recreation and social interaction. These spaces would support a sense of community and well-being while complementing the area's commercial and residential functions.

By embracing mixed-use development and TOD principles, 35th Street can become a model urban corridor that combines sustainability, inclusivity, and vibrant urban life. This vision exemplifies Jacobs' ideas, demonstrating how cities succeed when infrastructure and diverse uses work together to foster dynamic, interconnected communities.

Rethinking Abandoned Infrastructure



Seoulo 7017 Skygarden, MVRDV, Seoul, 2017

The transformation of abandoned infrastructure into vibrant public spaces has become a pivotal strategy in urban revitalization, enhancing community engagement and promoting sustainable development. A notable example is MVRDV's Seoulo 7017 Skygarden in Seoul, South Korea, which repurposed a disused highway overpass into a thriving pedestrian walkway and botanical garden.

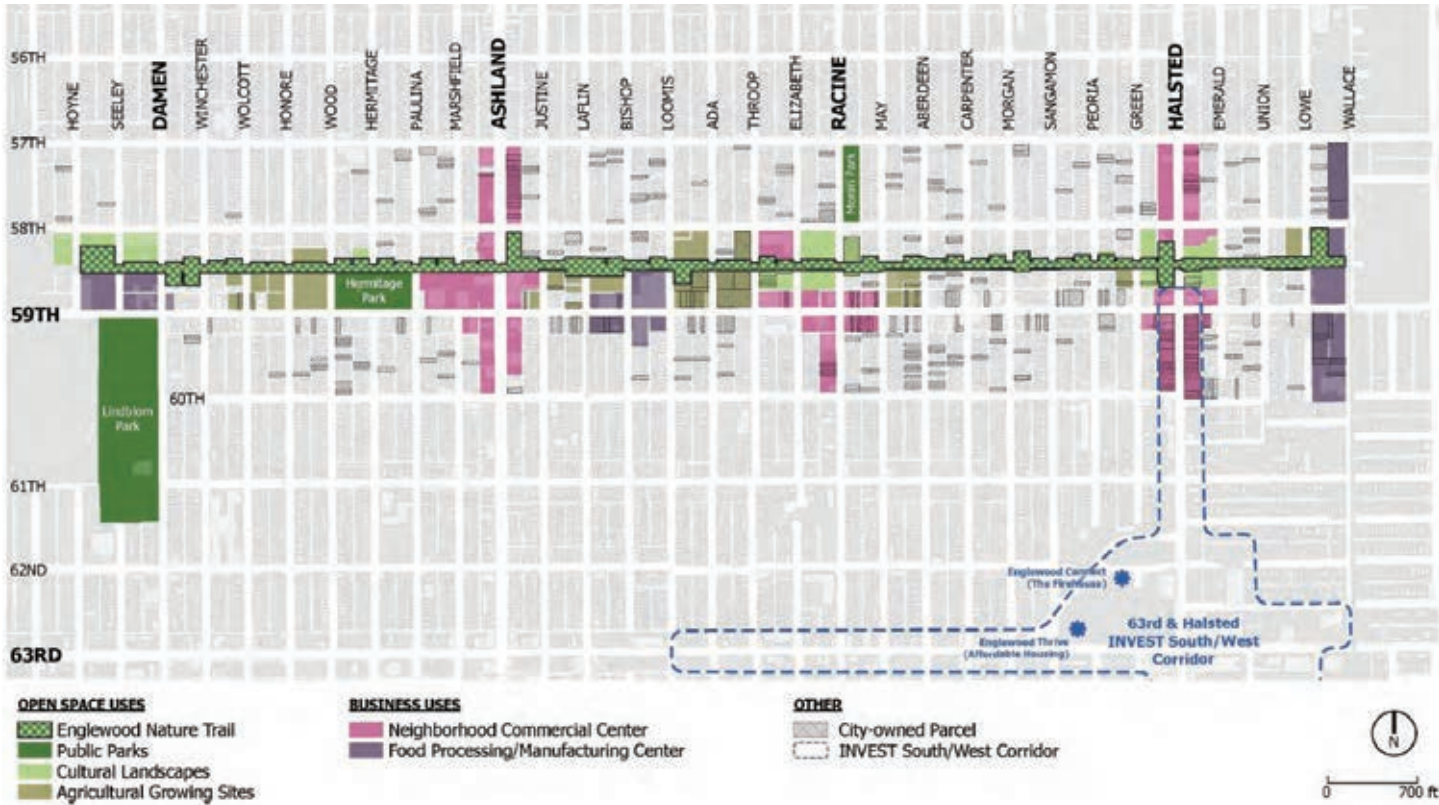
Originally constructed in 1970, the Seoul Station overpass was closed in 2015 due to safety concerns. Recognizing the potential for adaptive reuse, the Seoul Metropolitan Government collaborated with MVRDV to redesign the structure as an elevated park, which opened in 2017. The Skygarden features over 24,000 plants, representing 228 species, arranged in a linear sequence that reflects the Korean alphabet. This design not only introduces green space into the dense urban fabric but also serves as an educational arboretum for visitors.

The project's impact on urban mobility and community interaction is significant. By providing a safe and accessible pedestrian route, Seoulo 7017 enhances walkability in central Seoul, encouraging foot traffic and reducing reliance on automobiles. The park connects various neighborhoods, fostering social cohesion and offering a venue for cultural events, markets, and performances. This integration

of nature and urban life exemplifies how repurposed infrastructure can address environmental concerns while enriching the social and cultural landscape of a city.

Moreover, Seoulo 7017 contributes to economic revitalization by attracting tourists and stimulating local businesses. The increased footfall has led to the emergence of new commercial activities in the vicinity, boosting the local economy and creating job opportunities. This aligns with findings from the Urban Institute, which highlight that investment in repurposed infrastructure can produce good jobs for residents of disinvested communities.

In conclusion, Seoulo 7017 Skygarden demonstrates the multifaceted benefits of rethinking abandoned infrastructure. By transforming a neglected overpass into a dynamic public space, the project enhances urban walkability, strengthens community ties, and stimulates economic growth. Such initiatives serve as models for cities worldwide, illustrating how adaptive reuse can address contemporary urban challenges through innovative design and planning.



Englewood Nature Trail, Gensler, Chicago, 2022

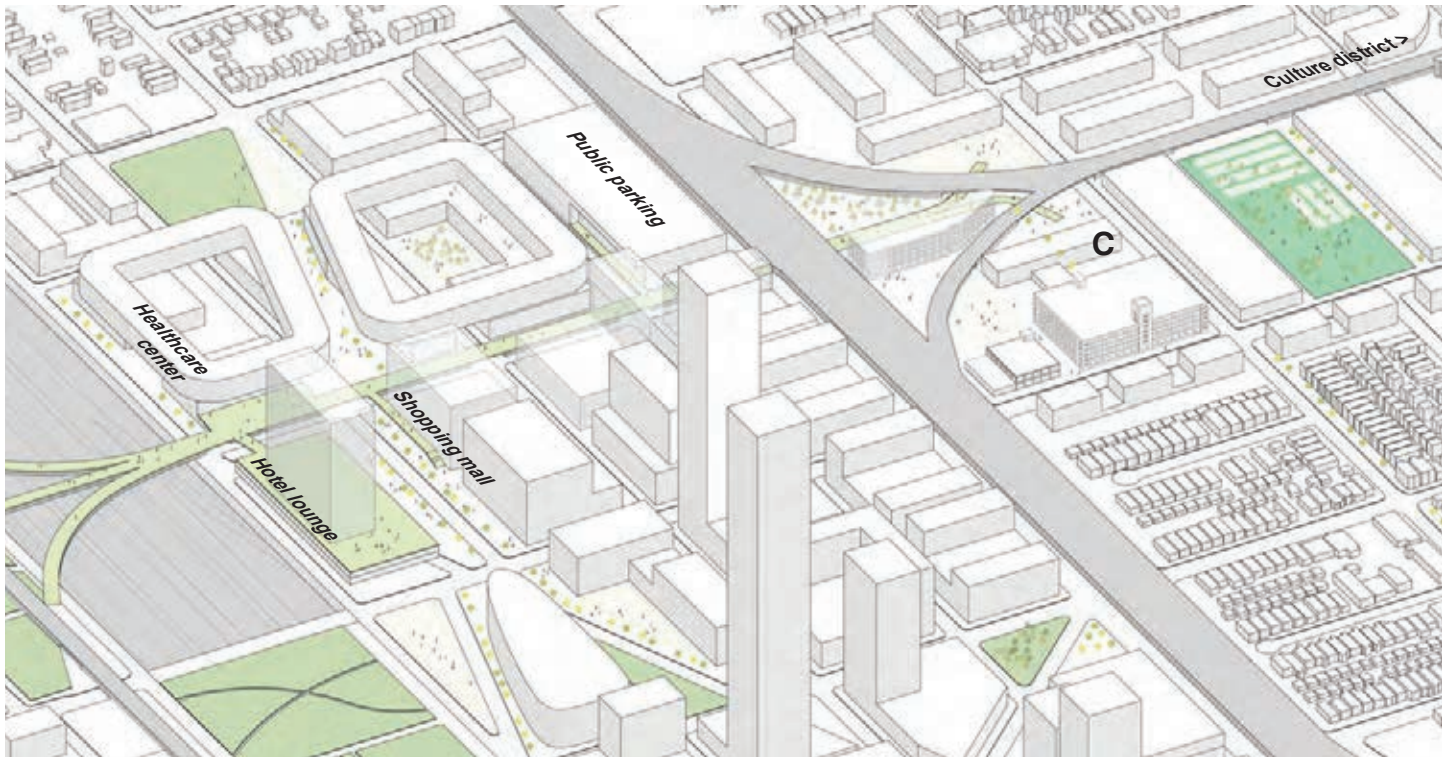
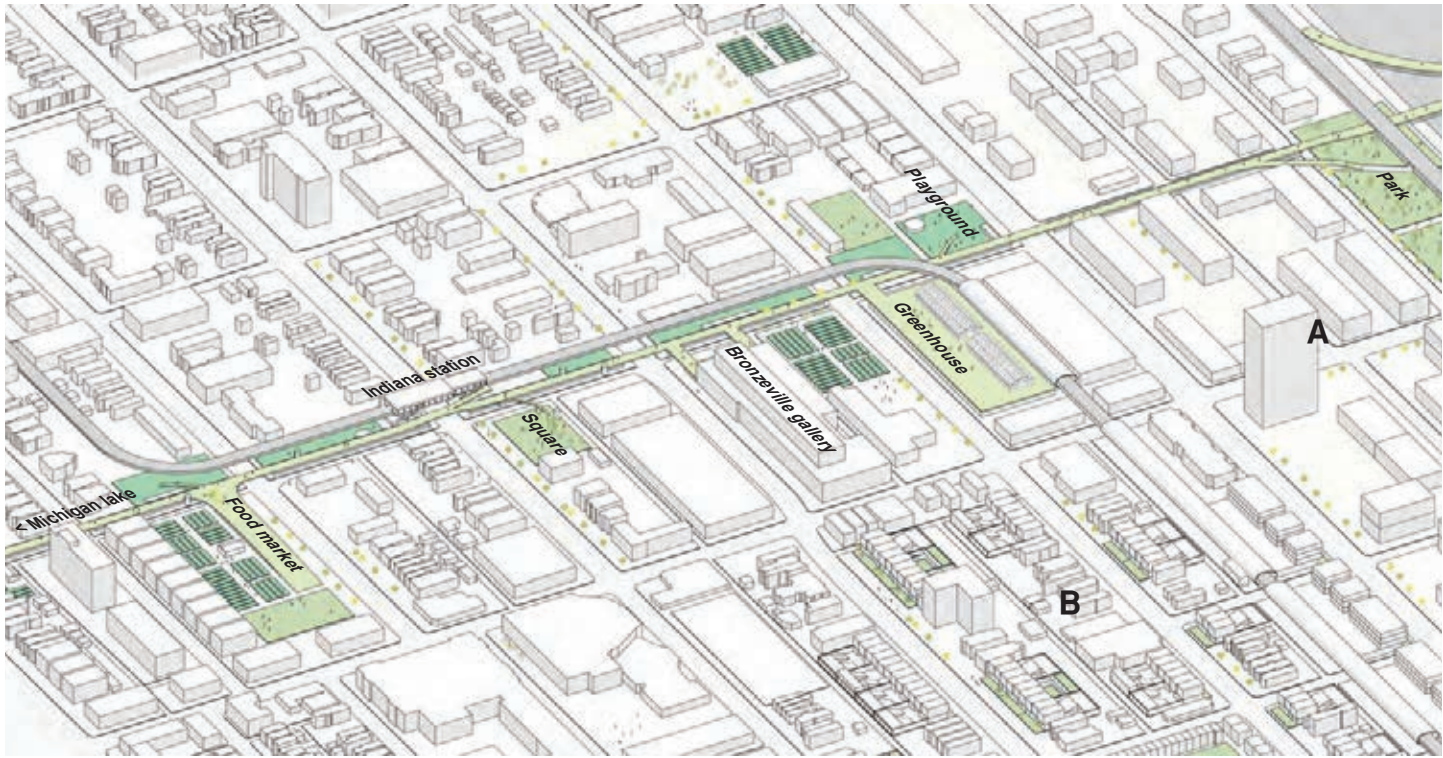
The Englewood Nature Trail, designed by Gensler, is a visionary project in Chicago that repurposes a disused railway corridor into a linear park and community space. Stretching approximately 1.75 miles through the South Side neighborhood of Englewood, the project aims to reconnect a historically underserved area while promoting sustainability, community engagement, and urban revitalization.

Once a symbol of industrial decline, the railway corridor is being transformed into a green infrastructure that fosters environmental restoration and community well-being. The trail features walking and biking paths, native landscaping, and recreational spaces that create opportunities for outdoor



activities and ecological education. Its design prioritizes accessibility and inclusivity, ensuring it serves as a resource for all residents, regardless of age or mobility.

Beyond its environmental impact, the Englewood Nature Trail serves as an economic catalyst, spurring investment in local businesses and creating job opportunities. By attracting visitors and connecting neighborhoods, the trail reinforces social cohesion and provides a platform for cultural events and community programs.



Bronzeville Community Trail

The proposed Bronzeville Community Trail seeks to transform an abandoned rail corridor facing the Green Line into a dynamic public space for the local community. Inspired by successful projects like the Englewood Nature Trail, this initiative aims to provide an elevated pedestrian environment that enhances connectivity and fosters community engagement. By linking the trail to the Indiana Station, accessibility is significantly improved, creating a seamless connection between the trail, transit, and surrounding neighborhoods.

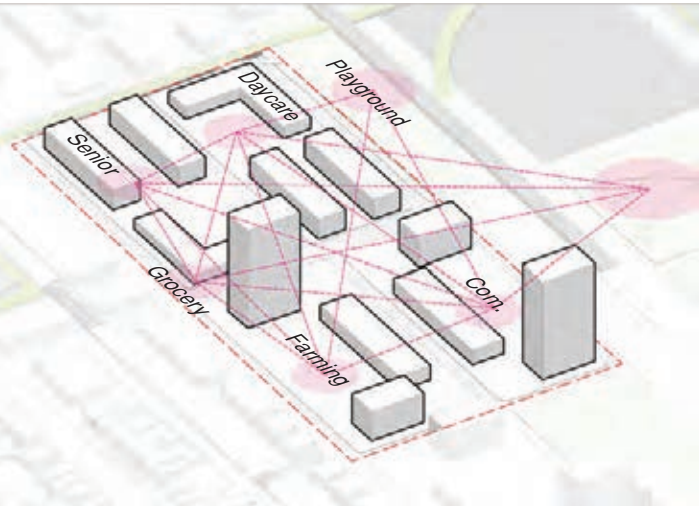
Along the trail, a variety of public programs will be introduced to serve the needs of the Bronzeville community. Local

food markets, plazas, galleries, greenhouses, parks, and playgrounds will be strategically distributed to encourage social interaction, cultural exchange, and recreational activities. These facilities not only activate the space but also contribute to the well-being of residents by creating a vibrant, walkable environment. Vacant lots and underused industrial buildings along the trail will be repurposed, supporting sustainable redevelopment and preserving the area's historical and architectural character.

The Bronzeville Community Trail will also connect directly to ongoing urban development projects in the Bridgeport area, creating a broader network of integrated urban spaces.

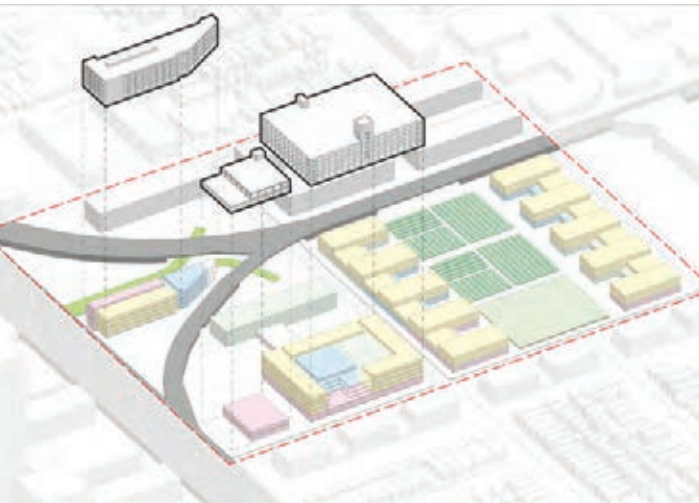
This includes connections to hotel lounges, shopping malls, and shared parking facilities, enabling economic and social synergies between the two neighborhoods. The integration of these facilities strengthens the trail's role as both a local asset and a regional connector, promoting economic growth and fostering a shared sense of identity across communities.

By reimagining the abandoned rail infrastructure as a multi-functional public corridor, the Bronzeville Community Trail not only revitalizes a neglected space but also positions itself as a catalyst for urban regeneration. This project exemplifies the potential of adaptive reuse and community-centered design to address urban challenges, creating a more inclusive, sustainable, and interconnected city.



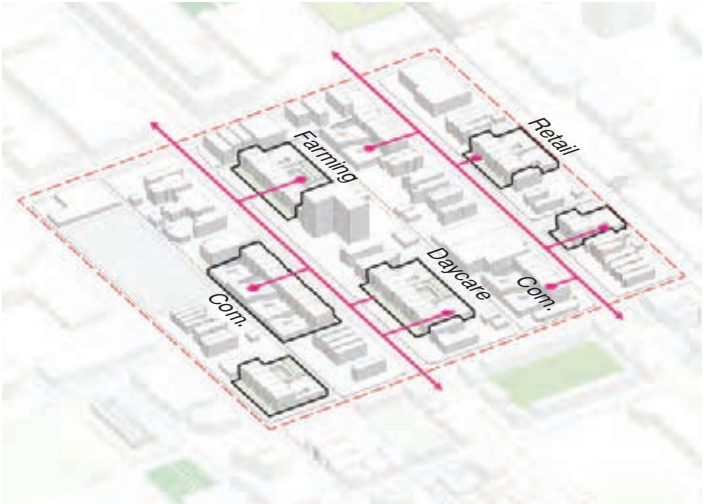
B Co-location Housing

Consolidating vacant lots in the area offers an opportunity to address social issues by creating colocation housing that integrates retail, community spaces, and urban farms on the lower floors. This approach not only provides affordable housing but also promotes social interaction, local commerce, and food security. By combining residential and communal functions in a single development, these projects can foster stronger community ties, improve access to essential services, and contribute to the neighborhood's economic and social revitalization.



A Mixed-Income Apartments

City-owned vacant lots present a valuable opportunity to develop mid- to high-rise mixed-income housing that integrates essential services such as senior housing, daycare facilities, community centers, and grocery stores. This approach promotes community integration by addressing diverse needs within a single development, fostering intergenerational connections and social cohesion. By utilizing these lots strategically, cities can enhance access to resources, support economic diversity, and create vibrant, inclusive urban environments that cater to residents of all ages and backgrounds.



C Industrial Building Reuse

In Bridgeport and Bronzeville, old industrial buildings are repurposed into mixed-use spaces that combine work and living environments. These redevelopments provide affordable housing, co-working spaces, and community hubs, fostering local jobs and stronger community ties. By preserving their historical character while supporting modern uses, the projects promote economic growth, social equity, and neighborhood revitalization.

05

Permeable Block

CLUSTERED INTERACTIONS

Mixed-income housing is a critical strategy for achieving social integration across income levels and ensuring long-term economic sustainability. However, its success depends on creating diverse and overlapping cluster spaces that enable meaningful interaction. This project examines a residential block along Pershing Road, exploring how public, semi-public, semi-private, and private spaces can be designed to weave users together, fostering connection and interaction.



1. Lot for Sale

The prevalence of vacant lots in Bronzeville creates gaps in the urban fabric, contributing to a sense of underutilization and disconnection. These lots vary in scale, from small privately owned parcels to large city-owned properties, scattered across the neighborhood. While their presence currently adds to the area's stagnation, they offer significant redevelopment potential for mixed-use housing, commercial spaces, and community amenities. With strategic planning, these vacant lots can become catalysts for revitalization, restoring vibrancy and economic activity.



2. Abandoned Manufacturing Facility

Bronzeville's historic industrial buildings serve as remnants of the area's manufacturing legacy. While some have been repurposed as storage facilities, many remain vacant and deteriorating, limiting their potential contributions to the neighborhood. These structures possess architectural significance and could be transformed into mixed-use developments, cultural spaces, or creative hubs. Through adaptive reuse, these buildings can contribute to economic revitalization while preserving the district's industrial character and historical identity.



3. Bronzeville Neighborhood Farm

The Bronzeville Neighborhood Farm serves as a vital community hub, fostering connections among residents and offering a space to alleviate mental stress through gardening activities. Beyond enhancing the neighborhood's vibrancy, the farm presents opportunities for local economic development by potentially linking with farmers' markets, thereby promoting access to fresh produce and supporting local agriculture.



4. Potential of Green Line Underpass

The underutilized space beneath the Green Line tracks presents an opportunity for urban activation. Currently vacant, this corridor could be transformed into public gathering spaces, such as outdoor seating areas, fitness zones, or markets, enhancing the community's recreational and social offerings. By repurposing the space, Bronzeville can introduce multi-functional urban programming, fostering a more dynamic, walkable neighborhood while maximizing land use efficiency.



5. TOD Development

Transit-oriented development (TOD) is actively shaping Bronzeville, particularly around the 43rd Green Line Station, where affordable housing and retail spaces are emerging. This walkable, transit-connected environment encourages sustainable urban living while supporting local businesses. By increasing density around public transit, TOD strengthens the neighborhood's economic viability and accessibility. The project exemplifies how investment in infrastructure can drive long-term community revitalization, attracting both residents and businesses while reducing car dependency.



6. Abandoned Railway

Once integral to industrial operations, the now-abandoned railway in Bronzeville has become overgrown and inaccessible, acting as a physical barrier that divides the community. However, this disused corridor holds promise for redevelopment into a nature or community trail, which could reconnect north and south sections of the neighborhood, provide recreational spaces, and promote environmental education.



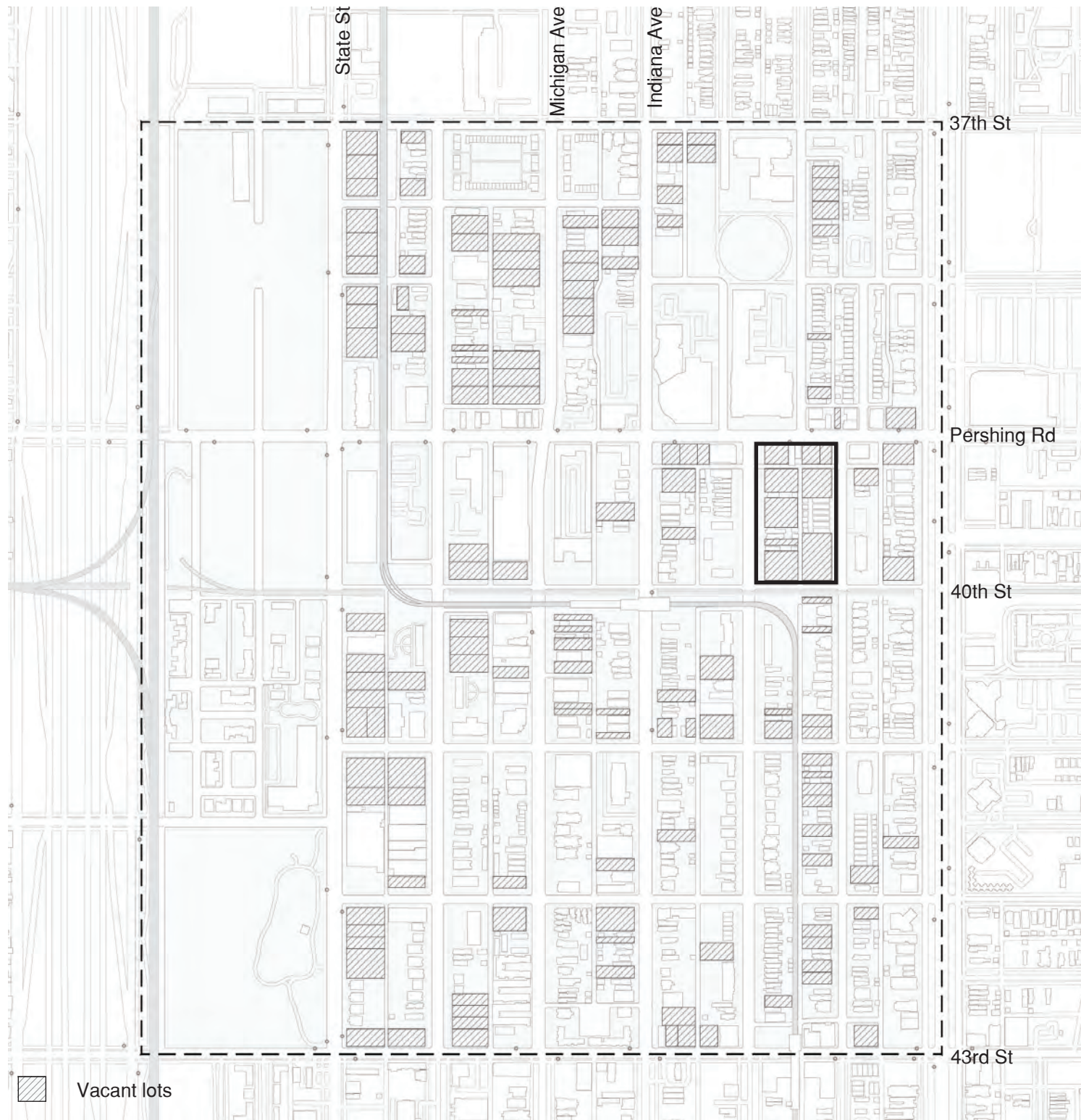
7. The Forum Bronzeville

Built in 1897, The Forum is a historic event venue at 318-328 E. 43rd Street in the Bronzeville neighborhood of Chicago. It served as a significant social and political meeting hall, reflecting the area's rich cultural heritage. Currently undergoing restoration, The Forum has the potential to once again become a central hub for community engagement and cultural events, contributing to the neighborhood's revitalization.



8. Pershing Road

Pershing Road stands as a major thoroughfare in Bronzeville with untapped potential. Currently lacking essential infrastructure such as bus shelters and bike lanes, the addition of these amenities, coupled with the integration of commercial activities, could significantly enhance urban vitality. Such improvements would promote public transit use, support local businesses, and create a more dynamic and connected community environment.



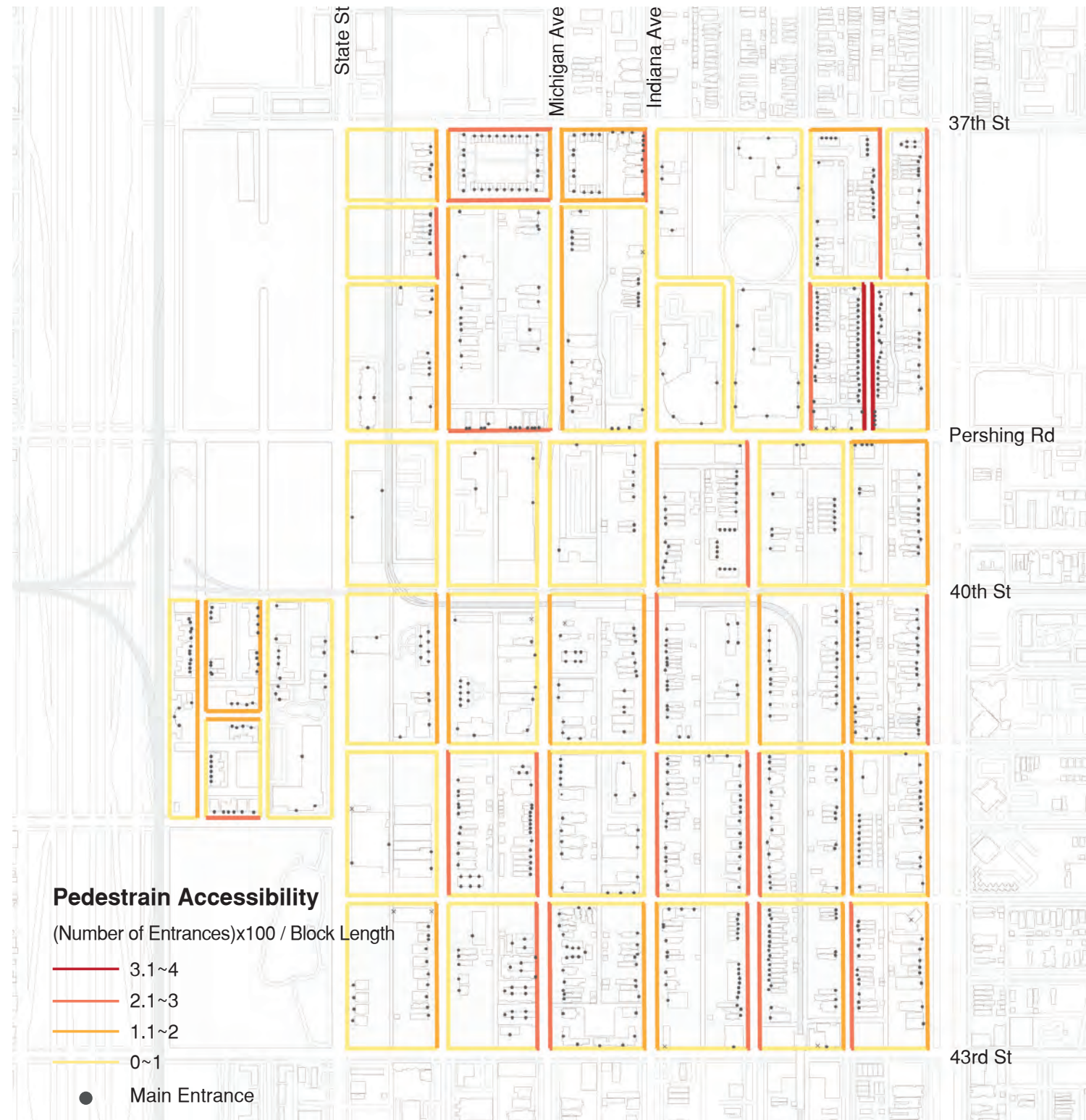
Vacant Land and Community Revitalization in Bronzeville

As of 2024, a significant portion of land in Bronzeville remains vacant, with both city-owned and privately owned lots scattered throughout the neighborhood. These empty spaces disrupt the continuity of the urban fabric, weakening street-level vibrancy, economic stability, and public safety. The prevalence of vacant land contributes to reduced foot traffic, lower investment potential, and increased perceptions of instability, negatively impacting neighborhood life.

Chicago's suburban and urban neighborhoods have long struggled with the consequences of vacant land, often experiencing higher crime rates, population decline, and economic stagnation. Without active redevelopment

strategies, these conditions continue to hinder social cohesion and local growth.

To address this challenge, this project prioritizes the strategic selection of vacant lots as testing grounds for housing alternatives and urban interventions. By reintroducing mixed-use housing, community amenities, and activated public spaces, the initiative seeks to revitalize neighborhood interactions, strengthen local economies, and create safer, more connected environments. This approach ensures that vacant lots are no longer seen as gaps in the city but as opportunities for design-driven, community-oriented regeneration.



Quantifying Street Activity Through Entrance Density

Walkable and vibrant streets depend on the frequency of pedestrian movement and street-level activity. One way to measure this is by quantifying the number of primary entrances per block length and visualizing the results through a color-coded analysis. This method highlights the relationship between urban form and pedestrian engagement, identifying areas with strong street presence and those lacking activation.

Chicago's residential grid system influences how entrances are distributed. Along the long sides of blocks, homes face each other, creating active frontages with frequent entry points. This increases pedestrian flow and street-

level interactions, contributing to a sense of security and engagement. However, along the short sides of blocks, primary entrances are typically absent, resulting in reduced foot traffic and lower street vitality.

By mapping and analyzing entrance density, this study aims to identify opportunities for intervention. Strategic design modifications—such as adding secondary entrances, integrating mixed-use functions, or activating underutilized spaces—can enhance pedestrian circulation and strengthen neighborhood vitality, ensuring that streets are engaging, accessible, and well-utilized throughout the day.



Chicago's Grid System: Origins, Benefits, and Urban Impact

Chicago's grid system is one of the most structured urban layouts in the world, originating from the Public Land Survey System (PLSS) implemented after the Northwest Ordinance of 1787. This system divided land into square-mile sections, which were further subdivided into uniform blocks, forming the foundation for Chicago's expansion in the 19th century. Following the Great Chicago Fire of 1871, the grid played a crucial role in rebuilding the city efficiently, providing an adaptable and organized framework for rapid development.

Advantages of the Grid System

- 1. Efficiency and Navigation** – The numbered street system, with its origin at State and Madison Streets, makes navigation intuitive, reducing confusion.
- 2. Scalability and Expansion** – The predictable layout allowed for seamless urban growth, accommodating industrial and residential expansion without disrupting infrastructure.
- 3. Zoning and Land Use Flexibility** – The modular grid supports diverse land uses, enabling a balance of residential, commercial, and industrial zones.
- 4. Public Transit and Infrastructure** – The grid simplifies public transit planning, ensuring direct, accessible routes for

buses and rail systems.

Challenges of the Grid System

- 1. Monotony and Lack of Organic Growth** – Unlike older cities with winding streets, Chicago's rigid layout lacks spatial diversity, making some areas feel repetitive.
- 2. Traffic and Pedestrian Flow** – The frequent intersections slow traffic movement, and inactive short block edges reduce pedestrian engagement in residential neighborhoods.
- 3. Auto-Oriented Development** – While efficient for transit, large-scale roads and long blocks in some areas prioritize cars over walkability and street life.

Cultural and Urban Impact

Despite its rigid structure, Chicago's grid fosters distinct neighborhood identities, influencing local economies, walkability, and social mobility patterns. The city's architecture, from bungalows to high-rises, has evolved within this framework, adapting to its benefits while mitigating its limitations. The grid remains a fundamental part of Chicago's urban fabric, requiring strategic interventions to enhance vibrancy and pedestrian experience in car-dominated areas.

Standardized Block Sizes in Bronzeville and Their Impact on Housing Typologies

As previously discussed, Chicago's urban layout follows a grid and block system, but block sizes vary slightly across different neighborhoods. In Bronzeville, the typical block measures 330 x 660 feet, accommodating a standardized residential lot size of 25 x 125 feet. This layout allows for 24 housing lots arranged in two rows, forming a consistent and repetitive residential pattern.

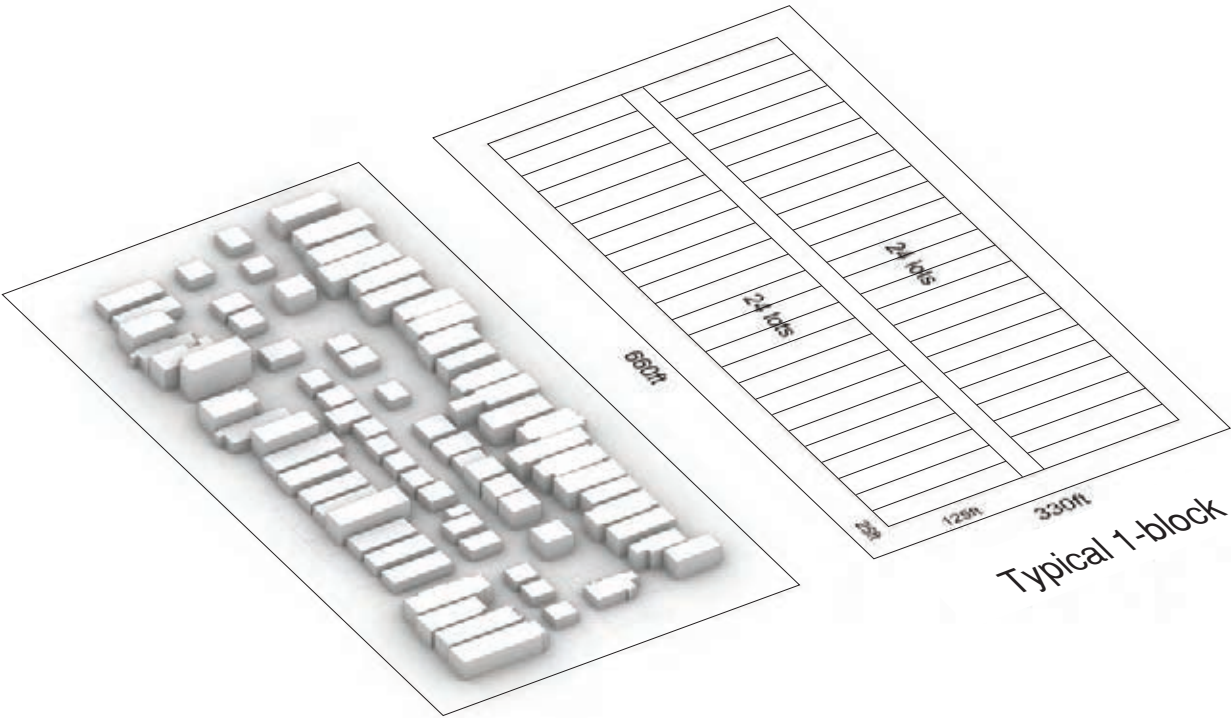
The standardization of lot sizes has historically been beneficial for construction efficiency, zoning regulations, and land management. It simplifies infrastructure planning, ensuring uniform street widths, alleyways, and utility layouts, making the development process more predictable. This system has supported the growth of Chicago's iconic housing types, such as bungalows, greystones, two-flats, and three-flats, which have long defined the city's residential neighborhoods.

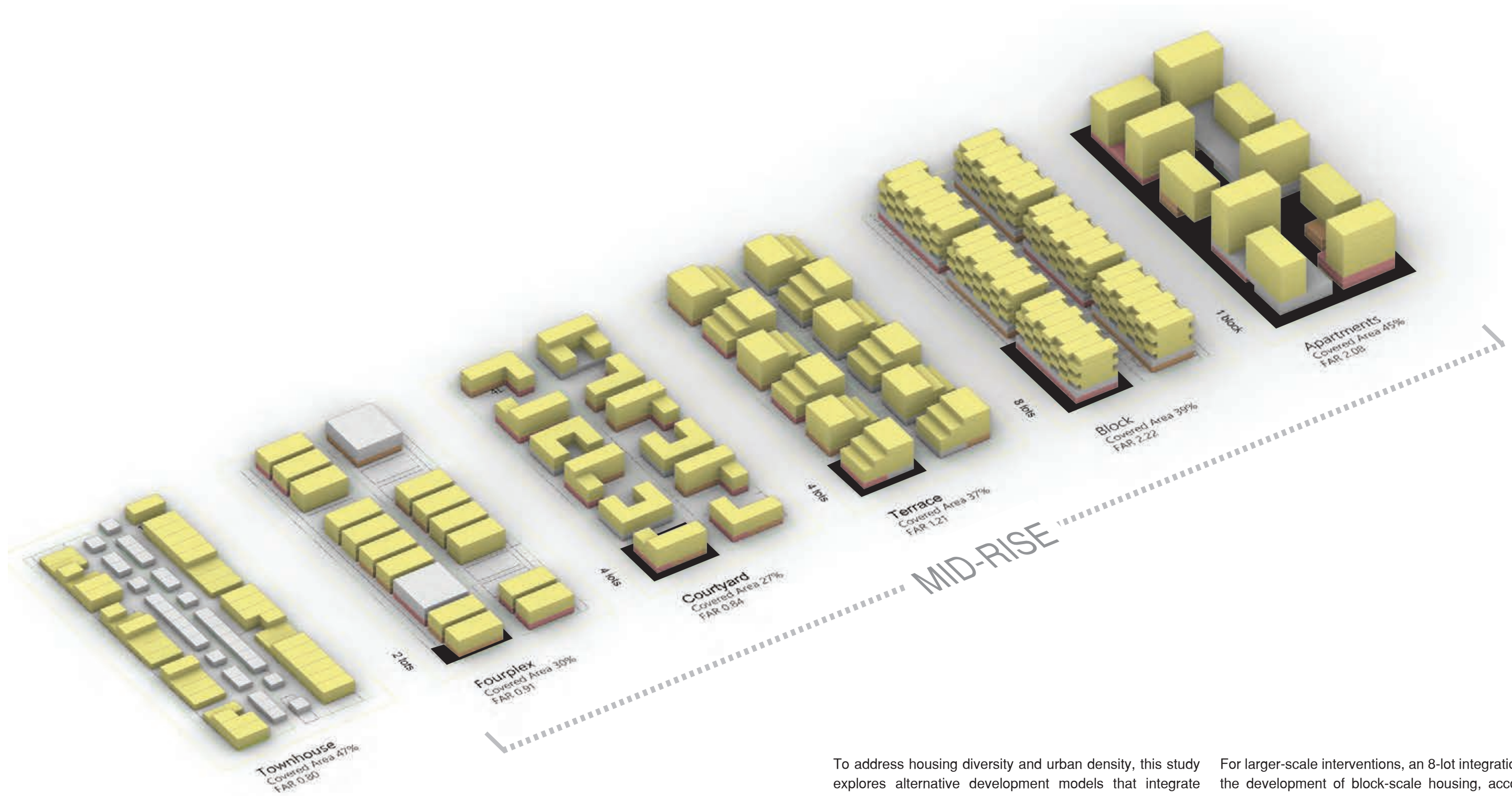
However, this rigid framework also imposes limitations on architectural diversity, urban adaptability, and social interaction. The repetitive lot sizes restrict the introduction of alternative housing models, such as co-housing communities, courtyard apartments, rowhouses, or mixed-use developments, which require different spatial configurations. As housing needs evolve, the fixed lot sizes can hinder efforts to create higher-density developments or

innovative residential solutions, particularly in areas aiming for greater affordability and social integration.

Additionally, the long 660-foot block lengths present a significant challenge to pedestrian connectivity and neighborhood interaction. Without mid-block crossings or passageways, it becomes difficult for residents to navigate across blocks efficiently, reducing the likelihood of casual encounters between neighbors. This lack of permeability limits spatial diversity, as fewer intersections mean fewer opportunities for small-scale commercial spaces, public gathering spots, or alternative urban forms.

This relationship between urban planning and architectural design highlights how block dimensions shape both housing typologies and social engagement. While the grid-based system ensures order and efficiency, it may not fully align with the demands of contemporary urban living. To foster greater architectural variety and pedestrian connectivity, strategies such as introducing mid-block passageways, flexible zoning policies, and mixed-use infill developments could be explored. By rethinking block structures, cities like Chicago can create more diverse, walkable, and socially interactive urban environments, improving housing accessibility, affordability, and neighborhood vitality.



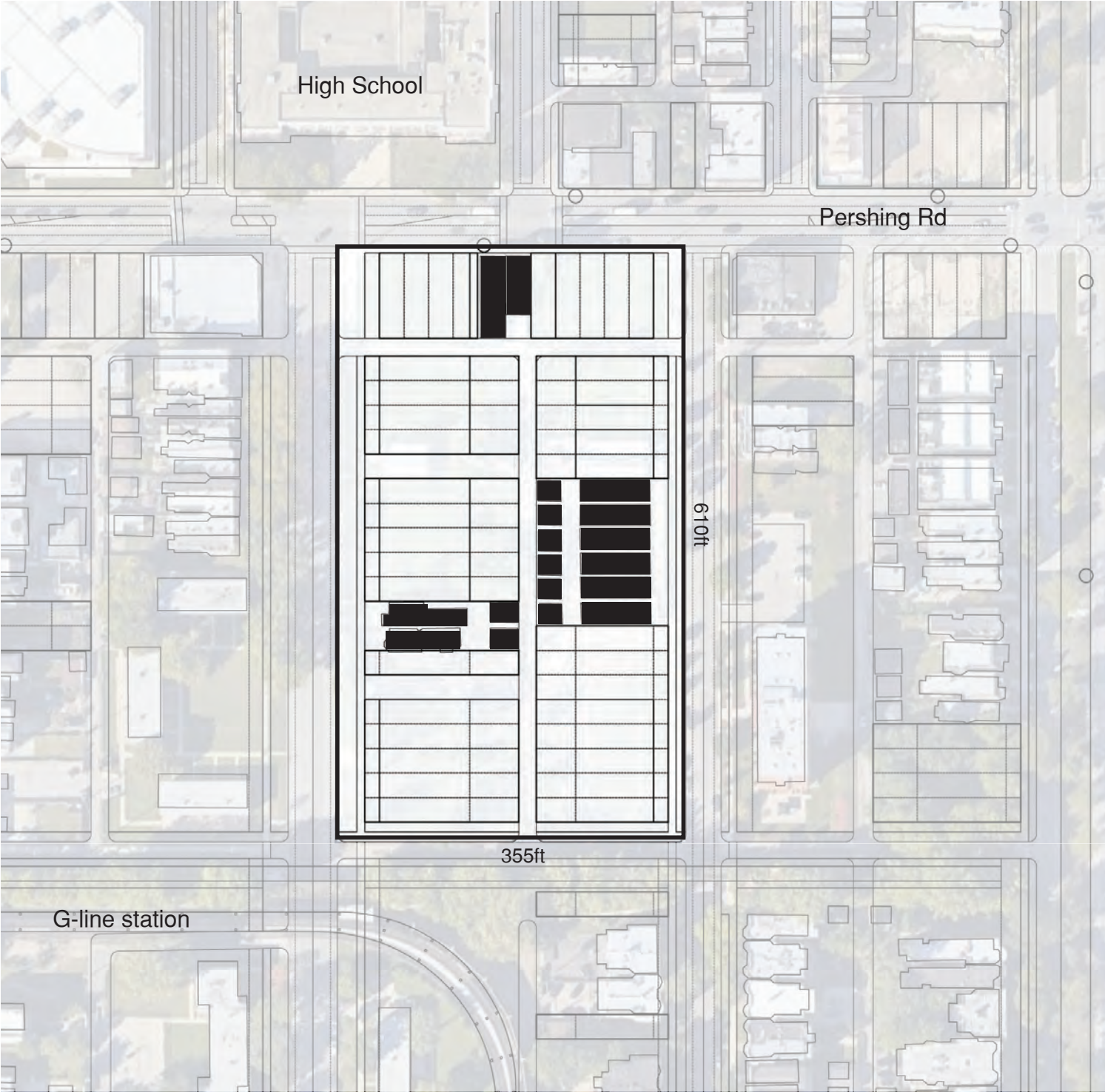


To address housing diversity and urban density, this study explores alternative development models that integrate multiple residential lots or entire blocks into cohesive housing solutions. Instead of maintaining strictly individual parcels, merging lots can create more efficient, community-oriented housing while optimizing land use.

One common strategy is combining two lots to create a fourplex, a model that increases floor area ratio (FAR) while reducing building coverage (site coverage ratio). This approach maintains a high residential density while incorporating shared outdoor spaces, balancing urban compactness with open-air interactions. Similar to European open-block housing, these configurations use courtyards and terraces to provide communal green spaces, fostering neighborly connections and improving residents' quality of life.

For larger-scale interventions, an 8-lot integration allows for the development of block-scale housing, accommodating affordable housing units, shared amenities, and parking facilities. This model supports a diverse mix of residents, offering both private living spaces and public resources, making it particularly effective for mixed-income developments.

At the largest scale, apartment complexes developed over a full city block introduce multi-functional urban housing that integrates shared amenities, community spaces, and various access points. This approach enhances social cohesion, maximizes public-private interaction, and ensures efficient land use, offering an urban living environment that balances density with livability.



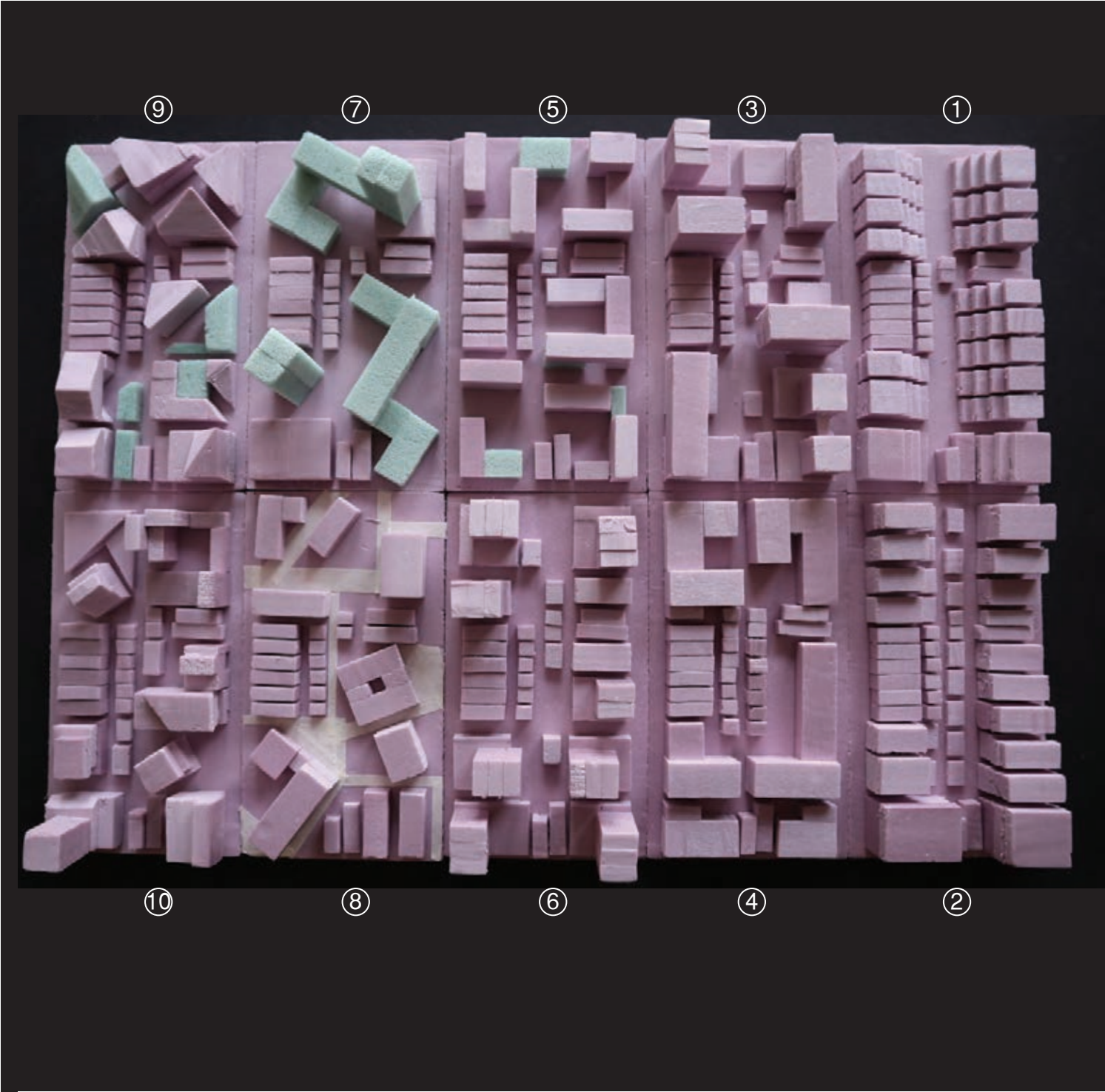
Site Context and Existing Conditions

The selected residential block measures 355 x 610 feet and currently consists of vacant residential lots, with the exception of a Black-owned business that remains in use. This underutilized land presents an opportunity for redevelopment while respecting the existing urban fabric.

The site is well-positioned within Bronzeville's urban framework, bordered on the north by Pershing Road. Across Pershing Road sits Wendell Phillips Academy High School, a historic institution that plays a significant role in the neighborhood's educational and cultural identity. The Green Line Station, located just a 2-minute walk south of the site, provides strong public transit access, supporting transit-oriented development (TOD) potential.

Like many traditional Chicago residential blocks, the site includes a service alley running through its interior, facilitating garage access, waste collection, and utility infrastructure. This alley structure is a key element of Chicago's urban planning, enabling rear-lot access and preserving pedestrian-friendly front streetscapes.

Given its strategic location and proximity to educational institutions, public transit, and commercial corridors, this site has strong potential for affordable and mixed-income housing, community-driven development, and walkable, transit-connected living. Redevelopment efforts can leverage these assets to create a more inclusive, active, and resilient urban environment.



Alternative Residential Block Strategies

To enhance neighborhood cohesion, street vibrancy, and diverse outdoor spaces, this project proposes alternative residential block models that explore varying densities, spatial configurations, and pedestrian accessibility.

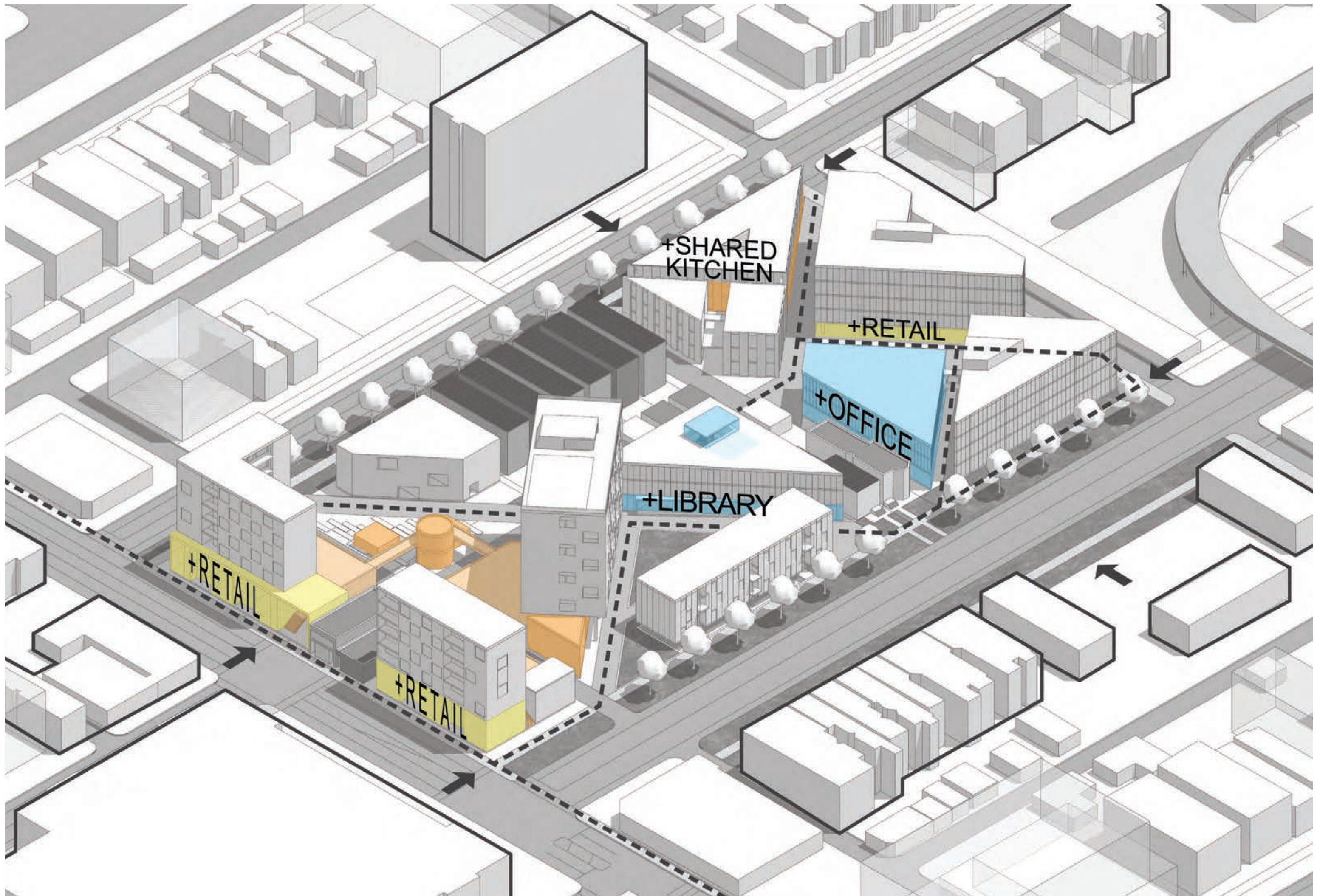
Options 1-2 respect the existing housing typologies while introducing lot consolidation to create diverse housing volumes with higher density. This approach maintains the architectural character of the area while optimizing land use.

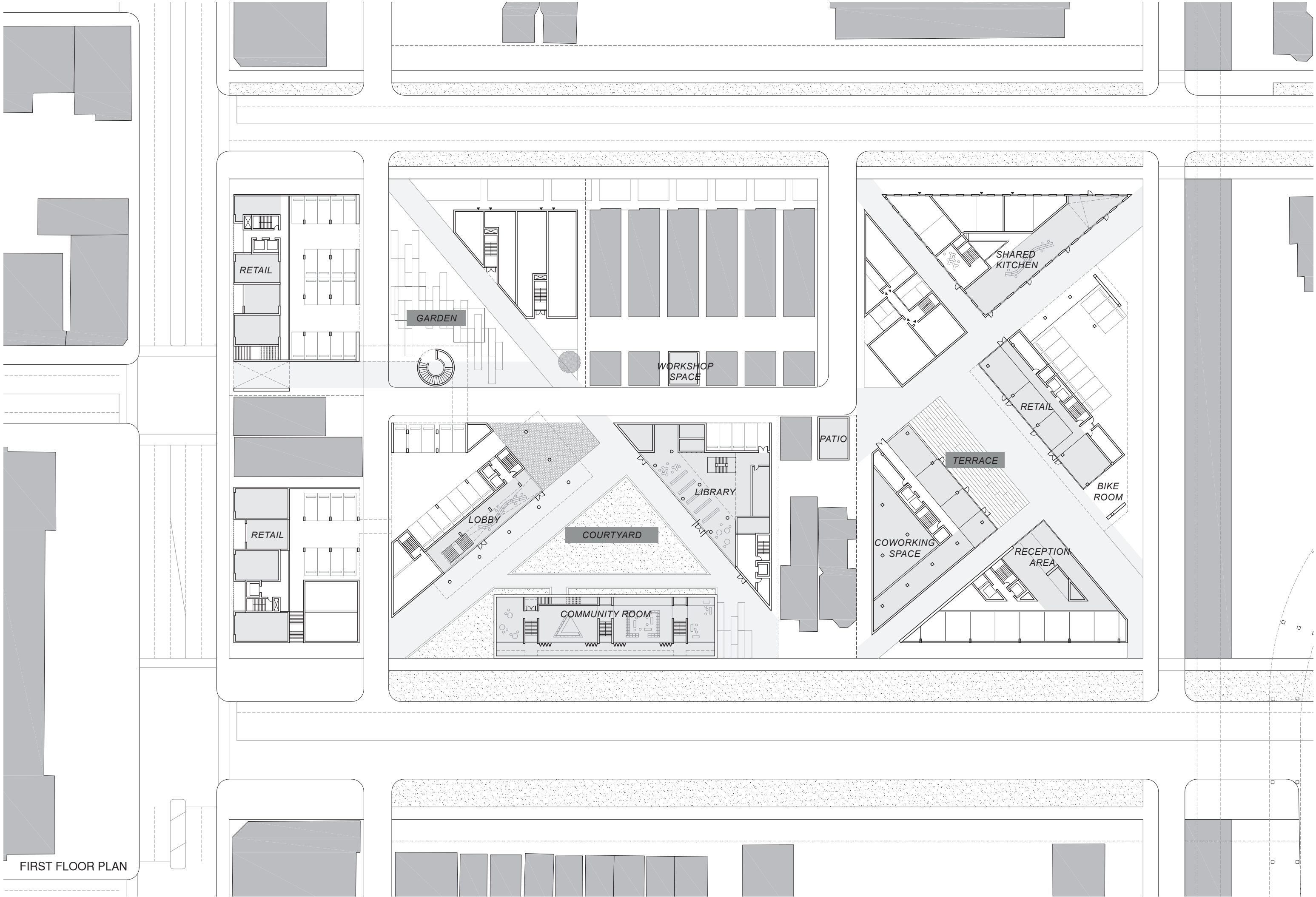
Options 3-6 adopt a courtyard block model, offering multi-scaled outdoor spaces that promote resident interaction. The design adjusts building heights to improve sunlight access for residential units, ensuring comfortable living

conditions while fostering communal engagement.

Options 7-10 intentionally shift building axes to create permeable residential blocks, integrating pedestrian pathways, alleys, and courtyards to enhance movement and connectivity. These strategies encourage spontaneous interactions, draw people into semi-public spaces, and redefine the neighborhood's residential landscape by turning housing clusters into social hubs.

Through these diverse block configurations, this proposal aims to reshape community living by balancing density, social interaction, and open space, fostering a dynamic and inclusive urban environment.

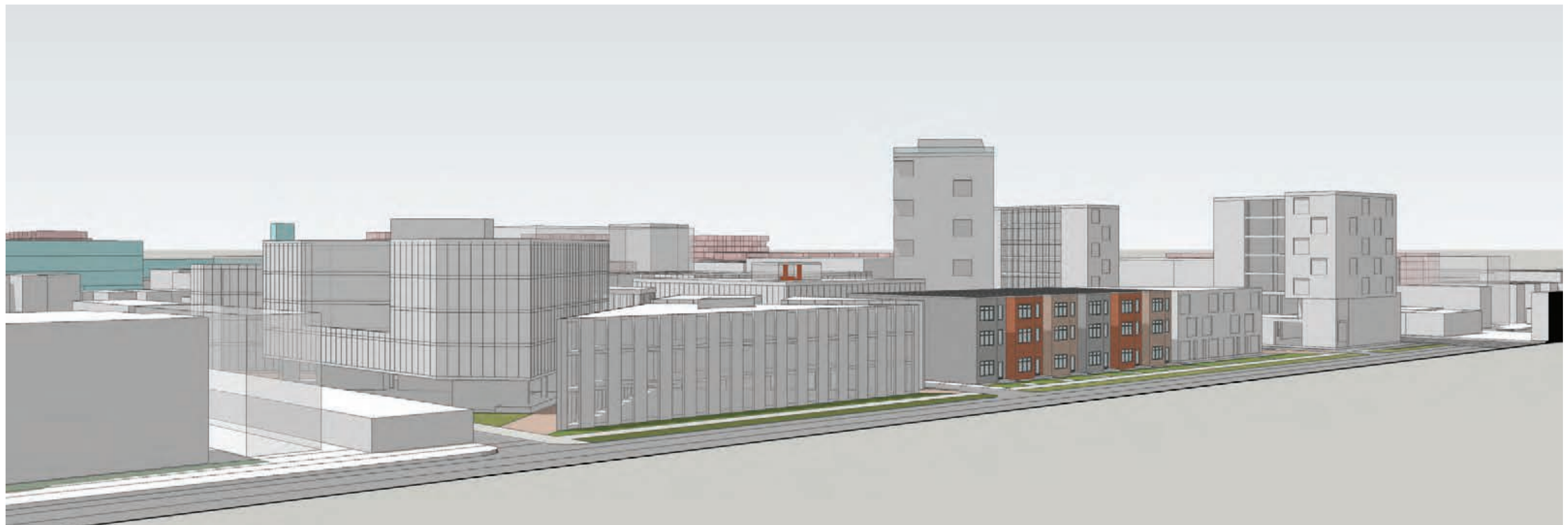




FIRST FLOOR PLAN



SECOND FLOOR PLAN



06

Conclusion

"The Interweaving City" explores strategies for fostering social integration through multi-layered spatial design, addressing challenges faced by marginalized communities and highlighting the limitations of existing urban policies. By examining urban form through three interrelated projects at the city, neighborhood, and block scales, this research proposes alternative approaches to revitalize underutilized areas, promote mixed-income integration, and encourage meaningful social interactions.

Through Exodus City, this thesis investigates how a floating urban design on Lake Michigan can provide displaced populations with a structured yet adaptable environment, using layered spatial configurations to cultivate community-building and belonging. New Bronzeville reimagines a functionally mixed neighborhood, integrating a renovated stadium, a transit hub, and affordable housing to bridge social and spatial divisions while enhancing urban connectivity. Lastly, Permeable Block explores clustered public and private spaces within a mixed-income residential framework, demonstrating how intentional spatial hierarchies can foster everyday interaction and neighborly relationships.

This thesis positions architecture and urban design as critical mechanisms for inclusivity, showing how layered, functionally mixed environments can strengthen communities, break socioeconomic barriers, and create more resilient urban spaces. By synthesizing design strategies with policy-driven approaches, this research offers a framework for building cities that are diverse, accessible, and socially cohesive, ensuring that urban environments serve as platforms for integration rather than segregation.

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